Ecology of American beech and sugar maple in an old-growth forest.

Ken Arii

A thesis submitted to the Faculty of Graduate Studies and Research in partial fulfillment of the requirements for the degree of Doctor of Philosophy

McGill University, Montreal

August 2002

Your file Votre référence
ISBN: 0-612-88415-5
Our file Notre référence
ISBN: 0-612-88415-5

General Abstract

The dynamics in *Fagus-Acer* forest have been frequently investigated, a particular interest being the replacement patterns and coexistence of the two dominant species, *Fagus grandifolia* Ehrh. and *Acer saccharum* Marsh. This thesis examines whether the community pattern and dynamics in a *Fagus-Acer* forest at Mont. St. Hilaire, Quebec, are consistent with predictions made by the disturbance hypothesis. This conceptual model explains the mechanisms underlying the coexistence and replacement patterns of the two dominant species. While the hypothesis suggests disturbance frequency and interspecific differences in growth under variable light conditions to be the key factors that determine the dynamics in *Fagus-Acer* forests, I demonstrate that edaphic factors and regeneration of *Fagus* by root sprouts can also play a significant role. Based on the findings, I propose an alternative model for the coexistence and replacement patterns in *Fagus-Acer* forest to better account for the influence of edaphic factors and of *Fagus* root sprouts.

Résumé général

Les dynamiques forestières de l'érablière à hêtre ont été fréquemment étudiées, en particulier pour les patrons de remplacement et de coexistence des deux espèces dominantes soit *Fagus grandifolia* Ehrh. et *Acer saccharum* Marsh. Cette thèse examine si ces patrons appliqués à une érablière à hêtre situé au mont Saint-Hilaire (Québec) sont consistants avec les prédictions du modèle basé sur l'hypothèse des perturbations. Ce modèle conceptuel explique les mécanismes à la base des patrons de coexistence et de remplacement des deux espèces dominantes. Alors que l'hypothèse suggère que la fréquence des perturbations et les différences interspécifiques de croissance sous différentes conditions de lumière sont les facteurs clés déterminant la dynamique de l'érablière à hêtre, je démontre que les facteurs édaphiques ainsi que la régénération du hêtre par drageon peuvent y jouer un rôle significatif. D'après ces résultats, je propose un modèle alternatif pour expliquer la coexistence et les patron de remplacement dans l'érablière à hêtre afin de mieux tenir compte de l'influence des facteurs édaphiques et du drageonnement du hêtre.

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Preface

In accordance with the rules and regulations set out in the "Guidelines for Thesis Preparation", I am obliged to note that all of the chapters in this thesis have been either accepted for publication or have been submitted for review in journals. Chapter 1 will be co-authored with Benoît Hamel and Martin Lechowicz. Chapters 2, 3, and 4 will be co-authored with Martin Lechowicz. Benoît Hamel assisted in the preparation of the GIS data for Chapter 1. The position of Martin Lechowicz as co-author in all chapters reflects the assistance provided in the collection of the data, discussion of ideas, and editing of the manuscripts. The plural form (we, our) that is used in the four chapters reflects the collaboration with the co-authors, but the large part of the research and writing is solely mine.

It is worth noting that the projects that are included in this thesis turned out quite different from what I had envisioned at the onset of my Ph.D. program in September, 1996. Initially, I intended to identify the differences in regeneration niche between *Fagus grandifolia* Ehrh. and *Acer saccharum* Marsh., and consider the consequences of the differences between the two species for the dynamics in *Fagus-Acer* forests. Based on the ideas put forward by Tilman (1988, "*Plant Strategies and the Dynamics and Structure of Plant Communties*", Princeton University Press), the initial project plans involved setting up experiments that examine the response (both morphological and physiological) of the two species under multiple resource limited conditions in controlled environments as well as in the field. These projects that were planned initially were affected significantly by a major ice storm that hit southwestern Quebec, southeastern Ontario, and the adjacent United States in January 1998. This storm is considered the worst ice storm on record in North America, and the study site, where all my studies were conducted, was one of the hardest hit locations in this extensive storm. While the storm caused major setbacks in terms of initial plans, it provided opportunities to investigate the potential role of ice storms in the ecology of northern hardwood forests. Consequently, initial plans had to be reorganized to accommodate new projects investigating the impacts of ice storms; this involved a major shift from experimental, manipulative studies to field surveys. Although some of the initial experiments were actually completed (e.g. growth response of 1st year *Fagus* and *Acer* seedlings under multiple resource limited conditions - done in a shadehouse), the results from these experiments are not included in the thesis. This thesis contains only studies that are based on field surveys. While the contents included in this thesis have changed considerably from the initial plans and proposals, the main thrust of the thesis, which is to better understand the mechanisms underlying the community pattern and dynamics in *Fagus-Acer* forests, remain unchanged. Also in this thesis, the potential role of ice storms on the dynamics of *Fagus-Acer* forests is discussed considerably, which is only natural given that the initial project plans were changed to accommodate the ice storm influence.

In addition to the four manuscripts (i.e. chapters) that are included in this thesis, two additional manuscripts related to ice storm damage on the trees at our study site have already been published, on which I am a co-author[1]. Two additional manuscripts, on which I will be the lead author, are also in preparation; these concern the changes in herb layer vegetation after the ice storm, and growth of *Acer* seedlings before and after the ice storm (no direct comparison with *Fagus* will be made in this manuscript).

1

Duguay, S., Arii, K., Hooper, M., and Lechowicz, M.J. 2001. Ice storm damage and early recovery in an old-growth forest. Environmental Monitoring and Assessment 67: 97-108.

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Acknowledgements

First and foremost, I thank my supervisor, Dr. Martin Lechowicz for his encouragement, insight, and guidance throughout my studies at McGill University. Due to unexpected turn of events, my research topic had gone through two major changes; his assistance and encouragement during these trying times was vital to the completion of this thesis.

I thank my committee members, Dr. Benoît Côté, Dr. Christian Messier, and Dr. Daniel Schoen, for providing advice and encouragement through the supervisory committee meetings. I am grateful to Dr. Pierre Dutilleul for providing statistical advice.

Collaboration with Stéphanie Duguay and Michael Hooper was essential in investigating the effects of the ice storm of 1998 on the forest at Mont St. Hilaire. Benoît Hamel helped with many of the logistical challenges that I encountered during the thesis write up. He has also translated the French abstract. Marilou Beaudet provided kind and helpful advice on the use of the GLA program and on the hemispherical photographs.

The data used in this thesis would not have been available without the help of a number of tireless, good-humored assistants. I am grateful to Pavel Parfenov, Julien Mercille, Julie Girouard, Jeff Yuen, Daniel Brunet, David Gregoire, Mariam Futer, Rachel Anderson, Regan Morris, and Suzanne Higgs.

I thank McGill University for access to the Gault Nature Reserve and Centre de la Nature du mont Saint-Hilaire for providing the forest cover data. I am grateful to

Marc-André Langlois, Martin Duval, and Denis Bergeron for providing logistical support.

I also thank the past and present members of the Lechowicz Lab for engaging me in insightful and intriguing discussions about ecology, world and local politics, religion, child care, and more. These discussions have made my stay here at McGill particularly a delightful one. Thanks to Dr. Zhang Ming Wang, Dr. Ernest Lo, Mark Vellend, Robin Naidoo, Alison Bond, Stéphanie Duguay, Fabiola Reygadas, Kae Miyazawa, Michael Hooper, Christian Marks, Justine Karst, Benjamin Gilbert, and Benoît Hamel. I am also grateful to the many people in the Biology Department who have taught me various aspects of ecology and evolution. In particular, I wish to thank Jessica Meeuwig, Murray Humphries, Carolyn Hall for their support and friendship.

While becoming an ecologist is certainly not one of the most common and respected career choices in Japan, my parents and grandparents gave me their full support when I decided to pursue my career in ecology. I am thankful for their understanding and their steadfast support. I am also grateful to my brother and sister, who took a few days off of their precious vacation time in Montreal to help me with some of the tasks related to my thesis.

Lastly, but certainly not the least, I would like to thank Chika, my wife. Without her immense support and encouragement, this thesis would not have completed. She has also added a delightful new dimension to my life by giving birth to our son, Ryo. Chika and Ryo are the reasons why I strive to become a better person each day.

My research was supported by grants from Natural Sciences and Engineering Research

Council of Canada (NSERC) and Fonds pour la Formation de Chercheurs et l'Aide à la Recherche (FCAR) to Dr. Martin Lechowicz, and Government of Canada Awards (Department of Foreign Affairs and International Trade) to myself.

General Introduction

With its roots in studies of natural history, human demography, biometry, and applied problems of agriculture and medicine, "ecology" began to emerge as a distinct branch of science by 1900 (Krebs 1994). One of the central topics in ecology at that time was how plant communities vary in space and through time. A Danish botanist J. E. B. Warming published influential books in 1895 and 1909, which raised questions about the structure of plant communities and the associations of species in these communities. In the same time period, the dynamics of vegetation change were documented and emphasized by a North American ecologist, H. C. Cowles, in a series of papers published in 1899. Much progress has been made in the century following these pioneering studies (McIntosh, 1985), but describing and comparing plant communities in space and through time remains an important topic in the field of plant ecology today.

One of the reasons why the topics of community composition and dynamics have persisted as important issues in plant ecology is because of the complexity and the enormity of the data sets being used in community-level studies (Begon et al. 1990). Simply describing and comparing communities can be a daunting task. As a consequence, progress made in the field of plant community ecology in the past century has been closely linked with developments made in methods to describe and compare patterns in plant communities (e.g. statistical methods such as ordination and cluster analysis; Barbour et al. 1987, Legendre and Legendre 1998). In the past quarter century, significant developments in the statistical methods have been made, accompanied by the technological advancements in computers necessary to perform these computationally intensive methods (see Jongman 1995). These advances have made description and comparison of plant communities less time consuming and more manageable and interpretable, and have lead to better recognition of patterns of plant communities in space and through time.

The recognition of patterns represents an important step in the development of all sciences. As patterns in plant communities are identified, the question inevitably arises as to the causes of these patterns. Understanding the processes that give rise to the patterns will allows us to predict, to a certain degree, the spatial and temporal variation in plant communities. Further, this understanding will better position us to assess the potential impacts of the rapidly changing environment on plant communities, and will provide better guidelines for conservation and management of natural habitats.

The pattern of *Fagus-Acer* forests in eastern North America

The terrestrial ecosystems on the earth's continents can be divided into large units based on their physiognomy (Beard 1973), which is defined by the appearance and the structure of dominant vegetation members (e.g. deciduous forests, coniferous forests, savannas, deserts, etc). Such units are referred to as *formation types* (Whittaker 1973). Of the various formation types, one of the most intensively studied is the temperate deciduous forest, which currently occurs in fairly restricted areas of North America, eastern Asia, and Europe (Fig. 1; note that smaller areas occur in the Near East and Patagonia, Chile, Röhrig 1991a). Deciduous forests once occupied more extensive continental areas in the temperate portions of the northern hemisphere in the Tertiary period (Röhrig 1991b), which is when a connection between the Eurasian continent and North America still existed ("Arcto-Tertiary Flora", Di Castri et al. 1981). However, subsequent environmental changes, such as movements of land masses (e.g. separation of North America from Eurasia) and climatic change, greatly reduced the extent of deciduous forests (Hicks and Chabot 1984). Additionally, several

Figure 1. Distribution of temperate deciduous forests.

glaciations during the Pleistocene gave rise to substantial differences in the floral composition of the current temperate deciduous forests (Röhrig 1991b). But, due to exchanges during the Tertiary period, strong floristic and ecological similarities still remain among the three major deciduous forest zones of the northern hemisphere (Graham 1972, 1999).

Similarities in the floristic composition are recognized by the fact that significant numbers of the same families and genera are found in all three major temperate deciduous forest zones. Of these families, Fagaceae (beech family) is particularly typical, and is the most important family that characterizes the forests in these regions (Röhrig 1991b). The genus *Quercus*, a member of the Fagaceae, has a particularly rich diversity of species (more than 250 species), and is spread mainly throughout the warmer portions of the broadleaved regions, although some species extend into the northern areas (e.g. *Quercus rubra*, *Quercus robur*, *Qurecus mongolica*) (Röhrig 1991b). Another member of the Fagaceae, the genus *Fagus* itself, is also an important constituent of the temperate deciduous forest zone. In contrast to *Quercus*, *Fagus* only has 11 species worldwide, five of which are found in southeastern China and Taiwan (Peters 1992, note that some *Fagus* species in southeastern China may yet remain to be identified). From the central to the northern latitudes of the temperate deciduous forest zone, three *Fagus* species are particularly known to become dominant components in the mesic forests in the region (Röhrig 1991b). These three species occur separately in the three major temperate deciduous forest zones: the three species are *Fagus sylvatica* L. (Europe), *Fagus crenata* Blume (east Asia - Japan), and *Fagus grandifolia* Ehrh. (eastern North America). In areas where they dominate the forest (i.e. beech forest), *Fagus sylvatica* and *Fagus crenata* typically form mono-dominant stands (Röhrig 1991b, Hara 1992, Peters 1992, Maycock 1994). However, in the deciduous forests of eastern North America, mono-dominant stands of *Fagus grandifolia* are rarely found, and in the northern latitudes, codominance with *Acer saccharum* Marsh. is typical (Hara 1992, Maycock 1994). This type of forest co-dominated by *Fagus grandifolia* and *Acer saccharum*, referred to as *Fagus-Acer* forest, is recognized as one of the major forest types found in eastern North America (Fig. 2; Braun 1950, Runkle 1996). In the United States, *Fagus-Acer* forest type occupies much of the Till Plains of Ohio and Indiana, the western end of Glaciated Allegheny Plateau in northern Ohio and western Pennsylvania, and the southern half of the Lower Peninsula of Michigan and into southeastern Wisconsin (Braun 1950, Hicks and Chabot 1984). In Canada, *Fagus* and *Acer* are the dominant trees in the Niagara, the Huron-Ontario, and the Upper St. Lawrence section (Halliday 1937, Rowe 1972).

Mechanisms underlying the dynamics in *Fagus-Acer* forests

The *Fagus-Acer* forest, which is a unique form of "beech forest" found only in North America, has intrigued many researchers, a particular interest being the replacement patterns and coexistence of the overstory trees (Forcier, 1975; Fox, 1977; Woods, 1979, 1984; Runkle, 1981, 1984; Cypher & Boucher, 1982; Brisson et al., 1994; Kupfer & Runkle, 1996; Poulson & Platt, 1996; Foré et al., 1997; Beaudet et al., 1999; Forrester & Runkle, 2000). One of the hypotheses to explain the coexistence and the replacement patterns in *Fagus-Acer* forests is reciprocal replacement, which suggests that seedlings and saplings of each species tend to grow and survive better under canopies of the other species than under conspecific adults (Fox 1981, Woods 1979, 1984, Runkle 1981). Thus, through time, *Fagus* and *Acer* dominate the canopies in turn, which gives its name, "reciprocal replacement". However this pattern has not been always observed (e.g. Forcier 1975, Brisson et al. 1994), and thus lacks general applicability. More recently, Poulson and Platt (1996) have proposed a conceptual

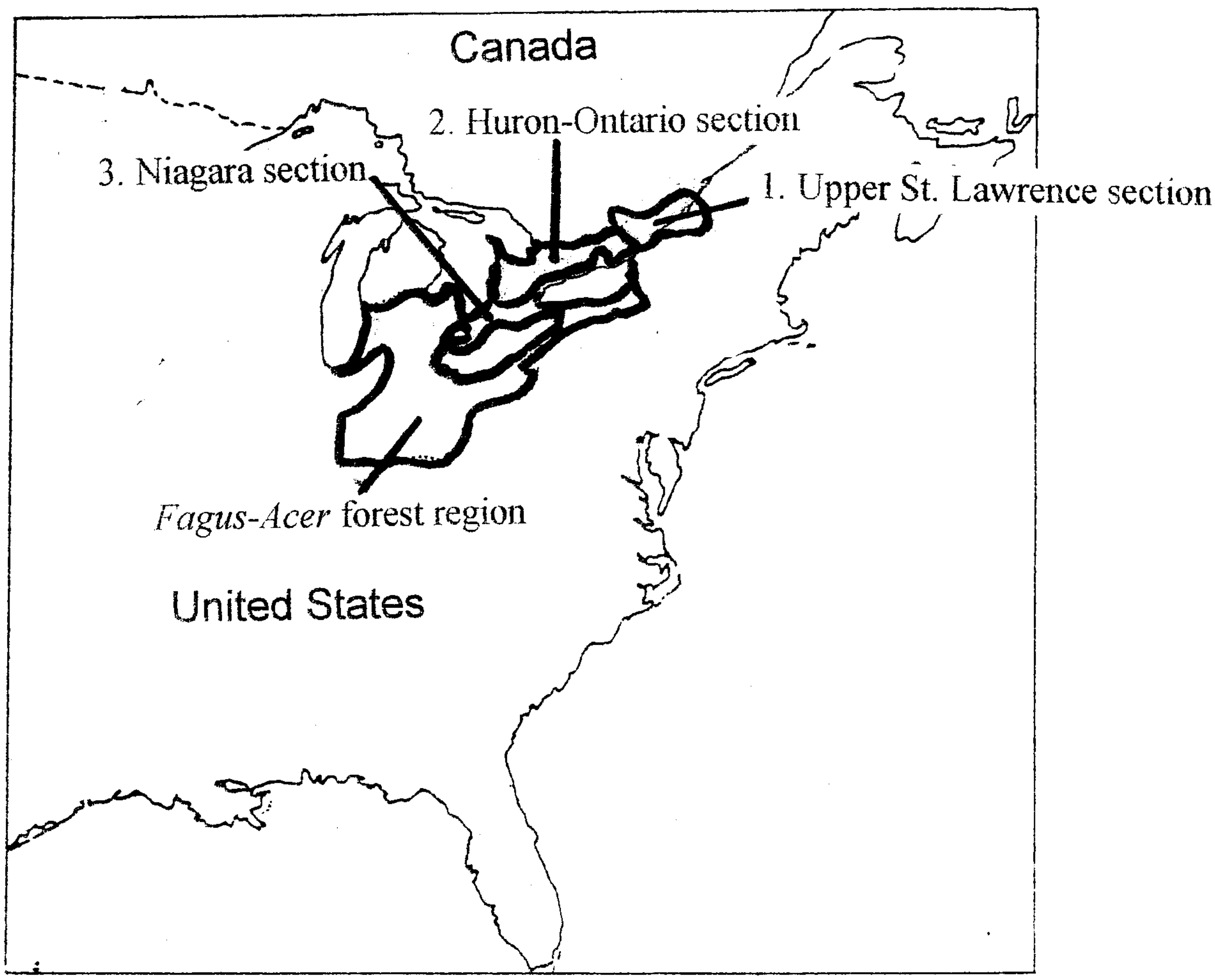

Figure 2. Distribution of *Fagus-Acer* forest region in the United States (Braun 1950), and Niagara, Huron-Ontario, and Upper St. Lawrence sections in Canada where *Fagus* and *Acer* dominate (Halliday 1937, Rowe 1972).

model that explains the mechanisms underlying the co-existence and the replacement patterns in *Fagus-Acer* forests (Fig. 3). In short, their model suggests that *Fagus* gains importance when the disturbance is infrequent (low light conditions in the understory persist for extended periods of time), while *Acer* increases importance when the disturbance is frequent (gaps are created and light availability temporarily increases). This prediction is based on species' differences in growth response under varying light conditions: *Fagus* survives for longer periods of time and grows faster upward in the shaded understory than *Acer*, while *Acer* grows more rapidly upward in gap conditions (Canham 1988; Poulson and Platt 1989, 1996). Thus this known difference in species' response to light, coupled with the frequency of disturbances, which creates temporal and spatial variations in understory light conditions, is the mechanistic basis of the model (hypothesis) suggested by Poulson and Platt (1996). Since disturbance plays the major role in this hypothesis, I will refer to this conceptual model as the "disturbance hypothesis".

While the disturbance hypothesis is quite convincing in explaining the coexistence and the replacement patterns of *Fagus* and *Acer*, it leaves out an important characteristic of *Fagus*, which is their ability to regenerate vegetatively by root sprouts. Interestingly, *Fagus grandifolia* is the only species in the entire genus that is known to regenerate by root sprouts (Hara et al. 1992). However, previous studies, including the study by Poulson and Platt (1996), have paid relatively little attention to *Fagus* sprouts in understanding the *Fagus-Acer* forest dynamics. This, in large part, is because *Fagus* sprouts do not occur frequently at most locations where previous studies were conducted. Held (1983) and Kitamura and Kawano (2001) show latitudinal and regional differences in the relative occurrence of *Fagus* root sprouts. Warren Woods (Michigan), the study site where Poulson and Platt conducted their study, is known to have very few *Fagus* individuals of sprout-origin (Ward 1961). Thus it is quite natural

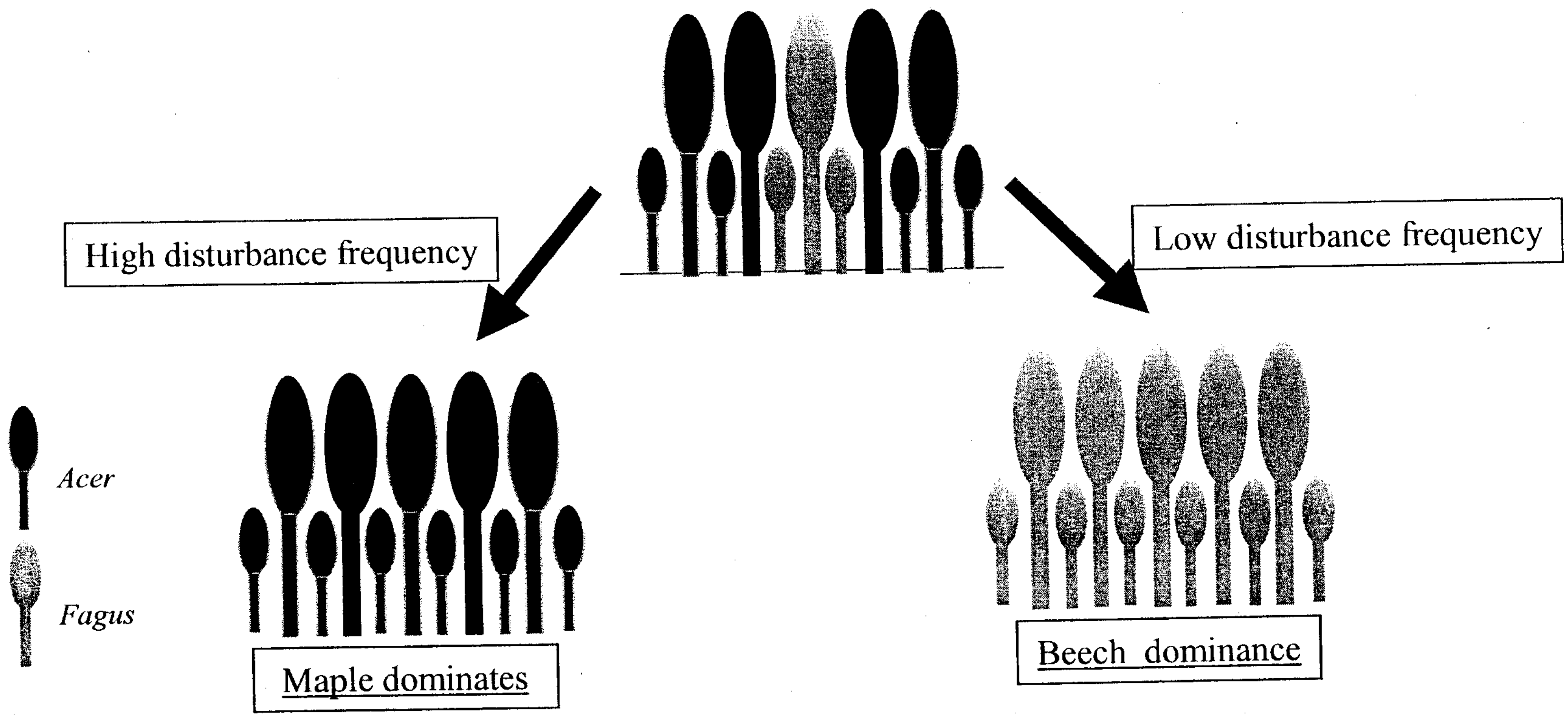

Figure 3. Illustration of the disturbance hypothesis (Poulson and Platt 1996). Under high disturbance frequency, *Acer* will dominate because of its ability to grow faster upward than *Fagus* under increased light conditions. Under low disturbance frequency *Fagus* will dominate because of its ability to survive and grow better than *Acer* in the shaded understory.

for them to leave *Fagus* sprouts out of their conceptual model. However, several authors have suggested the possibility of *Fagus* sprouts playing a significant role in *Fagus-Acer* dynamics (Forcier 1975, Melancon and Lechowicz 1987, Brisson et al. 1994, Beaudet et al. 1999), since sprout-origin *Fagus* can have different growth characteristics from the seed-origin *Fagus*. Obviously, the importance of sprouts will be more conspicuous in areas where the sprouts occur extensively, and in these areas, the community patterns and dynamics in the *Fagus-Acer* forest may be inconsistent with the predictions made by the disturbance hypothesis.

Another factor that the disturbance hypothesis does not consider is the effect of edaphic factors, such as soil nutrients and moisture. The consensus among foresters and plant ecologists is that forest succession and dynamics is driven largely by the interspecific differences in shade tolerance and by temporal and spatial variations in light availability (Kobe et al. 1995). The disturbance hypothesis is basically built on the same principle. However, numerous studies have shown that other factors, such as soil nutrient and moisture, affect growth, composition, and distribution of species in temperate deciduous forests (e.g. Boerner and Cho 1987, Walters and Reich 1996, Casperson and Kobe 2001), and there is no reason not to believe that edaphic factors influence the dynamics in *Fagus-Acer* forests. Gradient analyses of forest communities where *Fagus* and *Acer* are found have shown some evidence that these two species occur at different positions along a gradient of soil moisture or nutrient availability (e.g. Op de Beek 1972, Gauch and Stone 1979, Woods 2001). These findings suggest that edaphic factors may differently affect the growth and mortality of *Acer* and *Fagus*, which in turn can affect the coexistence and the replacement patterns of *Fagus* and *Acer*. Therefore, along with the influence of *Fagus* sprouts, edaphic factors may also exert significant influence on the dynamics of *Fagus-Acer* forests.

The objectives of the thesis and a brief overview

Held (1983) and Kitamura and Kawano (2001) suggest that *Fagus* root sprouts are found more extensively towards the northern limit of the distribution range of *Fagus*. The studies included in this thesis were conducted at a site near the northern limit of *Fagus*, and it has been shown previously that *Fagus* root sprouts occur extensively at this site (Kitamura et al. 2000). The study site also offers considerable variation in soil moisture and nutrient availability (e.g. Webber and Jellema 1965, Rouse and Wilson1969), partly due to its physiographic diversity (see Fig. 1 in Chapter 1). This thesis then investigates the community pattern and dynamics in a *Fagus-Acer* forest where *Fagus* regenerates primarily by root sprouts, and where there is considerable variation in edaphic factors. I consider whether or not the community pattern and dynamics in this old-growth *Fagus-Acer* forest at the northern edge of the deciduous forests in North America are consistent with predictions made by the disturbance hypothesis. The following paragraph provides a brief description of each chapter included in this thesis, and a more detailed literature review is given in the introduction of each chapter.

Chapter 1 investigates the current pattern of canopy tree composition at Mont St. Hilaire (MSH), the study site, and identifies abiotic factors that may give rise to this pattern. This chapter is intended to provide a general overview of the canopy tree species and the different canopy tree associations found at the study site. **Chapter 2** assesses the impact of the ice storm of 1998 on the forest canopy found at our study site (Fig. 4). Natural disturbances play an important role in determining the species composition, structure, and dynamics in deciduous forests of eastern North America (Lorimer 2001). The events that disrupt the forest canopy are also central to the disturbance hypothesis of *Fagus-Acer* forests. In the regions surrounding the study site, major ice storms are the most frequent form of natural disturbance, with a return time

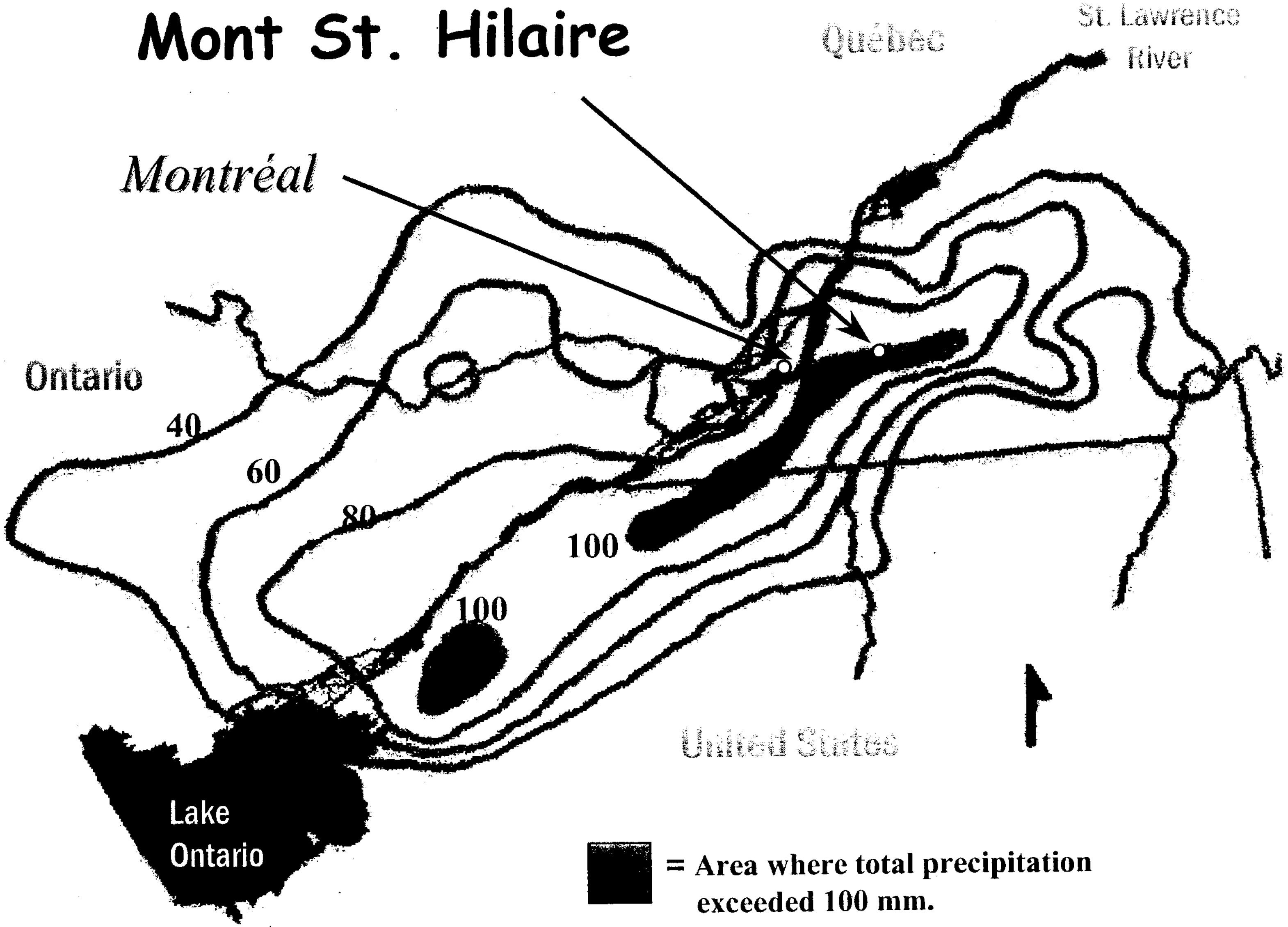

Figure 4. Distribution map of total precipitation (mm) during the ice storm of 1998 (Jan. 4-10), in the St. Lawrence River valley. Precipitation fell mainly as freezing rain

of approximately 20 years. With such frequent occurrence, ice storms are likely to have significant impact on the forests in this region. This chapter reports the changes in the light regime following the ice storm of 1998, and discusses the implications of the findings on the species composition and dynamics of forests found at the study site. **Chapter 3** attempts to identify the factors that determine the patterns of spatial variation and species composition of the sapling community at MSH. The effects of overstory trees and of edaphic factors, such as soil moisture and soil nutrients, are considered as potential factors that can determine the spatial variation and species composition. The distribution pattern of *Fagus* and *Acer* saplings will be examined if they are consistent with the predictions made by the disturbance hypothesis. **Chapter 4** compares the height growth of *Acer* and *Fagus* trees in gap light conditions following an ice storm. *Acer* seedlings and saplings are generally considered to have higher height growth than *Fagus* under increased light conditions: this difference in species' response is an important assumption of the disturbance hypothesis. However, this known difference between the two species is based on comparisons between *Acer* and seed-origin *Fagus*: *Fagus* is known to regenerate by root-sprouts, which may have different light response from seed-origin *Fagus*, as they receive "subsidies" from the parent tree. This chapter examines whether the previous notion that *Acer* grows better than *Fagus* under increased light conditions holds when sprout-origin *Fagus* is considered.

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Chapter 1

Environmental correlates of canopy composition in an old-growth forest.

Abstract

The environmental basis for variation in canopy composition was investigated in an extensive old-growth forest at Mont St. Hilaire, Quebec. Based on 146 permanent plots, spatial variation in canopy tree species and the effect of environmental variables on canopy composition were examined using canonical correspondence analysis (CCA). Slope and the amount of direct solar radiation received during the growing season, both of which are good indicators of soil moisture, were the main factors explaining plot-to-plot variation in canopy composition. *Quercus rubra*, *Betula papyrifera*, *Ostrya virginiana* and *Pinus strobus* predominated on plots with high insolation and steeper slope, while species such as *Acer saccharum*, *Fagus grandifolia*, *Betula alleghaniensis*, and *Tilia americana* occurred on sites with gentler slope and lower insolation during the growing season. Additionally, plots with greater dominance of *Acer saccharum* in the canopy had higher soil nitrogen availability, and plots with greater dominance of *Fagus grandifolia* had lower Ca availability.

Introduction

Mont St. Hilaire (MSH), located approximately 32 km east of Montreal, is a hill complex that rises prominently from the surrounding St. Lawrence River valley. It is one of eight monadnocks referred to as the Monteregian hills (Adams 1903), epizonal intrusions of Cretaceous age that penetrated Ordovician limestones and shales (Webber and Jellema 1965; Feininger and Goodacre 1995). MSH lies within the Upper St. Lawrence section of the Great Lakes-St. Lawrence Forest region (Rowe 1972), an area near the northern limit of the temperate deciduous forests of eastern North America.

European immigrants settled in the St. Lawrence Valley in the vicinity of Montreal beginning in the early 17th Century. Due to the continuous influx of settlers and their extensive agricultural practices, almost all of the original forest cover in this region has been cleared in the last 400 years (Rowe 1972; Ricketts et al. 1999). However, the Monteregian Hills were left as stranded forest "islands", in part due to their unsuitability as agricultural land, although they have been used to varying degrees for timber, maple sugar production and recreation (Maycock 1961). Among these forest islands, MSH is the least disturbed by human activities because of its relative inaccessibility and a long period of private ownership (Maycock 1961; Cook 1971). Many trees on MSH exceed 150 years in age and some are over 450 years old (Cook 1971). MSH is considered the largest remnant of primeval forests in this region; its scientific and cultural significance was recognized in 1978 when it became the first Canadian site to be designated an International Biosphere Reserve under the Man and Biosphere program (MAB) of the United Nations.

There have been several surveys of the flora and vegetation at MSH (e.g.

Marie-Victorin 1913; Maycock 1961; Cook 1971), two of which have focused on the compositional and spatial patterns of trees on the mountain (Phillips 1972; Enright and Lewis 1985). Phillips (1972) investigated the aspect-related differences in tree composition by sampling and comparing data obtained from eight geomorphic units that were defined prior to the survey (bogs, lowlands, summits, cols, north-slope, south-slope, east-slope, and west-slope). While the data collected by Phillips were quite extensive, site-specific environmental information was not recorded. Therefore, although the results are useful as a broad description of vegetation types found at MSH, relationships between species distribution and environmental conditions are obscure. Enright and Lewis (1985) attempted to identify the vegetation-environment relationships more rigorously by sampling quadrats with site-specific information on environment. The environmental variables they obtained included aspect and soil temperature, which were measured directly at each quadrat, and also estimated values of global radiation (direct and diffuse radiation) and evapotranspiration. While their method was more efficient in identifying the environmental basis of tree distributions at MSH, their sampling locations were restricted to a small corner of the mountain, and half of their localities were likely to have been affected by a fire that had occurred about 40 years prior to their survey.

Thus, although there have been attempts to identify the environmental correlates of tree distributions at MSH, we only have a limited and qualitative understanding of these relationships. This is unfortunate given the biological and ecological significance of the mountain in the region. In this study, we identify the environmental correlates of variation in canopy composition at MSH by using canonical correspondence analysis (CCA). Our data comes from a set of long-term monitoring plots that were established in the last few years. For the environmental variables, factors such as soil moisture and nutrient availability were measured directly at the plots and augmented by physical environmental variables derived from a digital elevation model (DEM) for MSH.

Methods

Study site

We assessed the effects of environmental factors on the distribution pattern of canopy trees in a 10 km^2 tract of old-growth forest at Mont St. Hilaire (MSH) in southwestern Québec (45°31'N, 73°08'W). The mean annual temperature in this region is 5.9°C (minimum and maximum monthly mean temperatures are –10.2°C (Feb.) and 20.6°C (Aug.), respectively), and mean total annual precipitation is 1017 mm (Environment Canada 2002; all climatic data are for St. Hubert (A) Climate Station (1928-1990), 45° 31'N, 73°25'W, on the valley floor approximately 20 km west of MSH). The mean annual freeze-free period is 140 days (Wilson 1971; based on data between 1931-1960). The soil type and soil texture at MSH is mainly ferro-humic podzol and sandy loam, respectively (Beaumont 1980; Grenon et al. 1999). The C horizon is believed to be mainly absent, except at lower elevation and in valley bottoms (Beaumont 1980; Grenon et al. 1999).

MSH lies in a region where the dominant forest community is composed chiefly of *Acer saccharum* Marshall and *Fagus grandifolia* Ehrh., with *Betula alleghaniensis* Britton, *Acer rubrum* L., *Tilia americana* L., *Fraxinus americana* L., *Populus grandidentata* Michx., and *Quercus rubra* L. (Rowe 1972, "Upper St. Lawrence section of the Great Lakes-St. Lawrence Forest region"), although there is a fairly diverse range of forest communities found in the St. Lawrence River Valley and adjacent areas (e.g. Dansereau 1943; Dansereau 1959; Walther 1963; Op de Beeck 1972; Bouchard and Maycock 1978; Bouchard 1979; Gagnon and Bouchard 1981; Gauvin and Bouchard

1983; Domon et al. 1986; St-Jacques and Gagnon 1988; Gauthier and Gagnon 1990; Lamontagne et al. 1991; Brisson et al. 1992; Ansseau 1993; Meilleur et al. 1994). Various tree communities are also found at MSH (Maycock, 1961; Phillips 1972; Enright & Lewis, 1985), but *Acer saccharum*, *Fagus grandifolia* and *Quercus rubra* are the three dominant species. Maycock (1961) provides a list of species that are found on the mountain; 41 species of trees on the mountain reach canopy or subcanopy positions.

Forest cover

To provide a better description of the different forest cover types found at our study site, we analyzed a digital map of forest cover for MSH. The digital map, created by DESFOR (Région Montérégie, Verchères, Québec), is based on an interpretation of infrared aerial photographs (scale 1:15 000) taken in 1994 (Centre de la Nature du Mont Saint-Hilaire 2002). The map identifies 128 different polygons (i.e. cover types), 119 of which are forested (minimum polygon surface area was set to 1 hectare). For each polygon, species that had less than 10 % cover were not recorded. Thirteen different species were recognized for the entire map. This data set of forested polygons (119 plots, 13 species) is analyzed with correspondence analysis (CA), which will be used to describe the broad variation in forest cover types found at MSH. CANOCO (ver. 4.0, ter Braak & Šmilauer, 1998) was used to perform the CA, and the "downweight" function in CANOCO was used to reduce the effects of rare species (weighted total frequency is used as weights instead of total frequency, see ter Braak & Šmilauer, 1998 for details).

Canopy tree-environment relationship

Collecting tree data

In the summers of 1997 and 2000, we established 146 permanent plots to sample forest variation over the mountain (Fig. 1). Each plot is 113 m^2 (6-m radius). We recorded the geographical coordinates of each plot using a Trimble GPS Pathfinder Pro XRS (Trimble Navigation Limited, Sunnyvale, CA, U.S.A.). At each plot, we recorded all woody individuals with DBH greater than 10 cm (species, DBH). We converted the data to importance values; for each plot, the importance value of species i (IV_i) is calculated as:

$$IV_i = (BA_i / BA_{tot}) + (DN_i / DN_{tot}) \qquad (0 \leq IV_i \leq 2)$$

where, BA_i = total basal area of species i in a plot, BA_{tot} = total basal area in a plot, DN_i = stem density of species i in a plot, DN_{tot} = total stem density in a plot. The tree species identified in the plots appear in Table 1, all of which are native species typically found in the forests of the St. Lawrence River valley (Walther 1963; Op de Beek 1972; Rowe 1972).

Collecting environment data

A list of abiotic environmental variables included in the analysis appears in Table 2.

Soil nutrients

In August 2000, we took soil samples to 10 cm depth at 8 different locations near the center of each plot, first removing the L-F-H layers. We pooled samples taken at each

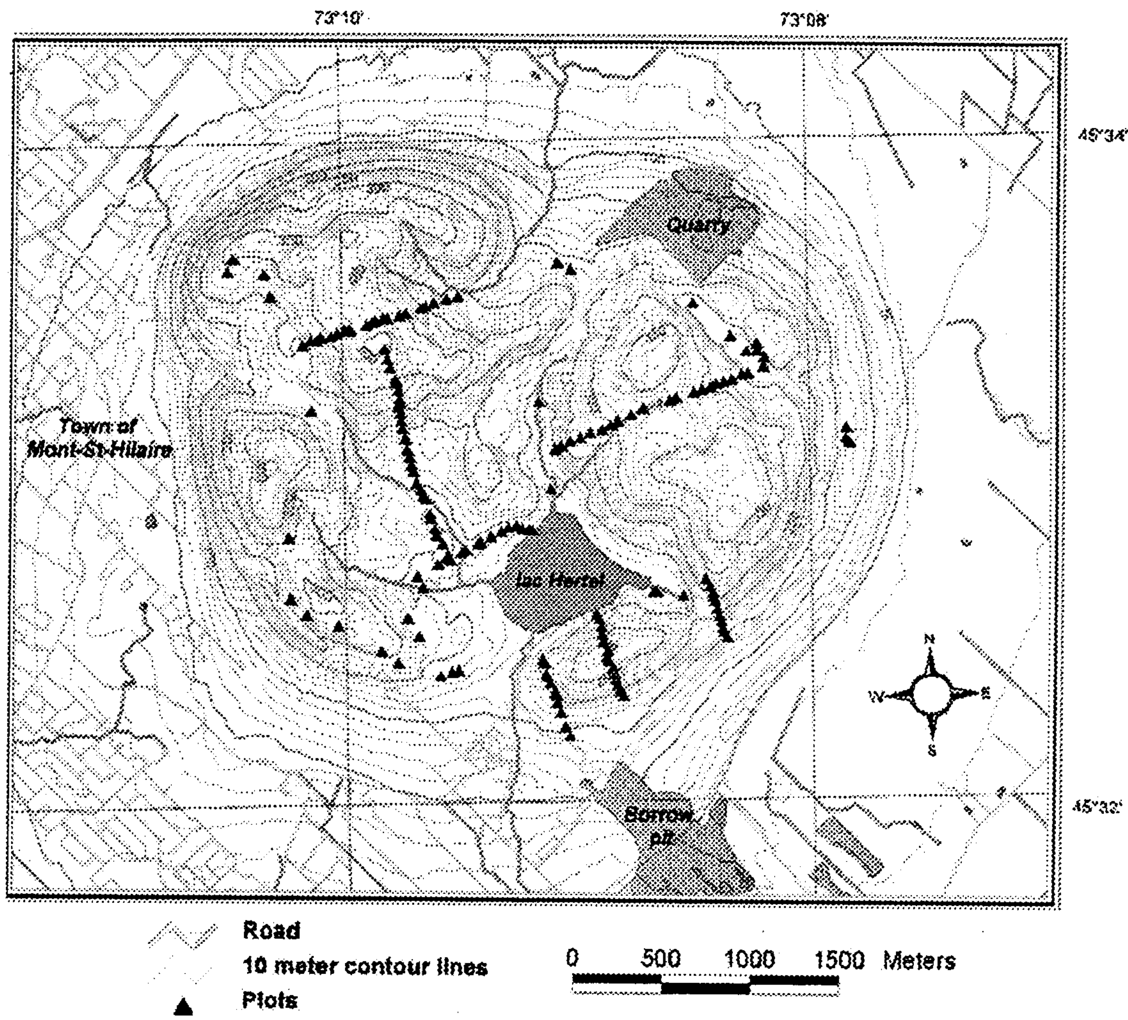

Figure 1. Map of Mont St. Hilaire. The triangles in the diagram indicate plot locations.

Table 1. List of species identified in the 146 permanent plots at Mont St. Hilaire. Their acronyms and basic statistics are shown. Nomenclature follows Gleason and Cronquist (1991).

Species	Acronym	Occurrence (out of 146 plots)	Stem density (per plot)				Basal area (m^2 ha^{-1})					Importance value				
			Max	Median	Mean	S.D.	Min	Max	Median	Mean	S.D.	Min	Max	Median	Mean	S.D.
Acer pensylvanicum L.	Apn	10	4	1	2.0	1.4	0.7	5.3	1.4	2.2	1.7	0.11	1.22	0.28	0.43	0.36
Acer saccharum Marshall	Asa	123	12	3	3.0	2.0	0.7	84.0	13.3	19.5	18.5	0.07	2.00	0.83	0.95	0.58
Acer spicatum Lam.	Asp	3	2	1	1.3	0.6	0.9	1.6	1.0	1.1	0.4	0.31	0.86	0.36	0.51	0.30
Betula alleghaniensis Britton	Bal	15	2	1	1.1	0.3	1.1	35.8	3.6	8.1	10.4	0.12	1.22	0.24	0.36	0.30
Betula papyrifera Marshall	Bpa	20	7	1	1.6	1.6	0.7	8.4	3.8	4.0	2.9	0.10	0.81	0.22	0.31	0.22
Carya ovata (Miller) K. Koch	Cov	2	1	1	1.0	0.0	0.8	1.2	1.0	1.0	0.3	0.15	0.30	0.22	0.22	0.10
Fraxinus americana L.	Fam	37	10	2	2.3	2.0	0.8	27.3	6.8	7.9	5.9	0.13	1.74	0.49	0.54	0.37
Fagus grandifolia Ehrh.	Fgr	67	11	2	3.1	2.2	0.8	66.3	12.2	15.0	14.7	0.14	2.00	0.66	0.82	0.57
Juglans cinerea L.	Jcn	1	1	1	1.0	n/a	3.7	3.7	3.7	3.7	n/a	0.25	0.25	0.25	0.25	n/a
Ostrya virginiana (Miller) K. Koch	Ovg	28	5	1	1.8	1.1	0.7	5.6	1.1	2.0	1.5	0.07	0.93	0.21	0.28	0.22
Populus grandidentata Michx.	Pgr	2	1	1	1.0	0.0	0.8	1.3	1.1	1.1	0.4	0.13	0.21	0.17	0.17	0.06
Pinus strobus L.	Pst	5	2	1	1.2	0.4	1.0	23.2	3.6	7.7	9.0	0.13	0.43	0.24	0.26	0.13
Prunus pensylvanica L.f.	Ppn	1	1	1	1.0	n/a	0.7	0.7	0.7	0.7	n/a	0.17	0.17	0.17	0.17	n/a
Prunus serotina Ehrh.	Pse	1	1	1	1.0	n/a	9.8	9.8	9.8	9.8	n/a	0.56	0.56	0.56	0.56	n/a
Prunus virginiana L.	Pvg	1	3	3	3.0	n/a	3.1	3.1	3.1	3.1	n/a	0.45	0.45	0.45	0.45	n/a
Quercus rubra L.	Qrb	56	15	3	3.4	2.8	0.7	48.8	17.0	19.2	12.6	0.15	2.00	0.96	0.97	0.51
Tilia americana L.	Tam	9	3	1	1.3	0.7	0.9	15.6	3.2	5.9	5.2	0.13	1.27	0.28	0.39	0.36
Tsuga canadensis (L.) Carriére	Tca	11	6	2	2.7	1.9	3.5	29.2	7.1	12.7	9.9	0.15	2.00	0.78	0.92	0.58
Thuja occidentalis L.	Toc	1	1	1	1.0	n/a	3.0	3.0	3.0	3.0	n/a	0.24	0.24	0.24	0.24	n/a
Ulmus americana L.	Uam	1	1	1	1.0	n/a	27.0	27.0	27.0	27.0	n/a	0.81	0.81	0.81	0.81	n/a

* Minimum, maximum, mean and S.D. are calculated based on non-zero values

Table 2. List of environmental variables included in the analysis. Their acronyms, basic statistics, and transformations used are shown. Note that aspect, a nominal variable, is not shown in the table.

Variable	Units	Acronym	Transformation	Mean	S.D.	Median	Range	Range reported in Walthers (1963) (n=34)	Range for samples taken in 1997 (n=92)
Total Nitrogen	ppm	TN	$Log_{10}(x)$	89	75	60	16-584	n/a	9-162
Mg	ppm	MG	$Log_{10}(x)$	167	121	124	21-632	28-840	159-924
Ca	ppm	CA	$Log_{10}(x)$	2748	2675	2027	179-14054	1420-12000	190-9942
K	ppm	K	$Log_{10}(x)$	174	89	153	37-487	160-1184	31-438
pH	-	PH	-	4.8	0.5	4.8	3.6-6.3	3.9-7.0	3.6-6.6
Moisture (Jul. 13)	$m^3\ m^{-3}$	MST1	$arcsin(x^{1/2})$	0.34	0.20	0.29	0.07-1.00	n/a	n/a
Moisture (Aug. 20)	$m^3\ m^{-3}$	MST2	$arcsin(x^{1/2})$	0.35	0.24	0.26	0.10-1.00	n/a	n/a
Elevation	m	ELV	-	249	64	239	126-400	n/a	n/a
Insolation (direct)	$kWh\ m^{-2}$	IDIR	-	658	55	670	419-728	n/a	n/a
Insolation (diffuse)	$kWh\ m^{-2}$	IDIF	-	690	26	699	591-741	n/a	n/a
Slope	degrees	SLP	-	14	9	13	0-37	n/a	n/a

plot and froze them within hours of collection. The samples were kept frozen before being air-dried and sieved (2 mm) in preparation for soil chemical analysis. Nutrient extractions and analyses of these extracts were carried out by the Soil Chemistry Laboratory at Université Laval. Mg, Ca and K were assayed in the Mehlich III extractant (Mehlich 1984), and were analyzed with ICP (Plasma 40; Perkin-Elmer Instruments, Boston, MA, U.S.A.). NH_4^+ and NO_3^- were assayed in a 2M KCl extract (Page et al. 1982) and analyzed with a QuikChem 4000 (Lachat Instruments, Milwaukee, WI, U.S.A.): NH_4^+ and NO_3^- were analyzed using the QuikChem Method 12-107-06-2-A (Lachat Instruments 1990) and the QuikChem Method 12-107-04-1-B (Lachat Instruments 1992), respectively. Total mineral N (NO_3-N + NH_4-N) was used in the statistical analysis.

Soil moisture

We measured volumetric moisture contents twice during the summer of 2000 (July 13, August 20) using frequency domain reflectometers (Theta Probe, Type ML2x, Delta-T Devices, Ltd., Cambridge, U.K.). It took two days to make measurements at all plots during each sampling period, and the sampling done in the same order on all sampling dates to minimize the time required to visit all plots. There was no rainfall during the sampling periods. At each plot, we determined average soil moisture from 5 random locations near the center of the plot.

Physical environmental factors

We calculated the physical environmental factors used in this study (elevation, slope, aspect, insolation) for each sampling plot using a digital elevation model (DEM) of MSH (minimum pixel size is 5 m x 5 m). To create the DEM, we interpolated a digital map (Ministère des Ressources Naturelles 1999; scale 1:20 000, 10 m contour line intervals) using Idrisi32 Release 2 (Clark Labs, Worcester, MA, U.S.A.). Arcview 3.2 and the Spatial Analyst extension 2.0a (Environmental Systems Research Institute, Inc., Redlands, CA, U.S.A.) were used to calculate elevation, aspect and slope for each sampling plot. Insolation was estimated with Solar Analyst extension 1.0 (Helios Environmental Modeling Institute, Vermont, KS, U.S.A.). Insolation is defined as cumulative energy (kWh) per square meter for every hour of sunshine received during the growing season (1 April to 30 September). Various factors are taken into account to calculate insolation, which includes latitude, elevation, aspect, shadows cast by surrounding topography, daily and seasonal shifts in solar angle, and atmospheric attenuation. Transmittivity of the atmosphere was fixed at 0.5. We divided global insolation into direct and diffuse insolation by fixing the diffuse insolation proportion to 0.53. This value is estimated using the equations provided in Frazer et al. (1999) and by using the hourly global radiation data measured at weather stations on the island of Montreal between 1961 and 2001 by Environment Canada (1961-1986 - Jean Brebeuf weather station 45°30'N, 73°37'W 1961-1986: 1987-2001 - Dorval Airport weather station 45°28'N, 73°45'W). We only used 34 years of data between 1961 and 2001, because some years had incomplete data.

We converted the aspect data (0-359°) to a nominal variable with five separate classes, which are defined as follows. The plots that had less than 5° slope were classified as "Flat (F)". Plots with more than 5° slope were classified as facing East (E), South (S), West (W), or North (N). "East" plots had aspect ranging between 45-135°, south plots between 135-225°, West plots between 225-315°, and North plots between 0-45° and 315-359°.

Statistical analysis

We used canonical correspondence analysis (CCA) to investigate the effects of environmental variables on the tree composition. CANOCO was used to perform the analyses, and the "downweight" function in CANOCO was used to reduce the effects of rare species. Environmental variables were transformed (Table 2) and standardized to zero mean and unit variance prior to analysis. This standardization makes the canonical coefficients comparable to each other, but does not influence other aspects of the analysis (ter Braak 1986). Significance of the sum of all canonical eigenvalues and the significance of the first two ordination axes were tested using Monte Carlo permutation tests (1000 permutations) (ter Braak and Šmilauer 1998; Legendre and Legendre 1999).

Results

Forest cover

Using the area (m^2) and the relative cover of the canopy trees in each polygon, we calculated the total area covered by each canopy tree species at MSH (Table 3). The three most abundant trees at MSH, *Acer saccharum*, *Fagus grandifolia*, and *Quercus rubra*, cover approximately 68 % of the study site. Figure 2 shows the CA-ordination diagram based on the forested polygon data set. The first and the second axes explain 35 and 16 % of the total variation, respectively. It should be noted that "unidentified species" were excluded from the analysis. Additionally, 2 out of the 119 forested polygons were excluded from the analysis, as they were dominated by species that do not occur (or were not identifiable) in any other polygons; one of the deleted polygons was dominated by *Alnus rugosa*, and the other by *Populus tremuloides*.

Table 3. Total area covered by the major canopy tree species based on interpretation of an aerial photograph of our study site.

Species	Total covered area (m^2)	Relative cover (%)	Cumulative cover (%)
Acer saccharum Marshall	3398293	32.1	32.1
Fagus grandifolia Ehrh.	2092012	19.8	51.9
Quercus rubra L.	1739734	16.4	68.4
Betula alleghaniensis Britton	509317	4.8	73.2
Fraxinus americana L.	308777	2.9	76.1
Pinus strobus L.	183644	1.7	77.8
Tilia americana L.	76346	0.7	78.6
Tsuga canadensis (L.) Carriére	72556	0.7	79.2
Ostrya virginiana (Miller) K. Koch	58958	0.6	79.8
Acer rubrum L.	44123	0.4	80.2
Betula papyrifera Marshall	41905	0.4	80.6
Populus tremuloides Michx.	21632	0.2	80.8
Alnus rugosa (Du Roi) Spreng.	9097	0.1	80.9
Unidentified species*	1385374	13.1	94.0
Others			
Rocky summits and cliffs	306367	2.9	
Ponds, lakes	324252	3.1	
Buildings	3528	0.0	
Total area	10575914		

* Species unidentified in the aerial survey include *Abies balsamea, Acer nigrum, Betula populifolia, Carya ovata, Carya cordiformis, Fraxinus americana, Fraxinus nigra, Juglans cinerea, Populus grandidentata, Picea glauca, Pinus resinosa, Populus deltoides, Populus balsamifera, Prunus pensylvanica, Prunus serotina, Thuja occidentalis, Ulmus americana, Ulmus rubra.*

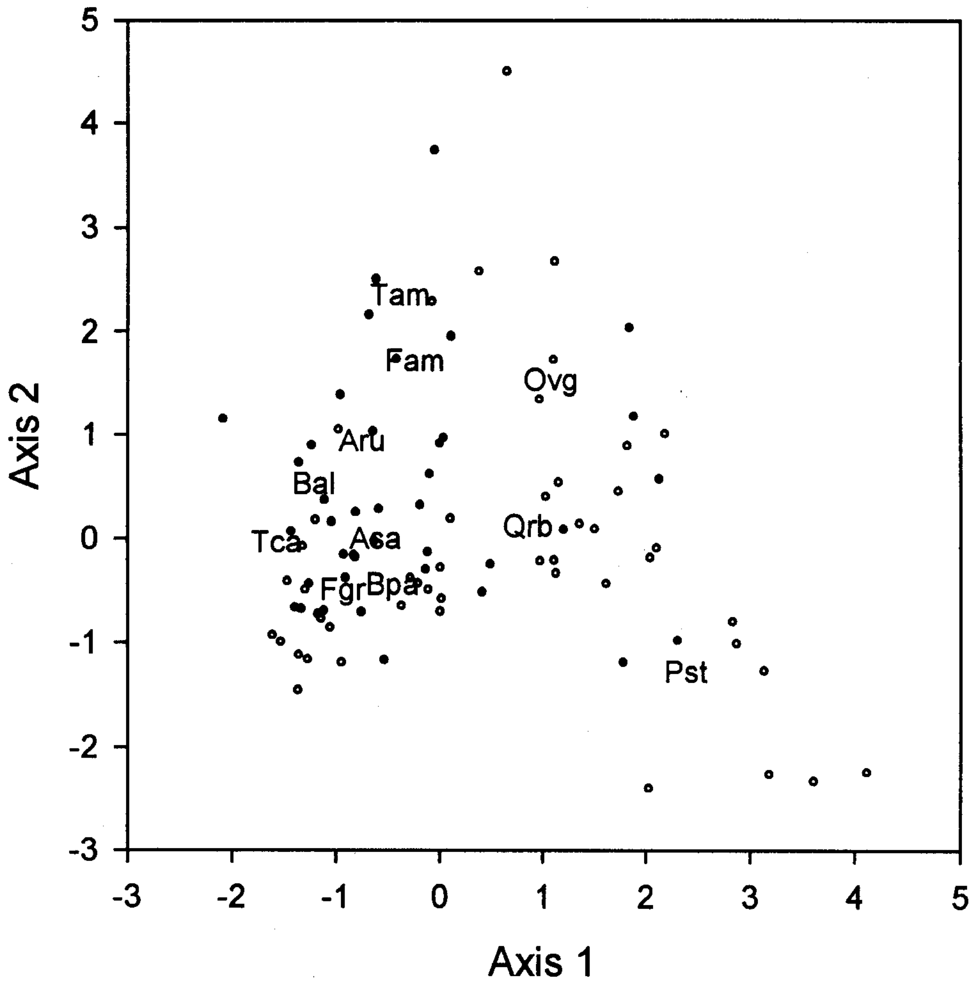

Figure 2. CA-ordination diagram of the forest cover data set. See Table 1 for acronyms. Site scores that are shown in open circles represent sites (polygons) in which the sampling plots were absent, and closed circles represent sites where the sampling plots were present (see text for details, p. 32).

In Fig. 2, the polygons that contain the plots sampled to investigate the tree-environment relationship (closed circles) are distinguished from the polygons that do not include the sampling plots (open circles). The site scores of polygons that contain the sampling plots are spread out in ordination space among the site scores of polygons that do not contain the sampling plots. This indicates that the plots sampled for investigating the tree-environment relationship covers representative forest cover types identified for MSH. Although the sampling plots are not found among the four polygons found in the lower right corner of the ordination space, these polygons only account for 2.5% of our study area.

Canopy tree-environment relationship

The total variation of the species composition data was 3.08, and the sum of all canonical eigenvalues (total explained variation) was 0.828. A Monte Carlo permutation test for the sum of all canonical eigenvalues showed that there is a significant relationship between plot-to-plot variation in the tree composition and the environmental variables ($p = 0.001$). The first two CCA axes were also significant based on the Monte Carlo permutation test (first axis eigenvalue = 0.450, $p = 0.001$; second axis eigenvalue = 0.167, $p = 0.001$). Canonical coefficients and intra-set correlation coefficients for the first two CCA axes are given in Table 4a, and the corresponding CCA ordination diagram is given in Fig. 3a. The first axis correlates positively with direct insolation and slope, and negatively with total nitrogen, diffuse insolation, and soil moisture (Table 4a). Species such as *Quercus rubra*, *Pinus strobus*, *Ostrya virginiana*, and *Betula papyrifera* are found on sites with higher axis 1 values, while *Acer saccharum*, *Fagus grandifolia* and *Betula alleghaniensis* are found at sites with lower axis 1 values. The second CCA-axis is

Table 4. Canonical coefficients and intra-set correlation coefficients of environmental variables for the first two CCA-axes. a) Results for CCA with all the environmental variables used as explanatory variables. The variables with asterisk (*) represent aspect, a nominal variable. The canonical coefficients for "West" are zero because it is used as a reference of the aspect classes. See ter Braak and Šmilauer (1998) for details. b) Results for CCA using the forward selection procedure in CANOCO to select four variables. Note: Total N is the sum of NH_4-N and NO_3-N.

a)

	Canonical coefficients		Correlation coefficients	
	Axis 1	Axis 2	Axis 1	Axis 2
Total N	-0.81	-0.20	-0.58	-0.44
Mg	0.12	0.88	0.19	-0.33
Ca	0.09	-1.32	0.18	-0.65
K	0.51	0.02	0.23	-0.23
pH	0.22	0.09	0.33	-0.20
Moisture (Jul. 13)	-0.02	0.08	-0.45	-0.09
Moisture (Aug. 20)	-0.08	-0.37	-0.43	-0.24
Elevation	0.00	-0.18	-0.08	-0.67
Insolation (diffuse)	0.09	-0.48	-0.45	0.07
Insolation (direct)	0.47	-0.01	0.55	0.10
Slope	0.47	-0.65	0.67	-0.29
Flat*	-0.15	0.29	-0.42	0.45
North*	-0.05	0.20	-0.34	-0.21
East*	-0.15	-0.01	-0.27	-0.19
South*	-0.07	0.30	0.65	0.06
West*	0.00	0.00	0.11	-0.22

b)

	Canonical coefficients		Correlation coefficients	
	Axis 1	Axis 2	Axis 1	Axis 2
Total N	-0.64	-0.60	-0.59	-0.69
Ca	0.45	-0.55	0.22	-0.85
Insolation (direct)	0.62	0.15	0.57	0.23
Slope	0.63	-0.44	0.73	-0.31

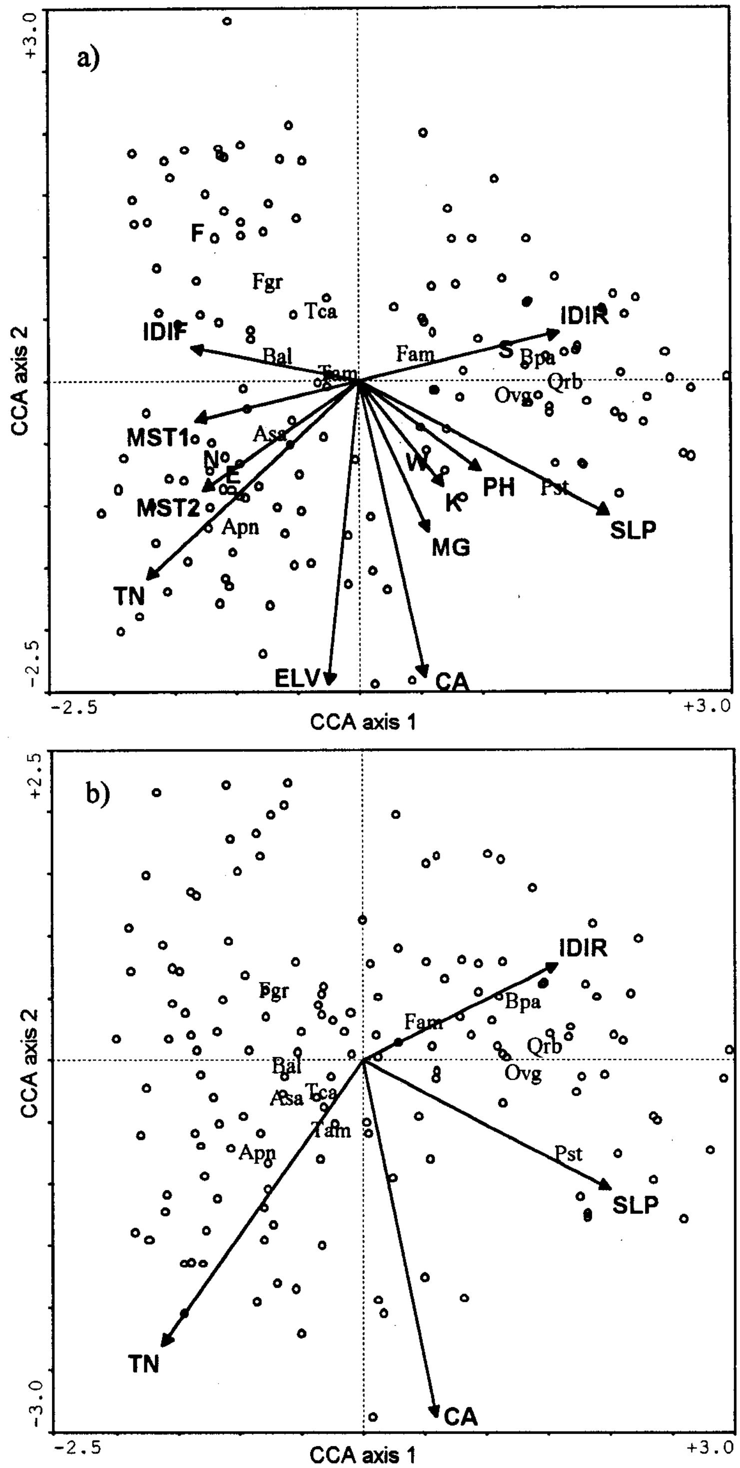

Figure 3. CCA-ordination diagram of the sampling plots. a) Diagram based on CCA with all environmental variables included in the analysis. Note that aspect classes are given as a point at the centroid (the weighted average) of the sites belonging to that class (ter Braak 1995, ter Braak and Šmilauer 1998). F = flat, N = north, E = east, S = south, W = west. b) Diagram based on CCA with the variables selected by forward selection. Species that occur in less than five sampling plots are not shown in the diagram. For acronyms, see Tables 1 and 2.

negatively correlated with Ca availability, elevation and total nitrogen (Table 4a). Of the three most commonly occurring species at MSH (Table 3), *Fagus grandifolia* is found at sites with low elevation and with low Ca availability, while *Acer saccahrum* is found on sites with higher total nitrogen content. Aspect is also a good indicator of different stand types (Fig. 3a). South facing plots are mainly dominated by *Quercus rubra*, north and east facing plots by *Acer saccharum*, and "flat" plots by *Fagus grandifolia*. West facing plots are codominated by *Acer* and *Quercus*.

To highlight the effects of the important environmental variables, forward selection of the environmental variables was done (select four variables; ter Braak and Šmilauer 1998). The procedure selected the following four variables: direct insolation, slope, Ca and total nitrogen. The sum of all canonical eigenvalues was 0.539 ($p = 0.001$), which is 65 % of the value obtained when all environmental variables are used as explanatory variables in the CCA. The first two CCA axes were significant based on the Monte Carlo permutation test (first axis eigenvalue = 0.399, $p = 0.001$; second axis eigenvalue = 0.097, $p = 0.002$). Canonical coefficients and intra-set correlation coefficients for the first two CCA axes are given in Table 4b, and the CCA ordination diagram is given in Fig. 3b. The relative coordinates of the species scores, as well as their relative position along the environmental gradients do not differ overall with the CCA ordination diagram based on all environmental variables. This indicates that the four variables selected have strong influence on the species composition of the sampled plots.

Discussion

The distribution of canopy trees at Mont St. Hilaire (MSH) is mainly organized on gradients of slope and direct insolation, both good indicators of soil moisture regimes.

Slope can affect soil moisture in two ways. First, steeper slope causes water to runoff, thereby causing low soil moisture on the slopes. At MSH, the sandy loam texture of the soil may enhance water infiltration. Second, steeper slope results in a thinner soil layer, which lowers the water holding capacity on the slopes. These two mechanisms together can lower the moisture availability on steeper slopes. Direct insolation can also have an influence on soil moisture; at MSH, snowmelt based on exposure has been shown to exert a strong influence on the surface soil moisture regime that is maintained throughout much of the growing season (Rouse and Wilson 1969). This can explain why soil moisture is lower at locations receiving higher amounts of direct solar radiation.

Previous studies have shown that soil moisture significantly influences the species composition of canopy trees in the St. Lawrence River valley, and in the surrounding Conifer-Northern Hardwood region (Maycock 1963; Walther 1963; Op de Beek 1972; Bouchard and Maycock 1978; Gauch and Stone 1979). Our results are consistent with the previous findings; species such as *Quercus rubra*, *Pinus strobus*, *Betula papyrifera*, and *Ostrya virginiana* are found on the drier sites (higher direct insolation, steeper slope), while the wetter sites (lower direct insolation, gentler slope) are mostly dominated by *Acer saccharum* and *Fagus grandifolia*, with minor components such as *Betula alleghaniensis*, *Tilia americana*, and *Tsuga canadensis*.

Aspect was also a good indicator of differences in canopy tree species composition at MSH. Our results show that south-facing plots are mainly dominated by *Q. rubra*, north- and east-facing plots by *A. saccharum*, and "flat" plots by *F. grandifolia*. West-facing plots are codominated by *A. saccharum* and *Q. rubra*. These results support the survey conducted by Phillips (1972) who investigated the aspect-related differences in tree composition at MSH. Both Phillips' study and our study found differences in species composition between west- and east-facing slopes, where east-facing slopes were more similar to species composition found on north-facing slopes. This may reflect the differences in soil moisture regime between the east- and the west-facing plots, which can be explained by the following two mechanisms: 1) Prevailing wind at our study site is from the west (Wilson 1971; Environment Canada 2002), which may increase evaporation on the west-facing plots. 2) East-facing plots receive solar radiation in the morning when the soil temperature is lower; this reduces evaporation from the east-facing plots compared to the west-facing plots, despite receiving similar amounts of solar radiation. As a consequence, east-facing plots have higher soil moisture, which is reflected in the differences in species composition with the west-facing plots (or similar composition with the north-facing plots).

While variables related to topography have significant influence on the differences in forest types, two edaphic factors, total nitrogen and Ca availability, were also found to explain the spatial variation in canopy tree species composition. At sites with high total nitrogen, *A. saccharum* is found to dominate the canopy. Previous studies have shown that nitrogen pool and mineralization rates are higher at sites mainly dominated by *A. saccharum* than sites dominated by *F. grandifolia* or *Q, rubra* (Boerner and Koslowsky 1989; Findzi et al. 1998a). *Acer saccharum* leaf litter has a lower C:N ratio than that of *F. grandifolia* or *Q. rubra*, which can lead to faster rates of decomposition (Pastor et al. 1984; Findzi et al. 1998a); the differences in the rate of litter decomposition spanning temporal scales of decades to centuries can lead to large differences in N content of soils (Parton et al. 1987). This mechanism may explain why sites dominated by *A. saccharum* have higher total nitrogen content. Our results also show that *F. grandifolia* is found more dominant at sites with low Ca availability, which is in agreement with previous studies (e.g. van Breeman et al. 1997; Finzi et al. 1998b; Woods 2000). *Fagus grandifolia* has been shown to modify the edaphic conditions through its leaf litter. *Fagus grandifolia* leaves have low pH (Finzi et al. 1998b), and as the pH is lowered, availability of Ca decreases (Ellis and Mellor 1995). Additionally, *F. grandifolia* leaf-litter is low in Ca concentration (Gosz et al. 1973; Côté and Fyles 1994a,b) and also has high levels of lignin and polyphenols (Melillo et al. 1982; Aber et al. 1990), which can complex with Ca and increase leaching of this element (Davies 1971). Thus the Ca gradient may be created by the greater dominance of *F. grandifolia* in the canopy. For both total nitrogen and Ca availability, we have suggested that species composition of the canopy trees may be affecting the availability of these elements, as opposed to soil nutrients affecting the canopy tree composition. Alternatively, successful recruitment of *A. saccharum* and *F. grandifolia* may have been limited to areas with high nitrogen and low Ca availability, respectively. Further investigation is necessary to examine to what extent total nitrogen and Ca availability at our study site is influenced by the relative dominance of *A. saccharum* and *F. grandifolia* in the canopy.

Some canopy tree species that are known to occur on the mountain but were not found in our sampling plots include *Acer rubrum*, *Pinus resinosa* Aiton., *Populus tremuloides* Michx., *Carya cordiformis* (Wangenh.) K. Koch., *Abies balsamea* (L.) Miller, *Populus balsamifera* L., *Betula populifolia* Marshall and *Fraxinus nigra* Marshall. The absence of species such as *Acer rubrum*, *Abies balsamea* and *Fraxinus nigra* indicate that wet and wet-mesic sites were underrepresented in our sampling plots. However, these habitats comprise only 1.7 % of the total area of MSH (Phillips 1972) and therefore will only represent a small fraction of the canopy trees found at MSH as a whole. *Pinus resinosa* is found to form dominant stands on west and south facing slopes at MSH

(Maycock 1961; Phillips 1972). However, the occurrence of *Pinus resinosa* is low at MSH (Phillips 1972), as well as in forest stands on other Monteregian Hills (Walther 1963) and in the St. Lawrence River valley (Op de Beek 1972). *Pinus resinosa* stands at MSH are eventually likely to be replaced by other species as is generally found in the Great Lakes region (Rudoff 1990), and in the absence of major disturbances, their presence at MSH will most likely to decline. However, given their lifespan, these *Pinus resinosa* stands are likely to persist at MSH for an extended period of time.

Overall, our results suggest that variables related to topography, good indicators of soil moisture, are the primary factors differentiating the major forest types found at MSH, from the sites dominated by *Quercus rubra* to the sites dominated by *Fagus grandifolia* and *Acer saccharum*. Additionally, high total nitrogen was associated with the greater dominance of *A. saccharum* and low Ca availability was associated with greater dominance of *F. grandifolia*. While we were able to identify the environmental correlates of canopy composition for the commonly occurring species, there are still pockets of unusual stand types found at MSH that were not assessed in the current study (e.g. stands of *Abies balsamea* and *Thuja occidentalis*). Identifying the factors that give rise to these very restricted forest types at Mont St. Hilaire merits further research.

Acknowledgements

This work was supported by grants from Natural Sciences and Engineering Research Council of Canada (NSERC) and Fonds pour la Formation de Chercheurs et l'Aide à la Recherche (FCAR). We thank McGill University for access to the Gault Nature Reserve. We thank Julien Mercille and Julie Girard for their assistance in the field, and the Centre de la Nature du mont Saint-Hilaire for providing the forest cover data. We also thank the

Soil Chemistry Laboratory at Université Laval for the nutrient extractions and analyses of soils samples.

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Chapter 2

Changes in understory light regime in an old-growth hardwood forest after an ice storm.

Abstract

We assessed gap fraction values (GF) in an old-growth beech-maple forest immediately before and in the 3 years after a severe ice storm. We estimated GF using hemispherical photographs taken at a height of 0.6 m above the soil surface in 101 permanent plots. There was a significant increase in GF immediately after the ice storm, but GF returned to pre-storm levels within three years after the storm. The relatively rapid gap closure we observed may arise in the high frequency of small gaps created by branch loss as opposed to the larger gaps associated with tree death or windthrow. Additionally, both GF prior to the storm and sapling species composition, significantly affected the change in GF after the storm. Plots with lower GF prior to the storm or with shade-tolerant species in the understory had greater change in GF (i.e. greater canopy damage). We discuss the implications of our results for the forest dynamics and species composition of northern hardwood forests in North America.

Introduction

Ice storms are one of the important agents of natural disturbance in the deciduous forests of eastern North America (Lorimer 2001). These storms occur when warm moist air passes over a ground-level layer of air at freezing temperature. Under these conditions, water droplets that are formed in the warm air mass fall through the layer of cold air and become super-cooled rain, which freezes on contact with ground-level objects (Lemon 1961). In forested landscapes, ice accumulates on trees and results in loss of twigs, branches, and occasionally the entire tree crown (Bruederle and Stearns 1985, Melancon and Lechowicz 1987, Hooper et al. 2001).

Studies investigating the effects of ice storms on the deciduous forests of eastern North America have mostly focused on assessing tree-level damage using arbitrarily chosen damage classes (e.g. Rebertus et al. 1997, Duguay et al. 2001), and quantifying the number, volume, or biomass of branch wood lost from the canopy (e.g. Bruederle and Stearns 1985, Melancon and Lechowicz 1987, Hooper et al. 2001). The damage inflicted on trees inevitably results in creation of gaps in the canopy, which are recognized as an important factor shaping the species composition, species diversity, structure, and the development of forest stands (Bazzaz and Wayne 1994). While damage reported in previous studies may provide a rough approximation of changes in gap fraction, quantifying the changes directly would enhance our understanding of the impact of ice storms on forest dynamics. We are not aware of any study that has quantified the changes in gap fraction caused by an ice storm. This, in large part, results from a lack of the pre-storm gap fraction values that are necessary to assess the changes in gap fraction caused by the storm. We were fortunate to have gap fraction values prior to the 1998 ice storm for 101 plots in an old-growth beech-maple forest near the northern margin of the deciduous forest biome in eastern North America. These pre-storm gap fraction values from summer 1997 provide a benchmark against which damage and recovery from the ice storm of 1998 can be gauged. This storm hit parts of southern Quebec, Ontario, and New Brunswick in Canada, as well as a large portion of northern New York and New England in the United States. More than 80 mm of freezing rain fell in the Montreal region between January 4-10, leading to accreted radial ice loads in excess of 45 mm (Irland 1998, Laflamme and Périard 1998). The last major ice storm at the site in December 1983 (Melancon and Lechowicz 1987) was of much lesser magnitude, with ice loads of only 15 mm. The main objective of this note is to report the changes in gap fraction values caused by the ice storm of 1998, as well as subsequent gap closure. We also identify factors that can explain spatial variation in changes in gap fraction values. We briefly discuss the implication of our findings for the dynamics and species composition of trees in northern hardwood forests.

Methods

Study site

We assessed the impact of the ice storm of January 1998 (Irland 1998) on a 10 km^2 tract of old-growth forest at Mont St. Hilaire, Quebec, Canada (45°31'N, 73°08'W), which lies within the region most hard hit by the storm. The forest at Mont St. Hilaire is the largest remaining remnant of the primeval forests of the St. Lawrence Valley: many of the trees exceed 150 years in age and a few are over 400 years old. Various tree communities occur on the mountain (Phillips 1972); *Acer saccharum*, *Fagus grandifolia*, and *Quercus rubra* are the common canopy dominants. The site is essentially undisturbed by human activities and is protected as an International Biosphere Reserve under the Man and Biosphere program (MAB) of the United Nations.

Data collection and analysis

In the summer of 1997, we established 101 permanent plots (6-m radius or 113 m^2) with a metal rod at the center of each plot so that sequential photographs can be taken at the exact same location. We recorded species and DBH of all trees and saplings (DBH ≥ 1 cm) at each plot.

Changes in gap fraction

In late summer from 1997 through 2000, we took hemispherical (fisheye) photographs from the center of each plot. We used a Samigon hemispherical lens mounted on a Pentax camera with ISO 400 color film (Fuji film). The camera was fixed on a tripod 0.5 m above the ground surface (lens height 0.6 cm above surface), leveled and oriented towards magnetic north. Previous studies have used various camera heights to quantify GF (e.g. 1.5 m (Canham et al. 1990), 0.6 m (Walter and Himmler 1996), 0.5 m (Valverde and Silvertown 1997)), but height differences of this magnitude have been shown to be insensitive to gap fraction estimation (Robinson and McCarthy 1999). We scanned each photograph (800 dpi), and used the digital images to estimate gap fraction values (GF, %) for each site and year with Gap Light Analyzer (GLA), version 2.0 (Frazer et al. 2000).

Because most of the sampling plots were placed on sloping terrain, some ground appeared on the periphery of many photographs. To avoid distorting the gap fraction estimates, we limited the calculation to zenith angles between 0° and 60° degrees (maximum slope angle in the sampling plots was 29°). While we lose some information by omitting zenith angles between 60° and 90°, their influence is believed to be small given that the surrounding vegetation obstructs most of the image areas near the horizon. Additionally, gap fraction values obtained for zenith angles between 0° and 60° have high correlation with gap fraction values obtained for 0° and 90° ($r = 0.97$, $P < 0.001$) in our data set.

Most of the hemispherical photographs were taken on an overcast day to reduce the overexposed region around the sun and the reflections on the leaves and bark, which could otherwise be construed as gap openings (Rich 1990). However, some photographs were unavoidably taken when there was direct sunlight. Reflection on the grey-silvery trunk of *Fagus grandifolia* posed a particular problem in these photographs. To reduce the effects of this reflection as well as others, we used Adobe Photoshop version 6.0 (Adobe Systems Inc.) to replace (i.e. fill in) the reflection regions with a darker color so that the program would not mistake these regions as openings. Additionally, recommendations described in the user's manual of GLA were used for photographs taken under non-overcast conditions (e.g. separate RGB planes, selecting the "region of interest").

We used parametric paired sample t-tests to compare GF for different years (SYSTAT ver. 9.0, SPSS Inc.). Bonferroni correction was used to account for multiple tests.

Factors influencing the variation in changes in gap fraction

For each plot, the changes in gap fraction values (GF_{chg}) are calculated as:

$$GF_{chg} = GF_{98} - GF_{97}$$

where, GF_{98} = gap fraction value in 1998, GF_{97} = gap fraction value in 1997. To explain the spatial variation in GF_{chg}, we examined the relationship between GF_{chg} and various factors, including slope, aspect, total basal area, total density and species composition of canopy trees (DBH $\geq$ 10 cm) and saplings (1 cm $\leq$ DBH < 10 cm) within the sampling plot. We used principal component analysis (PCA, basal area used as response variable) to differentiate species composition of canopy trees and saplings (CANOCO ver. 4.0, Microcomputer Power.); the ordination axis site scores for each plot were used to numerically rank the differences in species composition among the sampling plots. We only report the results for factors that showed significant linear relationships to GF_{chg} (SYSTAT ver. 9.0, SPSS Inc.).

Results

Changes in gap fraction

Figure 1 shows the distribution of percentiles for gap fraction values (GF) between 1997 and 2000. Parametric paired sample t-tests showed that differences in gap fraction values are significantly different among the years (all combinations showed $P<0.001$, except between 1997 and 2000, which had $P=0.048$). There is only a difference of 1 % between the mean values of 1997 and 2000 (Fig. 1). While GF increased significantly immediately after the ice storm, values return close to pre-storm conditions within three growing seasons after the storm (Fig. 1). However, the variation, or the standard deviation, in GF is higher after the storm, indicating that light availability in the understory has become more heterogeneous after the storm (Fig. 1).

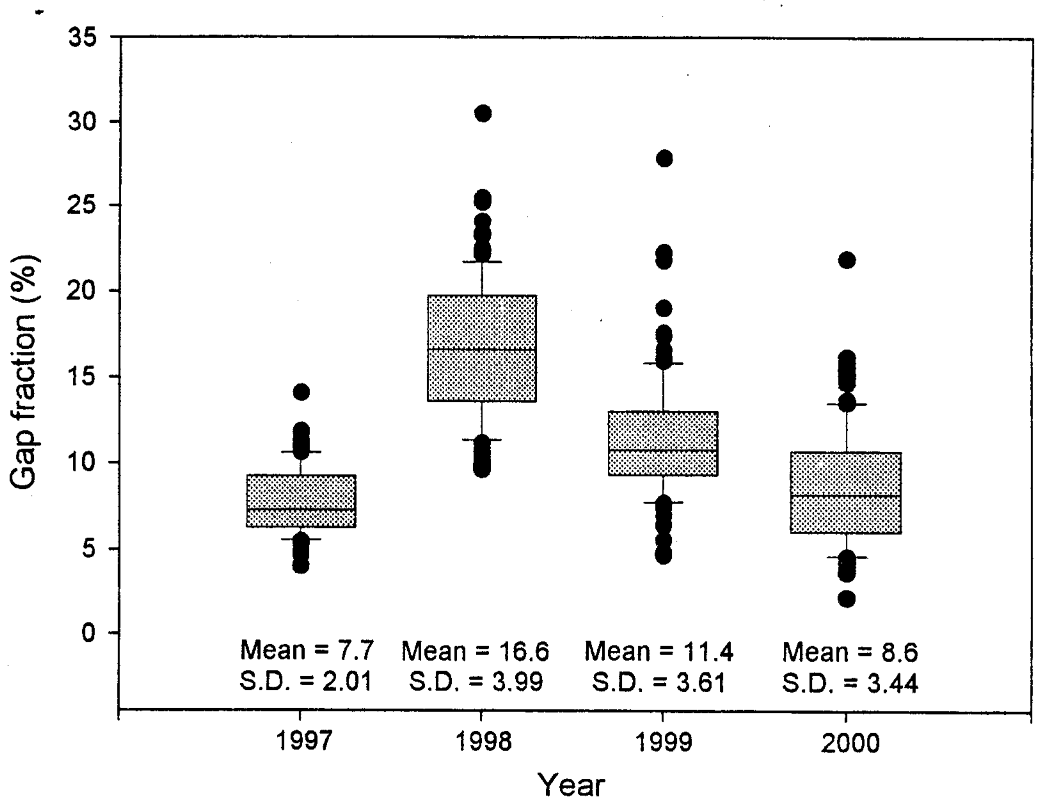


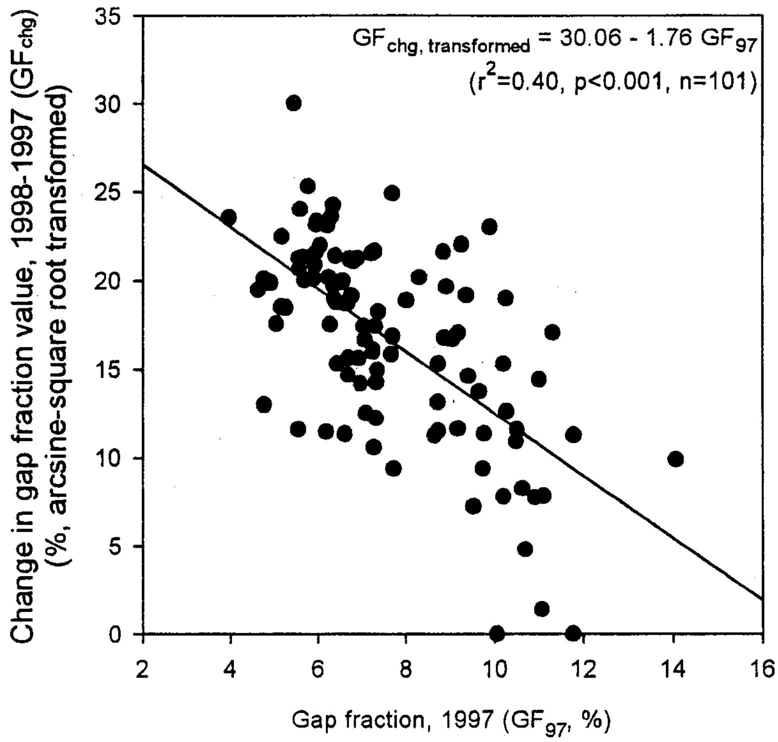
Figure 1. Boxplots showing the percentile distribution of the gap fraction values (GF) for each year (n = 101). The values at the bottom of the figure are means and the standard deviations of the gap fraction values for each year. The extreme outliers found for years between 1998 and 2000 are values for a single plot where a canopy tree lost an entire crown due to the ice storm. Note, however, that the extreme outlier found for 1997 is not a value from the same plot.

Factors influencing the variation in changes in gap fraction

Pre-storm gap fraction values in 1997 (GF_{97}) explain a significant amount of variation in changes in gap fraction values (GF_{chg}) (Fig. 2a). Greater GF_{chg} are found in plots that had lower GF_{97}. Sapling species composition in the plots, which was ranked numerically by the first PCA axis using the sapling basal area data (species scores for this axis are given in Table 1), also showed significant relationship with GF_{chg}. This first PCA axis, referred to as Sapling Compositional Index (SCI), can be considered a gradient from plots that consists mainly of saplings with low to moderate shade tolerance (e.g. *Ostrya virginiana*, *Fraxinus americana*, *Quercus rubra*, *Betula papyrifera* - low SCI values) to plots with saplings of high shade tolerance (e.g. *Fagus grandifolia*, *Acer pensylvanicum* - high SCI values). Note that *Acer saccharum* is found ubiquitously in our sampling plots, occurring in 91 plots out of 101 (Table 1). The significant relationship between SCI and GF_{chg} (Fig. 2b) shows that sampling plots that had shade-tolerant *Fagus grandifolia* and *Acer pensylvanicum* in the understory also had more canopy damage in the 1998 ice storm. Note that the two significant explanatory variables that explain the spatial variation in GF_{chg}, GF_{97} and SCI, showed high correlation ($r = 0.61$, $P<0.01$). Other factors that were considered (e.g. slope, elevation) did not show any significant relationship with GF_{chg}, alone or in combination.

Figure 2. a) Relationship between pre-storm gap fraction values (GF_{97}) and changes in gap fraction values (GF_{chg}, arcsine square root transformed). Note that we were unable to use the arcsine square root transformation for two samples that had negative GF_{chg} values (GF_{chg} = -0.4 % and -1.1 %). Because the two values were very close to zero, we treated the two values as no change (i.e. GF_{chg} = 0 %) and included them in the analysis. b) Relationship between sapling composition index (SCI) and the changes in gap fraction values (GF_{chg}). An apparent outlier above the regression line is from a plot with a tree that lost an entire crown (see Fig.1). We have no clear explanation for the three outliers below the regression line (two out of the three outliers are for plots that had negative GF_{chg} values). These four outliers (open circles) were excluded from the regression analysis (n=97).

a)


$GF_{chg, transformed} = 30.06 - 1.76\ GF_{97}$
(r^2=0.40, p<0.001, n=101)
Change in gap fraction value, 1998-1997 (GF_{chg})
(%, arcsine-square root transformed)
Gap fraction, 1997 (GF_{97}, %)
35
30
25
20
15
10
5
0
2
4
6
8
10
12
14
16

b)

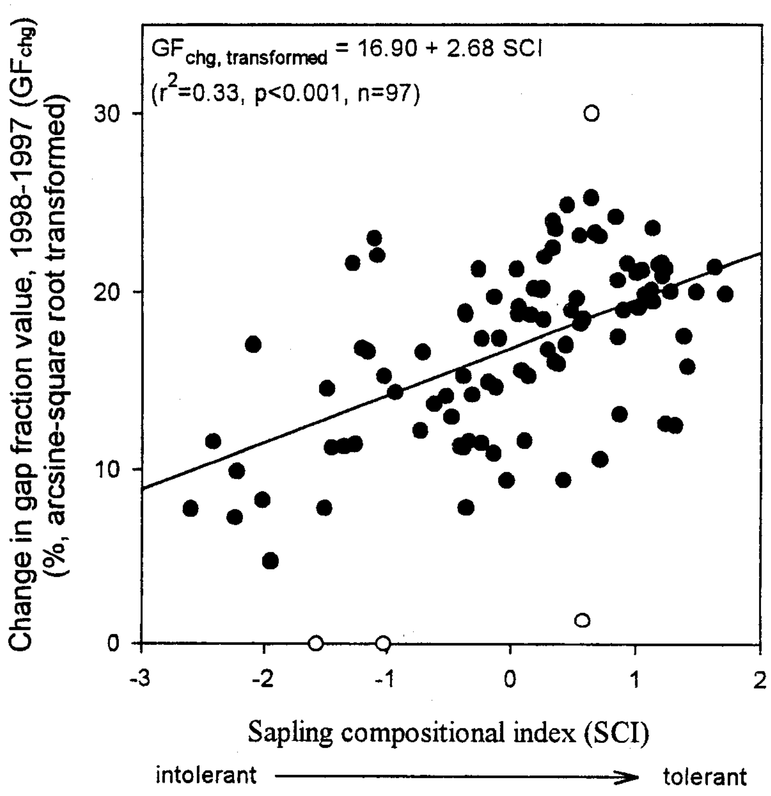

$GF_{chg, transformed} = 16.90 + 2.68\ SCI$
(r^2=0.33, p<0.001, n=97)
Change in gap fraction value, 1998-1997 (GF_{chg})
(%, arcsine-square root transformed)
30
20
10
0
-3
-2
-1
0
1
2
Sapling compositional index (SCI)
intolerant
tolerant

Table 1. Axis 1 species scores (adjusted for species variance) of PCA based on basal area of saplings in the sampling plots. The eigenvalue of the first axis is 0.341 (variation explained is 34.1 %). The axis can be considered as a gradient of plots where *Fagus grandifolia* mainly occupies the understory to plots that are occupied by various species, most notably *Ostrya virginiana, Fraxinus americana, Betula papyrifera*. Note that species that occurred in fewer than 5 sampling plots were excluded from the analysis (*Pinus strobus, Prunus pensylvanica, Prunus serotina, Tsuga canadensis*). The axis is used to numerically rank the sampling plots by species composition (Sapling Composition Index: SCI). See text for details.

Species	Axis 1 species scores	Occurrence (out of 101 plots)
Ostrya virginiana (Miller) K. Koch	-0.77	46
Fraxinus americana L.	-0.63	29
Quercus rubra L.	-0.61	14
Betula papyrifera Marshall	-0.54	7
Tilia americana L.	-0.51	27
Acer saccharum Marshall	-0.38	91
Prunus virginiana L.	-0.24	8
Betula alleghaniensis Britton	-0.16	16
Acer spicatum Lam.	0.17	21
Acer pensylvanicum L.	0.64	59
Fagus grandifolia Ehrh.	0.70	68

*nomenclature follows Gleason and Cronquist (1991)

Discussion

The substantial increases in gap fraction values (GF) in the summer immediately following the ice storm are not surprising given the amount of damage inflicted on the trees during the ice storm of 1998 at our study site (Duguay 2001, Hooper et al. 2001). Greater changes in gap fraction values (GF_{chg}) were found on sites that had lower pre-storm gap fraction values (GF_{97}) and also on sites where shade-tolerant species predominated the understory. The mechanisms in which these two factors affect GF_{chg} may be mediated by the same factor: the extent of branching in the canopy. Low GF_{97} suggests that leaves were displayed efficiently throughout the hemispherical view of the canopy, which is made possible by an extensive network of branches and twigs. The plots where shade-tolerant species predominate (or plots where shade-intolerant species are absent) indicate that these plots had been free of major disturbances that remove a significant portion of the canopy, which may have allowed branches and twigs to be formed extensively in the canopy. These suggest that sites that had more extensive branching prior to the ice storm resulted in greater GF_{chg}. More branches and twigs inevitably creates more surface area that will come in contact with freezing rain, resulting in greater ice loading, and hence causing more damage. The fact that the effects of GF_{97} and SCI can be explained by the same mechanism (i.e. extensive branching) may also explain why there is a significant correlation between the two factors.

The gaps created by the ice storm closed quickly after the storm, returning to pre-storm levels within three years after the storm. Two factors can contribute to this quick gap closure after the storm. The first is the height in which the photographs were taken (0.6 m above the ground). It is well recognized that old-growth stands in northern hardwood forests typically have an irregular, uneven-aged canopy with trees in various stages of development (Lorimer and Frelich 1994). A qualitative assessment of damage inflicted by the storm of 1998 at our study site found that both trees and saplings suffered severe damage, but a high number of the stems in the sapling stratum ($4 \leq$ DBH < 10 cm) were unscathed (Duguay et al. 2001). These relatively undamaged saplings have grown substantially after the storm, taking advantage of the heightened light availability in the understory. Therefore, by having the camera position at 0.6 m above the soil surface, we have captured the growth of these saplings, and this no doubt explains the rapid shading of the understory after the storm. Another important factor that can contribute to rapid gap closure is the way in which ice storms damage trees. Although an ice storm with a magnitude of the 1998 storm may occasionally remove an entire crown (Duguay et al. 2001), most gaps arise from loss of branches less than 10 cm in diameter (Melancon and Lechowicz 1987, Seischab et al. 1993, Hooper 2000). Thus unlike treefall gaps, gaps created by ice storms are smaller, and can be occupied readily by the extension of the intact branches; this may also explain the quick closure of the gaps after the storm.

The rapid closure of the gaps after the ice storm has significant implications for the role of ice storms as a disturbance agent in forested landscapes. It is generally accepted that forest gaps increase the light conditions in the understory, allowing less shade-tolerant species to establish, which in turn helps maintain the species diversity in old-growth forests (Denslow 1985). Previous studies have estimated that the time required for a small gap to close in temperate forests is approximately 10 years (e.g. Valverde and Silvertown 1997). Our results show that the gaps created by an ice storm can close in a much shorter time, suggesting that the duration of increased light levels may not be sufficient to allow successful establishment and survival of seedlings of the less shade-tolerant species. If the ice storm gaps do provide a window of opportunity for the shade-intolerant species to establish, we would expect these species to be found abundantly at our study site, given that 1) the return time of damage-causing ice storms in our region is about 10-20 years (Proulx and Greene 2001) and that 2) ice storms cause damage over a substantial geographical area. However, two of the most shade-tolerant species in the temperate deciduous forests, *Acer saccharum* and *Fagus grandifolia*, dominate the canopy as well as the understory at our study site (except on dry summits and steep slopes where *Quercus rubra* dominates), while the less shade-tolerant, gap-phase species such as *Betula papyrifera*, *Populus grandidentata*, and *Fraxinus americana* are found only very infrequently (Phillips 1972, Arii and Lechowicz 2002). Thus, ice storm gaps may not be facilitating the establishment of less shade-tolerant species.

The same mechanism may explain the low abundance of *Betula alleghaniensis* at our study site. *Betula alleghaniensis* is a species with intermediate shade-tolerance, and is often found to codominate with the shade-tolerant *A. saccharum* and *F. grandifolia* in the northern hardwood forests of eastern North America. Because *B. alleghaniensis* rarely establish under intact canopy (Erdmann 1990), it has been suggested that minor disturbances are necessary for the maintenance of *Fagus-Acer-Betula* stands (Forcier 1975). Again, while the frequent occurrence of ice storms is likely to favor the establishment of *B. alleghaniensis*, it is considerably less frequently encountered than *A. saccharum* and *F. grandifolia* at our study site, both in the canopy and in the understory (Phillips 1972, Arii and Lechowicz 2002). *Betula alleghaniensis* requires treefall gaps to establish (McClure 2000), and continued overhead light is necessary to grow and to survive, particularly during the early stages of its development (Erdmann 1990). The gaps created by ice storms may not last long enough to allow successful recruitment of *B.*

alleghaniensis.

While the quick recovery may not contribute to the establishment of shade-intolerant species, it can be advantageous for the shade-tolerant species. The shade-tolerant species, such as *Acer saccharum* and *Fagus grandifolia*, can tolerate low light levels under an intact canopy by adjusting their morphological and physiological traits (Canham 1990). However, it has been shown that both *Acer saccharum* and *Fagus grandifolia* require multiple gap encounters to reach the canopy (Canham 1990). Thus the frequent occurrence of ice storms enhances their successful recruitment into the canopy, and because the gaps close at a rapid rate after an ice storm, competition from the shade-intolerant species may be prevented. This may explain the dominance of *Acer saccharum* and *Fagus grandifolia* at our study site.

The creation of canopy gaps does not only influence the tree seedlings and saplings, but also the shrubs and herbaceous species in the understory. Leckie et al. (2001) have found that seed bank at our study site is rich in species when compared with that in most other temperate forests. This may be due to the frequent occurrence of ice storms that allows these species to reproduce aggressively during the brief gap phase, which in turn contributes to the maintenance of a diverse, viable seed bank. We are currently monitoring the changes in species richness and species composition of the herbaceous vegetation following the storm at our study site.

Conclusion

Undoubtedly, the ice storm of 1998 was an extreme weather event. It is estimated that the return time of ice storm of this magnitude is on the order of millennia in southern Quebec and adjacent New England (Laflamme and Périard 1996). However, it has been shown that ice storms of lesser magnitude cause substantial damage to the forest trees (Melancon and Lechowicz 1987, Seischab 1993, Bruederle and Stearns 1985, Rebertus 1997), and these ice storms have a return time of about 10-20 years in the St. Lawrence River valley (Proulx and Greene 2001), as well as in adjacent New England and New York (Lemon 1961). It is difficult to predict whether the changes in gap fraction values caused by the current ice storm were significantly more substantial than the ice storm of lesser magnitude, as we have no information from the previous storms. If the smaller ice storms result in smaller changes in the gap fraction values, the gaps will close at a more rapid rate, which will further prevent the successful establishment of shade-intolerant species. This will direct the forest towards dominance by later-successional, shade-tolerant species in the absence of other major disturbances (note that previous studies have also suggested that ice storms accelerate forest succession, although the mechanism in which they invoke differ (Abell 1934, Carvell et al. 1957, Lemon 1961)). Therefore ice storm disturbance may have a different impact on the forest dynamics and species composition of the northern hardwood forests than other types of disturbance agents.

Acknowledgements

This work was supported by grants from Natural Sciences and Engineering Research Council of Canada (NSERC) and Fonds pour la Formation de Chercheurs et l'Aide à la Recherche (FCAR) to MJL and Government of Canada Awards (Department of Foreign Affairs and International Trade) to KA. We thank McGill University for access to the Gault Nature Reserve.

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Chapter 3

The influence of overstory trees and abiotic factors on the sapling community in an old-growth *Fagus-Acer* forest.

Abstract

We examine the influence of overstory trees and abiotic environmental factors on the patterns of spatial variation and species composition in the sapling community of an old-growth *Fagus-Acer* forest in southwestern Québec, Canada. Our main focus was to identify differences in the sapling distribution patterns of *Fagus grandifolia* (Ehrh.) and *Acer saccharum* (Marsh.), the two codominant species in the overstory, as well as the factors that determine the differences. Using canonical correspondence analysis (CCA), we show that soil moisture has the strongest influence on the spatial variation and species composition of the sapling community. *Acer saccharum* occurred on a wide range of soil moisture conditions, while *Fagus grandifolia* saplings were absent in dry habitats. Another factor that differentiates the distribution patterns of *Fagus grandifolia* and *Acer saccharum* saplings is the relative dominance of *Fagus grandifolia* trees in the overstory, which correlates negatively with pH and Ca availability in the forest floor. *Acer saccharum* saplings were not found on sites where *Fagus grandifolia* trees dominate the overstory, while *Fagus grandifolia* saplings are mostly limited to sites where *Fagus grandifolia* trees have high representation in the overstory. These findings are discussed in light of previous hypotheses on canopy tree replacement patterns in *Fagus-Acer* forests.

Introduction

Fagus-Acer forest is one of the major forest types found in eastern North America (Braun, 1950; Runkle, 1996). The dynamics within this forest type have been frequently investigated, a particular interest being the replacement patterns and coexistence of the overstory trees (Forcier, 1975; Fox, 1977; Woods, 1979, 1984; Runkle, 1981, 1984; Cypher & Boucher, 1982; Brisson et al., 1994; Kupfer & Runkle, 1996; Poulson & Platt, 1996; Foré et al., 1997; Beaudet et al., 1999; Forrester & Runkle, 2000). Since suppressed shade-tolerant saplings in the understory usually replace the overstory trees in *Fagus-Acer* forests (Woods, 1984), one of the key factors that determines the overstory replacement patterns and future overstory composition is the sapling community in the understory. Investigating the patterns of spatial distribution and species composition of saplings in the understory, and also identifying the factors that give rise to these patterns, can help understand the dynamics of *Fagus-Acer* forests.

Studies have shown that environmental factors such as soil moisture, light, and nutrients affect the growth and mortality of species that are typically found in *Fagus-Acer* forests (e.g. Canham et al., 1996; Walters & Reich, 1996, 2000; Casperson & Kobe, 2001). These differences in species' response to abiotic factors, coupled with the heterogeneous understory environment, are considered to play a significant role in determining the spatial variation and the composition of the sapling community in the forest understory (Brisson et al., 1994; Poulson & Platt, 1996). Biotic factors have also been implicated as determinants of the sapling community in *Fagus-Acer* forests. Canopy trees may affect the sapling community directly by seed dispersal, or indirectly by modifying the abiotic environment in the understory (Abrell & Jackson, 1977; Fox, 1977; Woods, 1979, 1984).

While both abiotic and overstory tree effects may play a significant role in determining the sapling community in *Fagus-Acer* forests, these factors have usually been investigated in isolation from one another. Because the overstory trees influence sapling composition partly through modifications of abiotic environment in the understory (Forcier, 1975; Woods, 1979), it is possible to overemphasize or misinterpret the effects of these factors if they are investigated separately (Magnan et al. 1994). Additionally, such partial studies provide no information on the relative importance of abiotic and canopy tree effects and their interactions. Identifying the relative importance of different effects and their interactions will help us better understand the mechanisms underlying the organization of the sapling community and its potential influence on overstory dynamics.

The objective of this study therefore is to identify the factors that determine the patterns of spatial variation and species composition of the sapling community within an old-growth *Fagus-Acer* forest by considering both abiotic and canopy tree effects. For the abiotic effects, we pay special attention to edaphic factors; previous studies have mostly focused on only light regime (i.e. gaps) as the abiotic factor determining the organization of the sapling community in *Fagus-Acer* forests (e.g. Runkle, 1981, 1984, 1990; Forrester & Runkle, 2000). Little information is available on the effects of edaphic factors, such as soil moisture and nutrient availability, on the dynamics within *Fagus-Acer* forests. We will mainly focus on the saplings of the two canopy codominants, *Fagus grandifolia* (Ehrh.) and *Acer saccharum* (Marsh.).

Methods

Study site

We assessed the effect of abiotic factors and canopy trees on understory sapling composition in a 10 km^2 tract of old-growth forest at Mont St. Hilaire in southwestern Québec, Canada (45° 31'N, 73° 08'W). Mont St. Hilaire is a hill complex of plutonic origin (Feininger & Goodacre, 1995) with peaks that rise between 200-300 m above the level floor of the surrounding St. Lawrence River Valley. The forest at Mont St. Hilaire is the largest remaining remnant of the primeval forests in this region: many of the trees exceed 150 years in age and a few are over 400 years old (Cook, 1971). Various tree communities occur on the mountain (Maycock, 1961; Enright & Lewis, 1985), but *Acer saccharum* and *Fagus grandifolia* are the common canopy dominants over most of the area. The site is essentially undisturbed by human activities and is protected as an International Biosphere Reserve under the Man and Biosphere program (MAB) of the United Nations.

Sampling procedures

Collecting tree data

In the summer of 1997, 92 permanent plots located randomly along seven line transects were sampled; plot elevation range between 145 m to 400 m. Each plot is 113 m^2 (6-m radius). At each plot, all woody individuals with DBH greater than 2.5 cm were recorded (species, DBH). These data were split into "overstory trees" data and "sapling" data, based on the size categories used by Cain (1935) and Poulson & Platt (1996). Saplings are defined as individuals with DBH between 2.5 cm and 15 cm, and overstory trees are defined as individuals with DBH greater than 15 cm, which includes both the "canopy tree" and "pole" categories in Cain (1935) and Poulson & Platt (1996). For the species included in the analysis, the height of trees with DBH between 2.5 cm and 15 cm (i.e. saplings) is typically 2 – 10 m (unpublished data).

The data set that consists of sapling density for each species in each plot is referred to as the sapling matrix (SM; 92 plots x 10 species). The variation in this SM is what we hope to explain in this study. The list of species included in SM and their basic statistics are given in Table 1. Species that occurred in less than 5 plots were not included in the analyses. We focus on the density of the saplings because, for a given species, the likelihood of becoming a canopy component is likely to be higher if the sapling density is higher in the understory.

The data set that consists of overstory tree basal area (cm^2) for each species in each plot is referred to as the tree matrix (TM; 92 plots x 4 species). This matrix is used as an explanatory or a covariable matrix in the multivariate analyses. Species that are included in the TM matrix and their basic statistics are listed in Table 2. While more overstory trees species were recorded in the sampling plots (total of 12 species), only four species were included in the TM. This was done because the rare occurrences of the remaining 8 species, which result in many "zeros" in the matrix, make them unsuitable to be used as variables in the explanatory or covariable matrix. Of the species included in TM, *Fraxinus americana* is the species that occurred least frequently (Table 2; 17 plots); any species not included in TM occurred in fewer than 9 plots.

Table 1. List of species included in the sapling data matrix (SM). Their acronyms and basic statistics are shown. Note that stem density of each species in each plot is used in the SM. See text for details.

Species	Acronym	Occurrence (out of 92)	Stem density per plot				Total basal area per plot (m^2 / ha)	
			Max	Mean*	S.D.*	Median*	Max	Mean*
Acer pensylvanicum L.	Apn	49	29	6.4	6.4	3	7.1	1.0
Acer saccharum Marsh.	Asa	82	65	16.8	14.9	12	9.8	3.1
Acer spicatum Lam.	Asp	12	29	9.2	9.8	5	6.8	1.4
Betula alleghaniensis Britton	Bal	11	6	2.2	1.8	1	1.4	0.4
Betula papyrifera Marsh.	Bpa	8	29	7.4	9.2	5	8.0	2.8
Fraxinus americana L.	Fam	20	9	2.6	2.5	1	5.0	1.2
Fagus grandifolia Ehrh.	Fgr	58	26	7.7	6.9	5	8.4	1.8
Ostrya virginiana (Mill.) K. Koch	Ovg	45	37	6.3	7.6	3	6.4	2.1
Quercus rubra L.	Qrb	10	7	3.3	2.3	3	7.1	2.6
Tilia americana L.	Tam	16	4	1.5	1	1	1.0	0.2

* calculation based only on the non-zero values.

Table 2. List of species included in the overstory tree data matrix (TM). Their acronyms, basic statistics and transformations used are shown. Note that basal area of each species in each plot is used in the TM, but stem density is also reported. See text for details.

Species	Acronym	Occurrence (out of 92)	Total basal area per plot (m^2/ha)				Density per plot		Transformation
			Max	Mean*	S.D.*	Median*	Max	Mean*	
Acer saccharum Marsh.	Asa_{15}	70	83.9	24.5	19.2	20.6	7	2.7	$\log_{10}(x+1)$
Fraxinus americana L.	Fam_{15}	17	15.1	6.6	4.1	5.8	4	1.5	$\log_{10}(x+1)$
Fagus grandifolia Ehrh.	Fgr_{15}	48	66.3	16.3	15.3	12.8	7	2.5	$\log_{10}(x+1)$
Quercus rubra L.	Qrb_{15}	40	48.8	21.6	13.0	18.9	11	3.1	$\log_{10}(x+1)$

*calculated only with the non-zero values

Collecting abiotic environment data

A list of abiotic environmental variables included in the analyses is given in Table 3. The matrix containing these environmental variables will be referred to as the abiotic matrix (AM).

Soil nutrients

In August 1997, soil samples were gathered at 8 different locations near the center of each plot. After the L-F-H layers were removed from the soil surface, the samples were taken down to 10 cm in depth. Samples taken at each plot were pooled and frozen within hours of collection. They were kept frozen for approximately 3 months before being air-dried and sieved (2 mm) in preparation for soil chemical analysis. Nutrient extractions and analyses of these extracts were carried out by the University of Alberta Soil Chemistry Laboratory. Exchangeable NO_3 and NH_4 were assayed in a KCl extract (Kalra & Maynard, 1991); available P (PO_4) was assayed using an NH_4F-H_2SO_4 extract (method 2 of Bray & Kurtz, 1945); exchangeable potassium, calcium, and magnesium were determined on an NH_4OAc extract at pH 7 (Chapman, 1965); pH was measured on an H_2O extract. For nitrogen, total mineral N (sum of NO_3-N and NH_4-N) was used in the statistical analysis (Table 3).

Soil moisture

Volumetric moisture contents were measured five times during the summer of 1999 (June 14, June 29, July 15, August 12, August 25) and twice during the summer of 2000 (July 13, August 20). Soil moisture sensors were used to make the measurements (Theta Probe, Type ML2x, Delta-T Devices, Ltd., Cambridge, U.K.). It took two days to make

Table 3. List of abiotic environmental variables included in the abiotic variables matrix (AM). Their acronyms, basic statistics, and transformations used are shown.

Variable	Acronym	Units	Mean	S.D.	Max	Min	Median	Transformation
N (NO_3-N + NH_4-N)	N	ppm	45.3	29.0	162	9	36	$\log_{10}(x+1)$
P (PO_4-P)	P	ppm	12.1	17.1	105.0	0.1	5.3	$\log_{10}(x+1)$
K	K	ppm	167.0	80.4	438	31	158	$\log_{10}(x+1)$
Mg	MG	ppm	318.0	155.7	924	159	231	$\log_{10}(x+1)$
Ca	CA	ppm	2476	1917	9942	190	1780	$\log_{10}(x+1)$
pH	PH	-	4.98	0.55	6.6	3.6	4.9	$\log_{10}(x)$
Moisture	MST	$m^3\ m^{-3}$	0.07	0.06	0.31	0.01	0.05	arcsin ($x^{1/2}$)
Slope	SLP	%	14.3	9.43	36	0	13	none

measurements at all plots during each sampling period, and the sampling was done in the same order on all sampling dates to minimize the time required to visit all plots. No rainfall was recorded during the sampling periods. At each plot, moisture content was measured at 5 different locations near the center of the plot. From the five measurements, an average was calculated for each plot.

Although absolute moisture content may vary by year and season, we were interested in the relative moisture content of the plots. At our study site, Rouse & Wilson (1969) have shown that differing rates of snowmelt based on exposure exert an influence on the surface soil moisture regime that is maintained through much of the growing season. Our data are consistent with this expectation. Pairwise Spearman rank correlation coefficients between the moisture content on different sampling dates ranged between 0.68 and 0.93, and were all found to be highly significant ($p < 0.001$). This indicates that the relative moisture contents among the plots are conserved across year and season.

Physical factors

The slope at each sampling plot was determined using a clinometer (slope was measured for the width of the plot).

Statistical analysis

Canonical correspondence analysis (CCA) and partial canonical correspondence analysis (pCCA) were used to investigate the effect of overstory trees (TM) and abiotic environmental factors (AM) on the sapling composition (SM) in the understory (ter Braak, 1986, 1988). CANOCO (ver. 4.0, ter Braak & Šmilauer, 1998) was used to perform the analyses. Density values in the SM were square-root transformed prior to analysis and the "downweight" function in CANOCO was used to reduce the effects of rare species (weighted total density is used as weights instead of total density, see ter Braak & Šmilauer, 1998 for details). Transformation used for the variables included in TM and AM are given in Tables 2 and 3. We used unimodal models over linear models (i.e. redundancy analysis (RDA)) since the standard deviation in unimodal models exceeded 2 S.D. (ter Braak, 1995). For each CCA/pCCA run, significance of the sum of all canonical eigenvalues and the significance of the first two ordination axes were tested using Monte Carlo permutation tests (1000 permutations) (ter Braak & Šmilauer, 1998; Legendre & Legendre, 1999).

Because a high correlation was found among the moisture content measurements taken on different dates, preliminary analysis was done to avoid the redundancy among these measurements. Preliminary CCA was conducted using SM as the main matrix, and moisture measurements as the explanatory matrix. Note that moisture measurements taken on different dates are considered as separate environmental variables in the explanatory matrix. The results showed that all the environmental variables (moisture measurements taken on different dates) correlate positively with the first CCA-axis, which indicates that inclusion of all moisture variables may be superfluous. Thus, we chose only one soil moisture variable (i.e. measurement taken on a single day) for the analysis by the following two steps. First, moisture content variables with missing values were dropped. Second, CCA with forward selection was done to choose the best moisture content variable to be used in AM. This procedure resulted in choosing the measurements taken on August 25, 1999 (Table 3; MST). Note that when all the moisture measurements were used in CCA, the eigenvalue of the first canonical axis was 0.253; when measurement taken on August 25, 1999 was the only moisture variable, the eigenvalue was 0.213. This indicates that addition of moisture measurements taken on different dates in the analysis do not significantly increase the amount of variation explained. While the use of August moisture measurement was chosen statistically, August measurement may be the best measurement to use as an environmental variable from a biological perspective as well: soil moisture is most limiting at Mont St. Hilaire in the late summer months (Rouse & Wilson 1969).

Variation explained by all explanatory variables

CCA was performed on sapling data (SM) using all variables in TM and AM as explanatory variables. This was done to determine how much of the total variation is explained by all environmental variables combined and to examine which variables best explain the sapling variation. We will refer to the variation explained by both TM and AM simultaneously as "total explained variation". Note the difference between "total variation" and "total explained variation"; "total variation" is the sum of "total explained variation" and "unexplained" variation.

Variation partitioning

Total sapling composition variation ("total variation") was partitioned into four components using a series of CCA and partial CCA (pCCA) (ter Braak, 1986, 1988; Borcard et al., 1992; Legendre & Legendre, 1999). a) "Pure" overstory tree (TM) variation: fraction of the total variation that can be explained by TM variables independently of variables in AM. b) "Pure" abiotic (AM) variation: fraction of the total variation that can be explained by AM variables independently of variables in TM. c) "Shared" variation: fraction of the total variation explained by the interaction of the two factors d) "Unexplained" variation: the fraction of the total variation explained neither by TM nor by AM. The use of the terms "pure" and "shared" follows Magnan et al. (1994).

The procedure of partitioning the total variation into four components involves the following steps: (1) compute the variation accounted for by TM and AM together (total explained variation); (2) compute the variation accounted for by TM after removing the effects of AM (pure TM variation); (3) compute the variation accounted for by AM after removing the effects of TM (pure AM variation); (4) Subtract (2) and (3) from (1) to obtain the shared variation. (5) The unexplained portion of the variation is obtained by subtracting the total explained variation in step (1) from the total variation. (For more details on the partitioning procedure, see Borcard et al. (1992).)

Identifying the variables that best represent each variation component

By examining the ordination diagrams resulting from analyses (2) and (3) above, we identified variables that may best account for pure TM variation and pure AM variation. We also attempted to identify variables that best represent the shared variation component, however, this is not easily done because this variation is not derived directly from CCA or pCCA. To estimate the variables that may best account for the shared variation, we examined ordination results from analysis (1) and attempted to identify any relationship between abiotic and overstory tree variables. Abiotic variables that account for the shared variation component should correlate with one or more overstory tree variables, and vice versa.

Note on the percentage of variation explained

Økland (1999) has suggested that percentage of variation explained, which is a ratio of variation-explained-to-total inertia (or total explained variation divided by the total variation), may underestimate how much the variation components truly contributes to the total variation and has proposed a shift in focus from variation-explained-to-total inertia ratio to the relative amounts of variation explained. In this study, we mainly focused on the relative importance of different ordination axes and variation components on the total explained variation, rather than on the total variation.

Results

The total variation of the sapling composition data was 1.591 and the sum of all canonical eigenvalues (total explained variation) was 0.725. A Monte Carlo permutation test for the sum of all canonical eigenvalues showed that there is a significant relationship between plot-to-plot variation in the sapling composition data and the explanatory variables ($p = 0.001$). The first two CCA axes explain 78 % of the total explained variation and both axes are significant based on the Monte Carlo permutation test ($p = 0.001$). Figure 1 shows the ordination diagram resulting from this CCA; Figs. 2 and 3 show the density of saplings and basal area of canopy trees in ordination space, respectively.

The first CCA axis accounts for 48 % of the total explained variation (eigenvalue = 0.345) and correlates positively with Qrb_{15} and negatively with Asa_{15} and Fgr_{15} (Table 4, Fig. 1). The effects of these overstory tree variables on the first axis can be considered as combined effects of two gradients represented by one or two of the overstory tree variables that are also correlated with some abiotic environmental variables. The first gradient correlates positively with Qrb_{15} and negatively with Asa_{15}; this gradient is likely

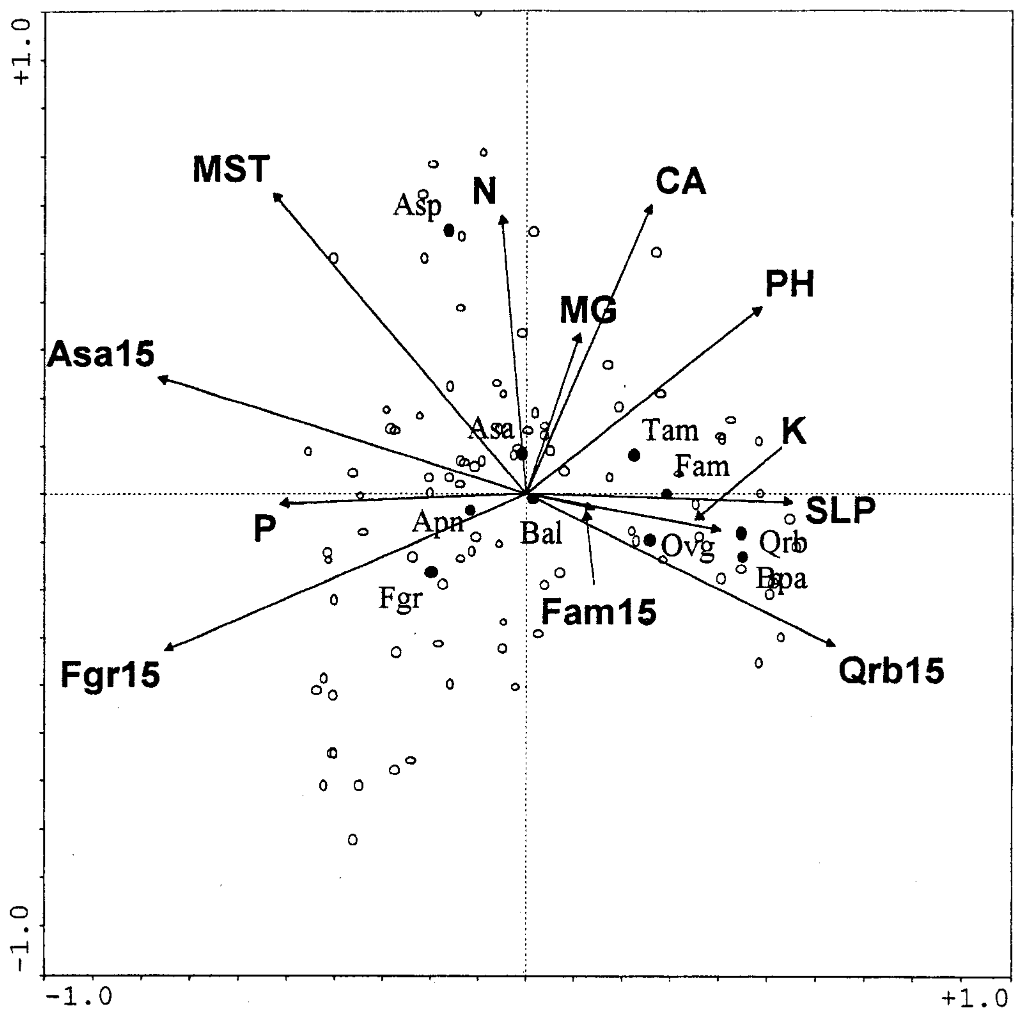


Figure 1. Ordination diagram (biplot scaling) based on CCA using all variables in overstory tree matrix (TM) and abiotic variable matrix (AM) as explanatory variables. Refer to Tables 1-3 for explanation of acronyms.

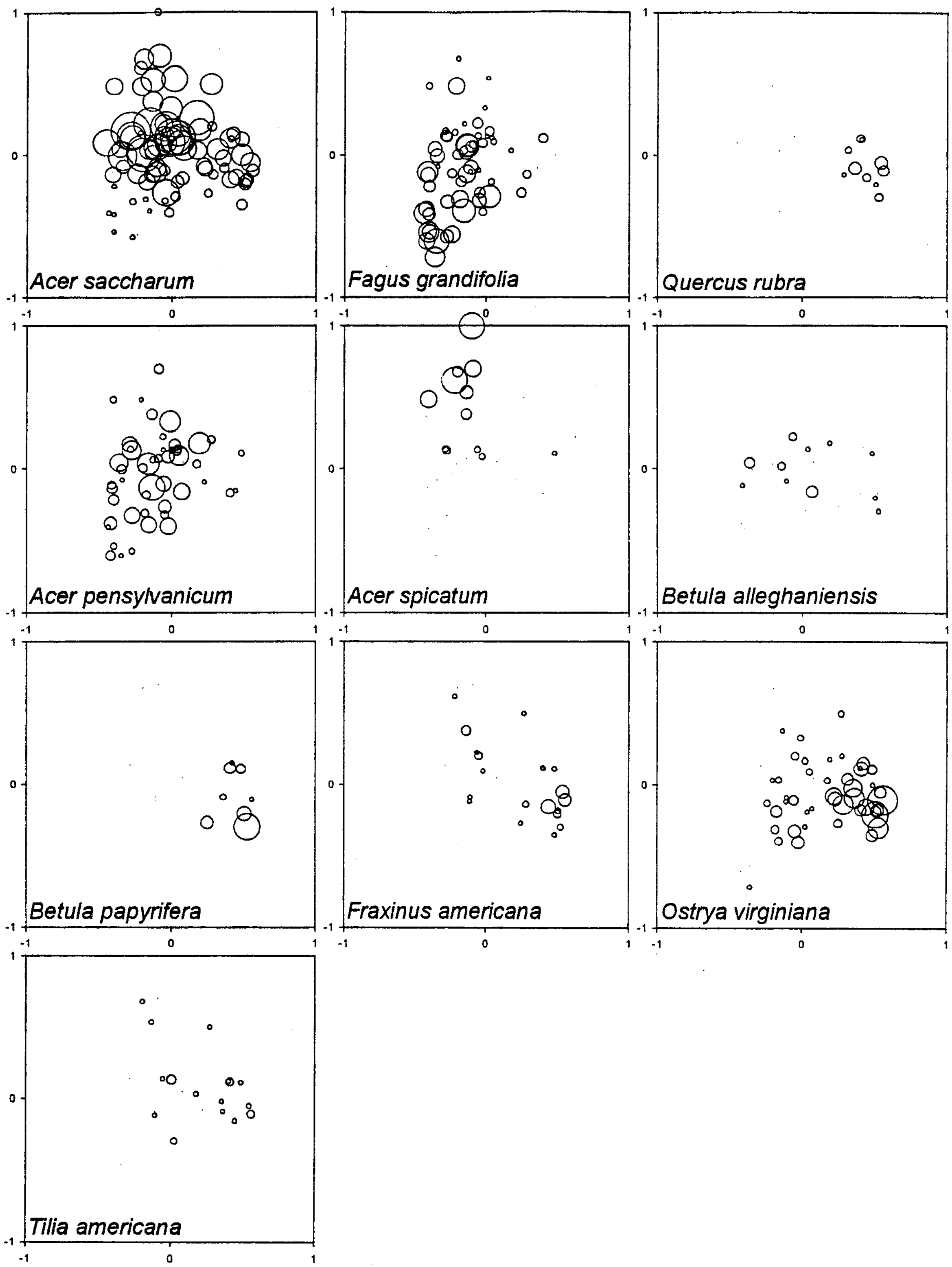


Figure 2. Density of saplings in ordination space for each species. Density (square-root transformed) of saplings at each site is superimposed on the site scores given in Fig. 1. Bigger symbols indicate greater stem density.

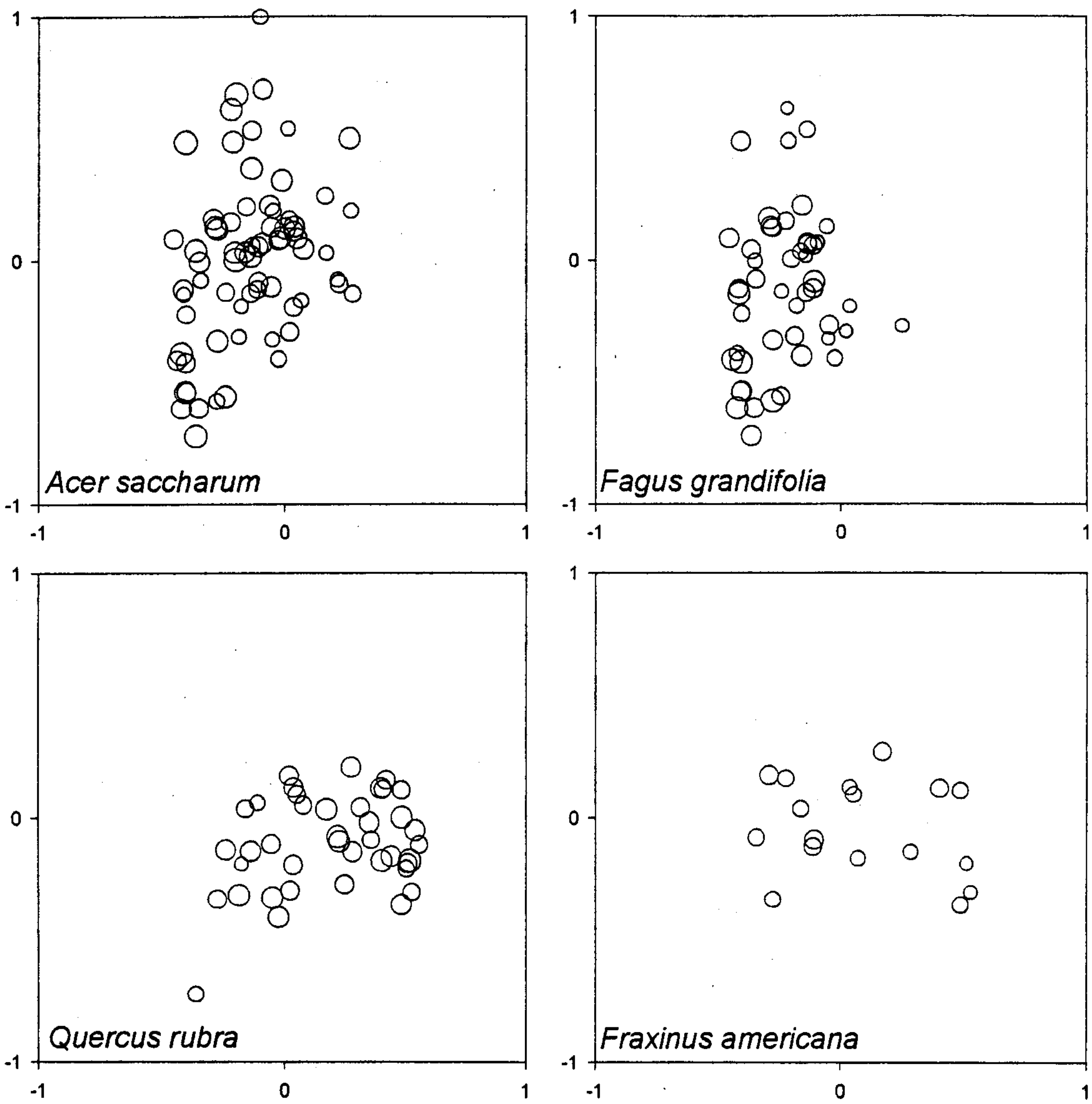


Figure 3. Basal area of canopy trees in ordination space for each species. Basal area (log-transformed) of trees at each site is superimposed on the site scores given in Fig. 1. Bigger symbols indicate greater basal area.

Table 4. Canonical coefficients and intra-set correlation coefficients of explanatory variables for the first two CCA-axes (all variables in TM and AM are used simultaneously as explanatory variables).

Variable	Canonical coefficients		Correlation coefficients	
	Axis 1	Axis 2	Axis 1	Axis 2
Asa_{15}	-0.38	0.08	-0.77	0.24
Fam_{15}	0.07	-0.03	0.14	-0.03
Fgr_{15}	-0.39	-0.24	-0.75	-0.33
Qrb_{15}	0.19	0.07	0.64	-0.32
N	-0.12	0.39	-0.05	0.58
P	-0.13	-0.08	-0.51	-0.02
MG	-0.47	-0.45	0.11	0.33
CA	0.39	0.88	0.26	0.60
K	0.23	-0.48	0.40	-0.08
PH	0.14	-0.05	0.49	0.39
SLP	-0.04	0.20	0.55	-0.02
MST	-0.03	0.48	-0.53	0.62

to reflect a soil moisture gradient as it is correlated negatively with MST (Fig. 1; this gradient will be referred to as the "MST gradient"). The second gradient is represented by Fgr_{15}, which correlates negatively with pH and Ca (Fig. 1; this gradient will be referred to as the Fgr_{15} gradient). Given the spread of the species scores along the two gradients, MST gradient appears to explain more sapling variation than the Fgr_{15} gradient.

Along the MST gradient, *Quercus rubra* (Qrb) and *Betula papyrifera* (Bpa) occur only on the drier plots, while *Acer spicatum* (Asp) occurs on the wettest (Figs. 1 and 2). The species scores of the two focal species, *Fagus grandifolia* (Fgr) and *Acer saccharum* (Asa), are both found near the origin of the diagram along the MST gradient (Fig. 1). However, *A. saccharum* saplings are found on a wider range of moisture conditions than *F. grandifolia* saplings (Fig. 2). Species scores for *F. grandifolia* and *Acer pensylvanicum* (Apn) are positioned at the higher extreme along the Fgr_{15} gradient (Fig.2), thus they are the two species that are found abundant at plots where *F. grandifolia* trees have higher representation in the canopy (Figs. 2 and 3). Saplings of other species rarely occur on sites dominated by *F. grandifolia*. Note that *A. saccharum* saplings, although they are abundant on most plot locations, are not found on sites where *F. grandifolia* trees are found in the overstory (Figs. 2 and 3).

The second CCA axis in Fig. 1 accounts for an additional 31 % of the total explained variation (eigenvalue = 0.224) and is represented by soil moisture (MST), Ca, and N (Table 4). Upon inspection of the species scores along the second axis, it is clear that *A. spicatum* has a significant influence on this axis (Fig. 1). To evaluate the influence of *A. spicatum*, we ran a CCA without *A. spicatum* in the data set. The results show that the second axis decreases its importance after *A. spicatum* is removed (eigenvalues of the second axis with and without *A. spicatum* are 0.224 and 0.153, respectively), and soil moisture (MST) is represented more weakly along the second axis (Fig. 4a). Despite the removal of *A. spicatum*, the general interpretation of other species along environmental gradients does not change (compare the relationship between the species scores and environmental gradients in Fig 1 and Fig. 4a). Our two focal species have opposed species scores along the second axis (Fig. 1). *Acer saccharum* saplings are found on sites with higher N- and Ca-availabilities than *F. grandifolia* saplings, however, Ca availability better separates the two species (Fig. 1).

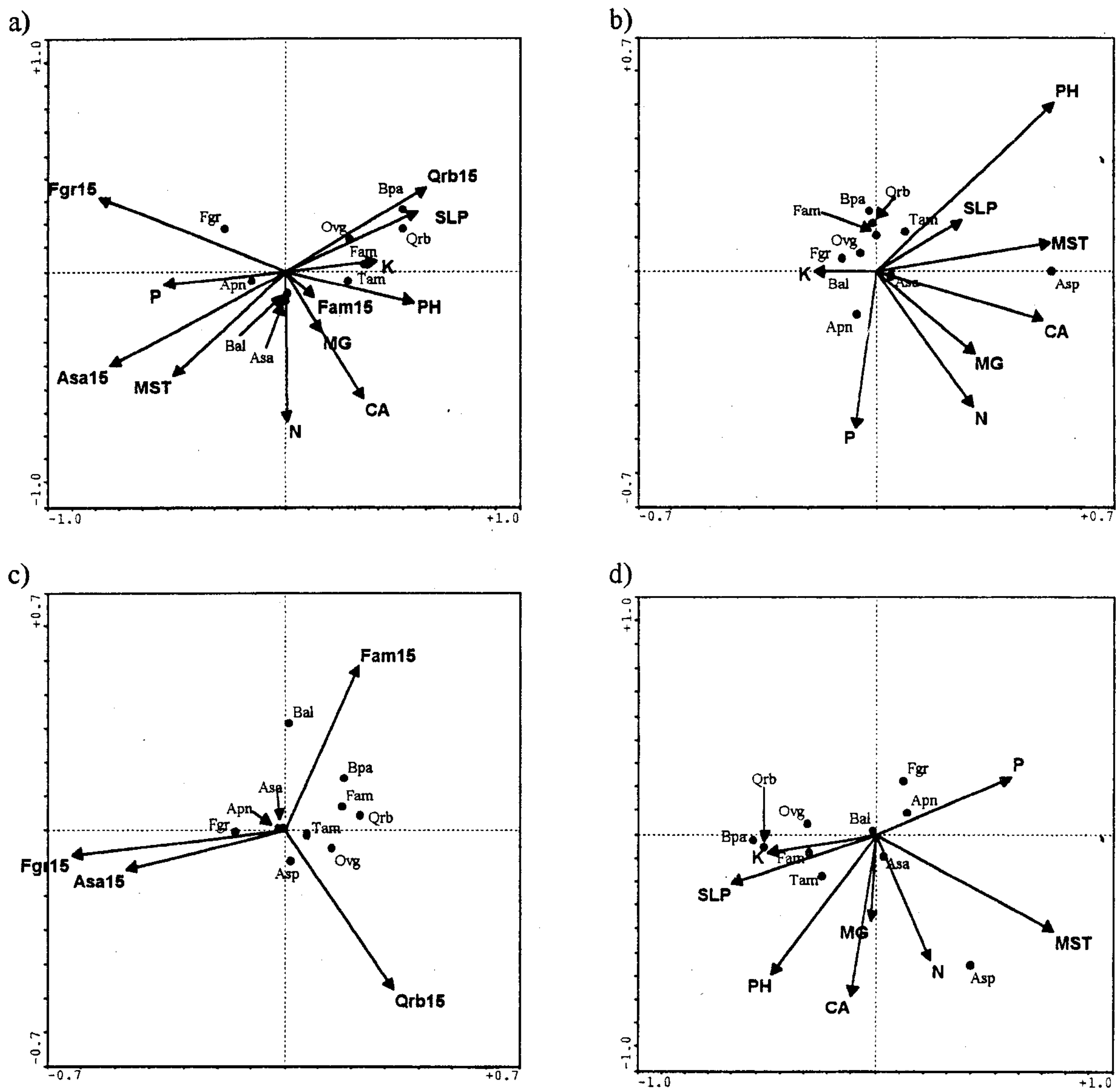
Variation partitioning

The result of variation partitioning procedure is given in Table 5. It shows that pure TM variation accounts for 20 % of the total explained variation ($p = 0.001$), while pure AM variation ($p = 0.001$) and shared variation each account for 40 % of the total explained variation.

Variables that represent the pure AM variation

Figure 4b shows the species-environment pCCA ordination diagram using AM as the explanatory matrix and TM as the covariable matrix (i.e. pure AM variation). The first axis accounts for 53 % of pure AM variation. The diagram shows that *A. spicatum* has an extremely high first axis score, which is represented well by soil moisture. However, if *A. spicatum* is removed from the analysis, much of the remaining species variation can be explained by a N-Ca gradient. This finding is similar to the results found for the second axis in Fig. 1, and the species occurrence along the first axis in Fig. 4b is similar to the second axis in Fig. 1. Because the effects of overstory trees were mainly reflected on the first axis in Fig. 1, the second axis, which is orthogonal to the first axis, may indeed be

Figure 4. (a) Species-environment ordination diagram (biplot scaling) based on CCA using all variables in overtory tree matrix (TM) and abiotic variable matrix (AM) as explanatory variables, but with *Acer spicatum* excluded from the sapling matrix (SM) (refer to text for details). (b) Species-environment ordination diagram (biplot scaling) based on a pCCA using variables in abiotic variable matrix (AM) as the explanatory variables and variables in overstory tree matrix (TM) as the covariable variables (accounts for pure AM variation). (c) Species-environment ordination diagram (biplot scaling) based on a pCCA using variables in overtory tree matrix (TM) as the explanatory variables and variables in abiotic variable matrix (AM) as the covariable variables (accounts for pure TM variation). (d) Species-environment ordination diagram (biplot scaling) based on CCA using all variables in abiotic variable matrix (AM) as explanatory variables. Refer to Tables 1-3 for explanations of acronyms.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PH
CA

Table 5. Variation explained by each variation component.

Variation component	Sum of all canonical eigenvalues	Percentage of total explained variation	Percentage of total variation
Pure TM variation	0.143	20	9
Pure AM variation	0.288	40	18
Shared variation	0.294	40	18
Total explained variation	0.725	100	45

represented mostly by the variables that explain the pure AM variation. Note that the best separation of species scores of our two focal species occurs on a Ca gradient.

Variables that represent the pure TM variation

Figure 4c shows the species-environment pCCA ordination diagram using TM as the explanatory matrix and AM as the covariable matrix (i.e. pure TM variation). The first axis represents approximately 69 % of pure TM variation and is negatively correlated with Fgr_{15}; the second axis was found to be non-significant. Although Fgr_{15} was also highly correlated with the first axis in Fig. 1, this effect of Fgr_{15} is correlated with abiotic variables. Therefore, the effect of Fgr_{15} in Fig. 4c, which is the effect of Fgr_{15} not correlated with abiotic variables, may account for considerable portion of the remaining variation that was not explained by the first two axes in Fig. 1.

Based on the ordination diagram in Fig. 4c, *F. grandifolia* saplings occur on sites where *F. grandifolia* trees are more present in the overstory, while *A. pensylvanicum*, *A. saccharum*, *A. spicatum*, and *Betula alleghaniensis* (Bal) occur on sites with intermediate occurrence of *F. grandifolia* trees. Based on the occurrence of species along this gradient, we believe this may represent a light availability gradient, a variable not included in our analysis. However, the variation explained by pure TM variation is small, relative to the other variation components.

Variables that represent the shared variation component

In Fig. 1, the overstory tree variables, which were highly correlated with the first ordination axis, are also correlated with some abiotic environmental variables. This implies that these abotic and overstory tree variables that are correlated may explain a significant portion of the shared variation component. Additionally, the fact that these variables were reflected on the first ordination axis may explain why the shared variation explained a high proportion of the total explained variation (40 %). The correlation between the overstory tree variables and abiotic variables can also be examined by comparing the ordination diagram in Fig. 1 with an ordination diagram based on a CCA that uses abiotic factors as the only explanatory variables (Fig. 4d). When the two ordination diagrams are compared, the relationships among the abiotic variables (relative directions of the arrows), and the occurrence of species (saplings) scores along the abiotic variables are almost identical (note that the two diagrams are rotated 180°). Additionally, the sum of canonical eigenvalues of the first two axes in Fig. 4d is 0.476, which is 85 % of the sum of canonical eigenvalues of the first two axes in Fig. 1. This suggests that much of the information in Fig. 1 is being preserved in Fig. 4d even though the overstory tree effects had been removed and that overstory tree effects are indeed correlated with some or a set of abiotic environmental variables.

Many of the previous studies attempting to identify the factors that give rise to the patterns of spatial distribution and species composition of saplings have been done 1) by examining the overstory-understory relationships or 2) by investigating the occurrence of saplings along one or a few environmental gradients. It is clear in the preceding results that some variables in the TM (overstory effects) and AM (abiotic environmental effects) are collinear, and that these collinear variables account for a considerable fraction of the total explained variation. We therefore emphasize the nature and the implications of this collinearity in the following discussion.

Discussion

While various factors may contribute to the patterns of spatial variation and species composition of the sapling community, our results show that soil moisture has the strongest influence on the overall pattern. Soil moisture is negatively correlated with *Q. rubra* in the overstory (Qrb_{15}) and positively with *A. saccharum* in the overstory (Asa_{15}). This relationship is in agreement with previous studies on the distribution patterns of adult trees along moisture gradients. *Quercus rubra* trees are often found on dry sites at the northern edge of its distribution, while *A. saccharum* trees are typically found on more mesic sites (Walther, 1963; Op de Beek, 1972; Anderson et al., 1990). The occurrence of saplings (i.e. species scores) along the soil moisture gradient agrees with a study that investigated the distribution patterns of saplings along moisture gradients in the forests of the St. Lawrence Lowlands in Quebec and Ontario (Op de Beek, 1972); this study included stands on Mont St. Hilaire.

Soil moisture also partially explains the differences in spatial patterns of our two focal sapling species, *F. grandifolia* and A. *saccharum*. While *A. saccharum* tends to occur across the full range of soil moisture conditions we have sampled, *F. grandifolia* saplings are completely absent on the drier sites. Both species are generally known to be drought intolerant, but *F. grandifolia* is known to be more sensitive to reduced moisture than *A. saccharum* (Tubbs & Houston, 1990), which may explain their absence at the drier sites.

However, on mesic sites, moisture does not distinguish the distribution patterns of *F. grandifolia* and *A. saccharum* saplings. On these sites, relative dominance of *F. grandifolia* trees in the overstory (Fgr_{15}) appears to have considerable influence; *F.*

grandifolia saplings are found on sites with more *F. grandifolia* trees in the canopy, while *A. saccharum* saplings are mostly absent on such sites.

One of the hypotheses to explain the coexistence and the replacement patterns in *Fagus-Acer* forests is reciprocal replacement (Fox, 1977; Woods, 1979; Runkle, 1981), which suggests that seedlings and saplings of each species tend to grow and survive better under canopies of the other species than under conspecific adults. We found no evidence of such reciprocal replacement patterns: *Fagus grandifolia* saplings are found only at sites where *F. grandifolia* overstory trees are abundant, suggesting "self-replacement". *Acer saccharum* saplings are found abundant at most locations at the study site, except at sites where *F. grandifolia* trees are found more abundant in the overstory; these are the sites where the reciprocal replacement hypothesis predicts *A. saccharum* saplings should be abundant.

Our results show that the effects of *F. grandifolia* overstory trees are correlated with pH and Ca availability. While a naturally occurring pH and Ca availability gradient may have given rise to the changes in relative dominance of *F. grandifolia* in the overstory, *F. grandifolia* trees are also known to influence edaphic conditions in the understory. Leaf-litter of *F. grandifolia* has low pH (Finzi et al., 1998), and as the pH is lowered, availability of Ca decreases (Ellis & Mellor, 1995). Additionally, *Fagus* leaf-litter is low in Ca concentration (Gosz et al., 1973; Côté & Fyles, 1994a,b) and also has high levels of lignin and polyphenols (Melillo et al., 1982; Aber et al., 1990), which can complex with Ca and increase leaching of these elements (Davies, 1971). Therefore, pH and Ca availability may be decreased in stands with frequent occurrence of *F. grandifolia* trees. From this study, we are unable to determine whether the edaphic conditions gave rise to changes in relative dominance of *F. grandifolia*, or if *F. grandifolia* overstory trees influenced the edaphic conditions. However, what our results do show is that low pH and Ca availability are found at sites where *F. grandifolia* trees are dominant in the overstory, as also has been reported in other studies (van Breeman et al., 1997; Finzi et al., 1998).

Kobe et al. (1995) and Kobe (1996) found that mortality of *A. saccharum* saplings was higher in non-calcareous soils than calcareous soils under deep shade (less than 5 % of the full sun). *Acer saccharum* has been found to achieve maximum dominance at sites with high pH and Ca content in northern Michigan (Woods 2001), and deficiencies in Ca have also been linked to the wide spread dieback of *A.saccharum* in southern Quebec and adjacent regions (Liu et al., 1997). Thus the low pH and Ca availability under *F. grandifolia* trees may decrease the growth and survivorship of *A. saccharum* saplings, and therefore may explain the lack of regeneration of *A. saccharum* saplings at sites where *F. grandifolia* trees are frequnetly found.

It is often argued that light is the main controlling factor in determining the patterns of understory vegetation in eastern deciduous forests of North America (Shugart, 1984; Kobe et al., 1995; Finzi & Canham, 2000), and in *Fagus-Acer* forests, temporal variation in understory light conditions has been considered to be a key factor in explaining the patterns of canopy replacement and coexistence of *F. grandifolia* and *A. saccharum* (Runkle, 1984; Poulson & Platt, 1996). Poulson & Platt (1996) have suggested that *F. grandifolia* tends to gain importance when the disturbance is infrequent (low light conditions persist for extended periods of time), and *A. saccahrum* tends to increase importance when it is frequent (gaps are created and light availability temporarily increases). This hypothesis is based on the fact that *F. grandifolia* survives for longer periods of time and grow faster upward in the shaded understory than *A. saccharum*, while *A. saccharum* grows more rapidly upward in gap conditions (Canham, 1988;

Poulson & Platt, 1989). This known difference in species' response to light coupled with the frequency of disturbances, which creates temporal and spatial variations in understory light conditions, is the mechanistic basis of the "disturbance" hypothesis.

At our study site, relatively large-scale ice storms have occurred approximately every 20 years for the past 60 years (major storms have been recorded in 1942, 1961, 1983, 1998). The ice storm damage creates gaps in the canopy, and while there is spatial variation in the gaps created, the damage is wide spread, which is unlike the gaps created by treefalls (Runkle, 1984; Hooper et al., 2001). Based on the "disturbance" hypothesis, our study site should be advantageous for *A. saccharum* saplings given the frequency of disturbance, and as expected, most of the study plots have high density of *A. saccharum* saplings in the understory. However, this pattern does not apply at all sites; *A. saccharum* saplings are almost absent at sites where *F. grandifolia* trees are dominant in the overstory. This exception suggests that edaphic conditions under an overstory with *F. grandifolia* (low pH and Ca availability) may indeed have a strong influence on the distribution pattern of *A. saccharum* saplings. There may be two reasons why this pattern may not be detected at the study site where Pouson and Platt (1996) conducted their study. First is the fact that the soil at their study site is highly calcareous, and the second is the possibility of "climatic harshness" amplifying the influence of low pH and Ca availability on the growth and survival of *A. saccharum* toward the northern range of its distribution.

Another factor that may be important in explaining the abundant occurrence of *F. grandifolia* saplings (or the absence of *A. saccharum* saplings) at sites where *F. grandifolia* trees dominate is their ability to propagate clonally by root sprouts. The occurrence of root sprouts is considered to gain importance with increasing latitude (Held, 1983; Melancon & Lechowicz, 1987), and indeed *F. grandifolia* sprouts have been found to occur extensively at our study site (Kitamura et al., 2000; Arii, unpublished data), as well as in *Fagus-Acer* forests also in southwestern Québec (Beaudet et al., 1999). Beaudet et al. (1999) have shown that *F. grandifolia* sprouts (up to 3 m) grow considerably faster than the adjacent *A. saccharum* saplings in an old-growth *Fagus-Acer* forest in Québec. This suggests that, during the "inter-disturbance" periods, *F. grandifolia* saplings of sprout origin could gain considerable height advantage over *A. saccharum* saplings in the understory, which may benefit *F. grandifolia* saplings when a gap opens. Additionally, we found that *F. grandifolia* sprouts grow just as fast as *Acer* saplings after gap openings created by an ice storm (Arii, unpublished data). Thus, at sites where root sprouts occur in greater density (i.e. sites where *F. grandifolia* trees mainly dominate the overstory), *A. saccharum* saplings may not benefit from having frequent disturbances. At the study site where Poulson & Platt (1996) conducted their study, root sprouting was not found to be important. Therefore, in conjunction with the edaphic conditions, the extensive occurrence of root sprouts may explain why *A. saccharum* saplings are not found at sites where *F. grandifolia* trees dominate the overstory, despite the frequent occurrence of disturbance at our study site.

Conceptual Model for *Fagus-Acer* dynamics

In summary, we provide a conceptual model explaining how *F. grandifolia* saplings can "self-replace" despite a frequent occurrence of canopy disturbance. The model can be viewed as an addendum to the model proposed by Poulson & Platt (1996) for locations with frequent disturbances. Consider an early successional forest where *F. grandifolia* and *A. saccharum* begin to establish (from seed). Because the light level is still relatively high beneath the early successional canopy, *A. saccharum* seedlings/saplings generally will perform better than *F. grandifolia* seedlings/saplings, except at sites where the conditions are less favorable for *A. saccharum* (e.g. more acid soil). At sites where *A. saccharum* saplings perform well, they will eventually dominate the overstory, and frequent canopy disturbance will ensure continuous regeneration of *A. saccharum* at these locations (Poulson & Platt, 1996). At locations where *A. saccharum* seedling/sapling growth is retarded, *F. grandifolia* saplings will capture a greater proportion of the overstory, which will make the conditions less suitable by modifying the edaphic conditions. Furthermore, adult beech trees will produce root sprouts. These factors lead to local dominance by *F. grandifolia* saplings/trees, and ensure "self-replacement" of *F. grandifolia* at these sites.

It should be noted that even at locations where conditions are favorable for *A. saccharum*, if several *F. grandifolia* seedling/saplings of seed-origin are able to establish by chance and grow into overstory trees, they may be able to slowly replace the neighboring stands over time by root sprouts. At our study site, we have observed numbers of small patches consisting of almost pure *F. grandifolia* stands, both in the overstory and understory, in an area where *A. saccharum* dominates the overstory. These patches may suggest the rare occasions where *F. grandifolia* of seed-origin are able to establish by chance and grow into overstory trees at sites more favorable for *A. saccharum*.

Conclusion

Our study has shown that the soil moisture gradient, which correlates negatively with relative dominance of *Q. rubra* overstory trees and positively with *A. saccharum* overstory trees, contributes the most in explaining the patterns of spatial variation and species composition of the sapling community in an old growth *Fagus-Acer* forest. *Acer* *saccharum* was found to occur on a wide range of soil moisture conditions, while *F. grandifolia* saplings were not found on the dry sites. The relative dominance of *F. grandifolia* in the overstory, which correlates negatively with pH and Ca availability, also explained the differences in the spatial distribution patterns of *A. saccharum* and *F. grandifolia* saplings at our study site. *A. saccharum* saplings were not found on sites where *F. grandifolia* dominates the overstory, while *F. grandifolia* saplings are mainly limited to sites dominated by *F. grandifolia* overstory trees. These finding do not support existing hypotheses for canopy tree replacement patterns in *Fagus-Acer* forests.

Acknowledgements

This work was supported by grants from Natural Sciences and Engineering Research Council of Canada (NSERC) and Fonds pour la Formation de Chercheurs et l'Aide à la Recherche (FCAR). We thank McGill University for access to the Gault Nature Reserve. We thank Julien Mercille and Julie Girouard for their assistance in the field and Pierre Dutilleul for statistical advice.

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Chapter 4

Seedling and juvenile height growth of *Acer saccahrum* and sprout-origin *Fagus grandifolia* after an ice storm.

Abstract

Seedling (height < 1 m) and juvenile (1<height<2 m) height growth of *Acer saccharum* Marsh. and spout-origin *Fagus grandifolia* Ehrh. were compared in gap light conditions following a severe ice storm. Sprout-origin *Fagus* grew faster upward than *Acer*, and *Acer* growth was retarded at locations where *Fagus* saplings were found abundantly in the understory. The fact that sprout-origin *Fagus* grew faster upward than *Acer* does not agree with the common idea that *Acer* is able to better exploit increased light conditions. The importance of *Fagus* sprouts previously has been argued in terms of their ability of grow faster upward and survive for longer periods of time under closed canopy conditions. Our finding that sprout-origin *Fagus* can also grow faster than *Acer* in elevated light conditions further emphasizes the potential importance of *Fagus* sprouts in the dynamics of *Fagus-Acer* forests.

Introduction

Fagus-Acer forest is one of the major forest types found in eastern North America (Braun, 1950; Runkle, 1996). The dynamics within this forest type have been frequently investigated, a particular interest being the replacement patterns of the two codominant species, *Fagus grandifolia* Ehrh. and *Acer saccharum* Marsh. (Forcier, 1975; Fox, 1977; Woods, 1979, 1984; Runkle, 1981, 1984; Cypher and Boucher, 1982; Brisson et al., 1994; Kupfer and Runkle, 1996; Poulson and Platt, 1996; Foré et al., 1997; Beaudet et al., 1999; Forrester and Runkle, 2000). The replacement patterns of the two species have been previously explained by temporal variation in understory light conditions (Runkle, 1984; Poulson and Platt, 1996). Poulson and Platt (1996) suggested that *Fagus* gains importance when the disturbance is infrequent (low light conditions in the understory persist for extended periods of time), while *Acer* increases importance when the disturbance is frequent (gaps are created and light availability temporarily increases). This hypothesis is based on species' differences in growth response under varying light conditions: *Fagus* survives for longer periods of time and grows faster upward in the shaded understory than *Acer*, while *Acer* grows more rapidly upward in gap conditions (Canham 1988; Poulson and Platt 1989, 1996). This known difference in species' response to light, coupled with the frequency of disturbances, which creates temporal and spatial variations in understory light conditions, is the mechanistic basis of the "disturbance" hypothesis (Poulson and Platt, 1996).

At our study site, which is mainly composed of *Fagus-Acer* forest (Arii and Lechowicz, 2002a), major ice storms occur frequently (about every 20 years) and damage the forest canopy over a large area. Based on the disturbance hypothesis, this condition should favor *Acer* dominance in the understory, and as expected, the study site does have a high density of *Acer* saplings at most locations. However, previous study has shown that, despite the frequent occurrence of ice storms, there are some localities at our study site where *Acer* saplings are almost absent; the understory at these sites is dominated by *Fagus* saplings (Arii and Lechowicz 2002b). Because the presence of these sites is inconsistent with the predictions made by the disturbance hypothesis, we may first question the validity of the assumption of the hypothesis that *Acer* grows more rapidly upward in gap light conditions than *Fagus*. We suggest two possible explanations that may contribute to *Fagus* having greater growth than *Acer* even in higher light conditions. Firstly, because *Acer* saplings were almost absent only at sites where *Fagus* saplings were found abundantly in the understory, we may, by first approximation, hypothesize that *Acer* growth is somehow restricted at sites dominated by *Fagus*. The slower growth of *Acer* at these sites may allow *Fagus* to attain height growth advantage even in increased light conditions. Secondly, the ability of *Fagus* to regenerate by root sprouts (Ward, 1961) may contribute to *Fagus* having greater growth than *Acer*. Previous studies that have shown that *Acer* saplings grow better than *Fagus* in increased light conditions are based on comparisons between *Fagus* and *Acer* saplings of seed-origin (Poulson and Platt 1989, 1996). The sprout-origin *Fagus* seedlings and saplings, with "subsidies" from the parent tree, may potentially grow faster upward than *Acer* even in gap light conditions.

The objective of this study is to reexamine the generality of the notion that *Acer* seedlings and saplings grow better than *Fagus* under increased light conditions. Two specific questions are asked: 1) Does *Acer* have a height growth advantage over sprout-origin *Fagus* under increased light conditions, 2) Under gap light conditions, is *Acer* growth retarded at locations where *Fagus* saplings are found abundantly in the understory. We test this by comparing the height growth of *Acer* juveniles/seedlings with that of sprout-origin *Fagus* under increased light conditions created by an ice storm, and by examining the height growth of *Acer* seedlings/juveniles in relation to the relative occurrence of *Fagus* saplings in the understory. If *Fagus* seedlings/juveniles are found to grow faster than *Acer*, an important assumption of the disturbance hypothesis is violated. This would have significant implications for our understanding of the mechanisms underlying the dynamics of *Fagus-Acer* forests.

Materials and methods

In January 1998, an extensive ice storm hit southwestern Quebec, southeastern Ontario, and the adjacent United States (Irland 1998, Laflamme and Périard 1998); our study site was one of the hardest hit locations, receiving more than 100 mm of freezing rain. The storm caused extensive damage to the trees (Duguay et al. 2001, Hooper et al. 2001), which created significant canopy gaps (Arii and Lechowicz 2002c). For the subset of plots used in this study, gap fraction values in the summer immediately after the ice storm ranged between 15-45%; these values are significantly greater than the pre-storm conditions (gap fraction values were estimated by using hemispherical photographs taken at 0.6 m above the soil surface, before and after the ice storm: Arii and Lechowicz 2002c).

We took advantage of this event to assess the differences in height growth between *Fagus grandifolia* and *Acer saccharum* in gap light conditions. The study was conducted in a 10 km^2 tract of old-growth forest at Mont St. Hilaire in southwestern Québec, Canada (45°31'N, 73°08'W). Mont St. Hilaire is a hill complex of plutonic origin (Feininger and Goodacre, 1995) with peaks that rise between 200-300 m above the level floor of the surrounding St. Lawrence River Valley. The forest at Mont St. Hilaire is the largest remaining remnant of the primeval forests in this region: many of the trees exceed 150 years in age and a few are over 400 years old (Cook, 1971). Various tree communities occur on the mountain (Maycock, 1961; Enright and Lewis, 1985), but *Acer saccharum* and *Fagus grandifolia* are the common canopy dominants over most of the area (Arii and Lechowicz 2002a). The site is essentially undisturbed by human activities and is protected as an International Biosphere Reserve under the Man and Biosphere program (MAB) of the United Nations. The data for this study were collected in 117 permanent plots (6-m radius or approximately 113 m^2) established randomly along seven line transects at Mont St. Hilaire (Arii and Lechowicz, 2002b). At each plot the number and the species of saplings ($1 \leq$ DBH$<$ 10 cm) in the understory are recorded. As it will be clear in the following section, only a subset of these permanent plots was used in this study.

Comparison of seedling growth

In late summer of 2000, we sampled sprout-origin *Fagus* and *Acer* seedlings (height less than 1 m) found within 2 m of the plot center. Height of a seedling is defined here as the length of the main stem from the ground surface to the apical bud, but because we limited our sampling to individuals that showed little sign of deformities, the actual height from the ground surface and the length of the stem does not differ by much. To determine if a *Fagus* seedling was of sprout-origin, we examined whether or not the stem was attached to a lateral root on a nearby *Fagus* tree. In most cases, the lateral root was at or near the ground surface, so the identification of sprouts was straightforward. Nonetheless, there were a few individuals where we were not certain if they were of sprout-origin (e.g. we could not find the lateral roots at the top 10 cm of the soil, could not trace the lateral roots back to the larger tree); these individuals were excluded from the analysis. Of all the permanent plots, we recorded data only in plots that had both *Fagus* and *Acer* seedlings, so that direct comparison is possible (24 plots out of 117); the total number of seedlings that met the selection criteria was 51 for *Fagus* and 66 for *Acer*. For each seedling, we measured current year height growth (HG_{00}) as the length of the most recently formed annual node from the terminal bud scar, to the base of the terminal bud (cm). We also measured the height growth for the previous growing season (1999) by measuring the length between the two most recently formed annual nodes (HG_{99}).

We used linear regression to examine whether the number of *Fagus* saplings in a plot would influence the height growth of sprout-origin *Fagus* and *Acer* seedlings (SYSTAT, ver 9.0). We also used height at the end of the previous growing season as an explanatory variable to control for the influence of size on height growth. Height at the end of the previous growing season is estimated by subtracting current year height growth from the height at the end of the current growing season.

Comparison of juvenile growth

In late summer of 2001, we located 40 pairs of *Fagus* and *Acer* juveniles in 29 permanent plots. Juveniles are individuals that have height ranging between approximately 1 to 2 m. We use the term juveniles to distinguish from the beech "saplings ($1 \leq$ DBH$<$ 10 cm)" that were recorded for all the permanent plots (saplings are larger than juveniles, with height exceeding 2 m). Each pair of *Fagus* and *Acer* juveniles had similar height in 2001, and the paired juveniles were located within 6 m from each other within a plot. Sprout-origin *Fagus* juveniles were distinguished from the seed-origin *Fagus* in the same manner as the *Fagus* seedlings, but most of the individuals we considered as a potential target individual turned out to be of sprout-origin. For all the sampled juveniles, we measured the height growth for the past three years (1999-2001), and estimated the height at the end of each growing season from 1998 to 2000. We compared the height of the each paired individuals for each year between 1998 and 2001 to examine the relative growth of the juveniles after the ice storm.

Results

Comparison of seedling growth

Fagus seedlings (sprout-origin) have greater height growth than *Acer* seedlings overall, across the entire range of variation in sapling height and local abundance of *Fagus* saplings (Fig. 1). Multiple linear regression, done separately for the two species and for the two years, show that height growth of *Acer* seedlings is significantly affected by height at the end of the previous growing season, as well as the number of *Fagus* saplings in a plot (Table 1). Larger seedlings have less height growth, and a greater number of *Fagus* saplings in a plot reduces the height growth of the *Acer* seedlings. In contrast, height growth of *Fagus* seedlings did not show any significant relationship with the two explanatory variables (Table 1, Fig. 1). The lack of any effect of size (i.e. height at the end of previous growing season) on *Fagus* height growth may indicate that these seedlings are indeed receiving "subsidies" from the parent tree. When interpreting these results, it should be noted that the number of *Fagus* saplings in a plot does not correlate with the total number of saplings in a plot: in fact, the total number of saplings is typically greater at sites where *Fagus* saplings are less abundant. Thus the effect of *Fagus* saplings on the growth of *Acer* seedlings is not merely a density dependent effect.

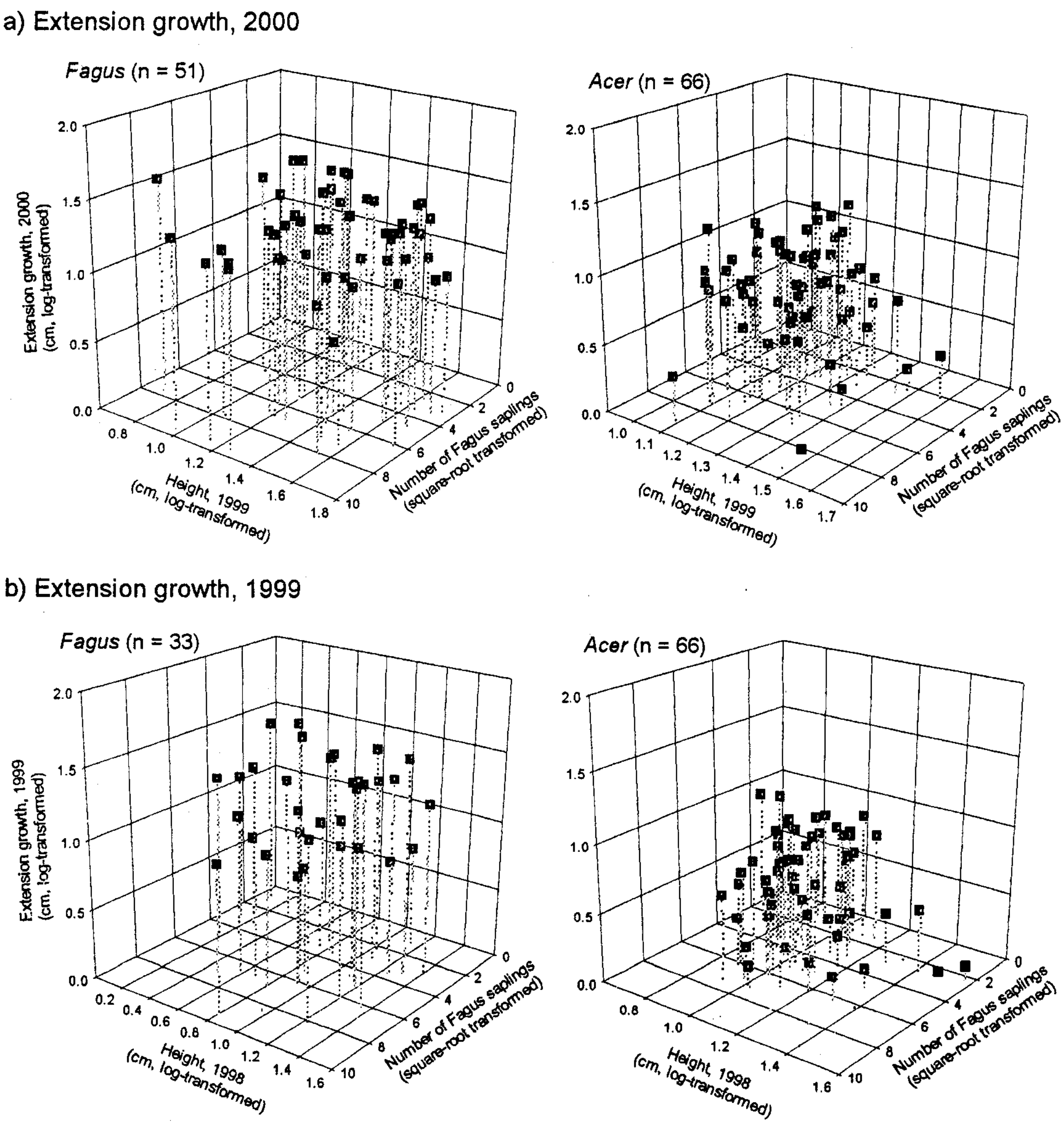

Figure 1. Height growth of *Fagus* and *Acer* seedlings plotted against the number of *Fagus* saplings in a plot and height at the end of the previous growing season: a) height growth for 2000, b) height growth for 1999. Note that the sample size for *Fagus* in 1999 is (n=33) fewer than that of 2000 (n=51). We were not able to determine with certainty the location of the annual node needed to measure 1999 height growth for some of the *Fagus* seedlings.

Table 1. Results of multiple linear regression: height growth of *Fagus* and *Acer* seedlings are used as dependent variables, while the number of *Fagus* saplings in a plot and height at the end of the previous growing season are used as independent variables. Also look at Fig. 1. a) *Acer* height growth 2000, b) *Acer* height growth 1999, c) *Fagus* height growth 2000, d) *Fagus* height growth 1999

a) *Acer* extension growth, 2000 (log transformed; $r^2 = 0.32$, $P<0.001$, $n = 66$)

Effect	Coefficient	Std. Error	Standardized coefficient	P (2 tail)
Number of *Fagus* saplings (square-root transformed	-0.037	0.017	-0.231	0.030
Height, 1999 (log-transformed)	-1.377	0.288	-0.498	<0.001
Constant	2.528	0.352	0.000	<0.001

b) *Acer* extension growth, 1999 (log transformed; $r^2 = 0.58$, $P<0.001$, $n = 66$)

Effect	Coefficient	Std. Error	Standardized coefficient	P (2 tail)
Number of *Fagus* saplings (square-root transformed	-0.059	0.015	-0.344	<0.001
Height, 1998 (log-transformed)	-1.148	0.168	-0.585	<0.001
Constant	1.961	0.174	0.000	<0.001

c) *Fagus* extension growth, 2000 (log transformed; $r^2 = 0.05$, $P=0.282$, $n = 51$)

Effect	Coefficient	Std. Error	Standardized coefficient	P (2 tail)
Number of *Fagus* saplings (square-root transformed	0.005	0.014	0.052	0.717
Height, 1999 (log-transformed)	-0.192	0.128	-0.213	0.141
Constant	1.471	0.191	0.000	<0.001

d) *Fagus* extension growth, 1999 (log transformed; $r^2 = 0.03$, $P=0.616$, $n=33$)

Effect	Coefficient	Std. Error	Standardized coefficient	P (2 tail)
Number of *Fagus* saplings (square-root transformed	-0.020	0.023	-0.161	0.397
Height, 1998 (log-transformed)	0.089	0.124	0.134	0.479
Constant	1.210	0.131	0.000	<0.001

Comparison of juvenile growth

Because paired *Acer-Fagus* juveniles were sampled to have similar height in the year in which they were measured (i.e. 2001), the data points for 2001 fall approximately on a one-to-one line when the height of the paired juveniles are plotted against each other (Fig. 2a). However, most of the points are found above the one-to-one line in 1998, indicating that *Acer* juveniles had greater height than the *Fagus* juveniles in the summer immediately after the ice storm. This implies that most *Fagus* juveniles grew upward faster than their *Acer* counterparts between 1999-2001. To assess this possibility statistically, we used paired sample t-tests to compare the height of *Acer* and *Fagus* juveniles separately for each year. The results show that the heights of *Acer* and *Fagus* do not differ significantly in 2001 and 2000, but are significantly different in 1999 and 1998 ($P<0.05$). This shows statistically that *Acer* juveniles had greater height in 1998 and 1999, but *Fagus* juveniles "caught up" with the *Acer* juveniles by growing more rapidly during the years after the ice storm.

We also examined whether the total number of *Fagus* saplings in a plot would influence the relative growth of *Acer* and *Fagus* juveniles. For this comparison, total height growth between 1999 and 2001 was calculated for each juvenile sampled; total height growth was obtained by summing all the height growth during this period. Then, for each *Acer-Fagus* pair, the difference between *Acer* juvenile total growth and *Fagus* juvenile total growth was obtained (*Acer* total growth – *Fagus* total growth). If the index value is near 0, it indicates that the paired individual of the two species grew equally well during this period. If the value is above 0, it indicates that *Acer* juveniles had grown more, and if below 0, *Fagus* juveniles had grown more. Once the differences for each pair were calculated, they were plotted against the number of *Fagus* saplings in a plot (Fig. 3). The

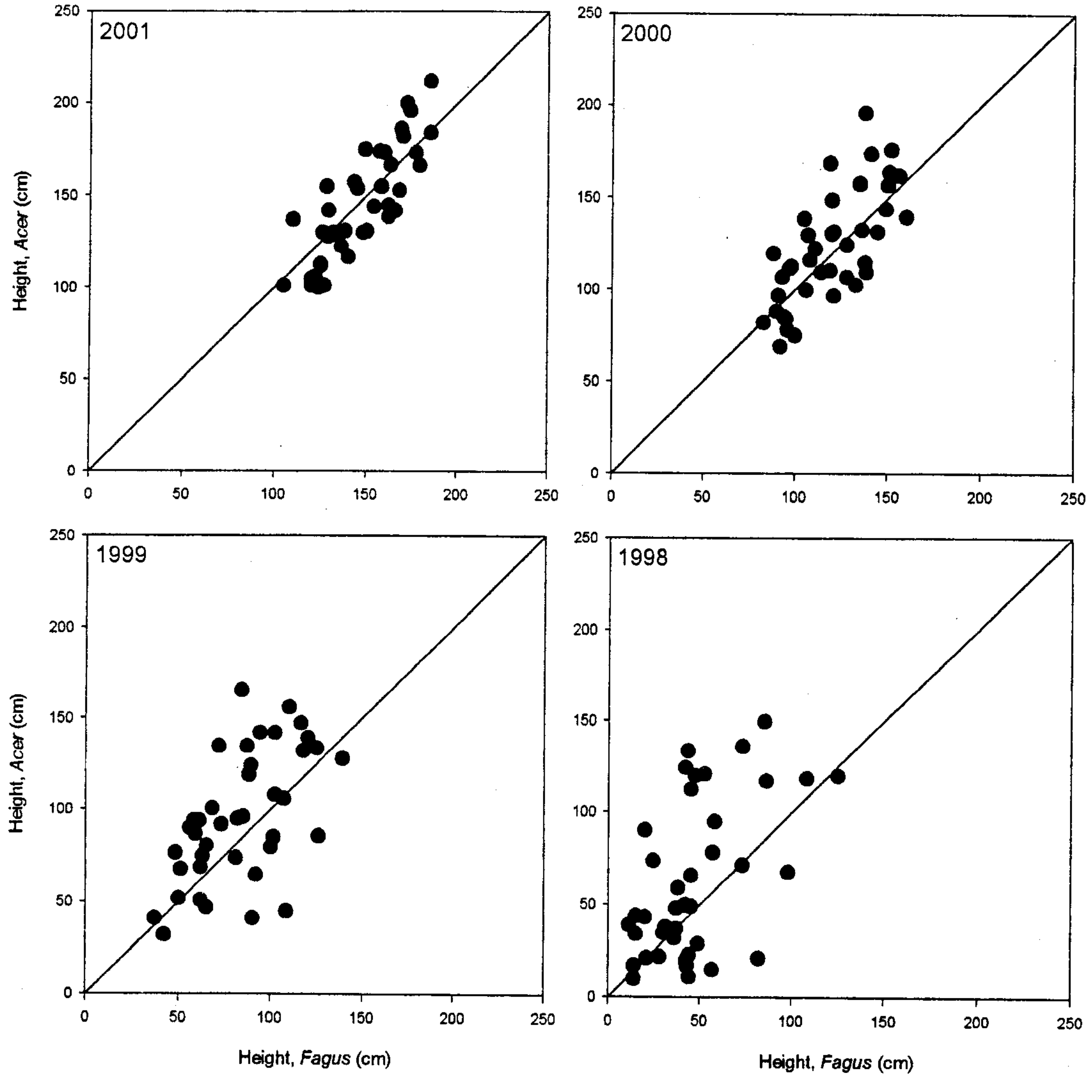

Figure 2. Height of *Acer* juveniles are plotted against the height of *Fagus* juveniles for the years between 1998 and 2001 (each point represents a paired *Acer-Fagus* juveniles).

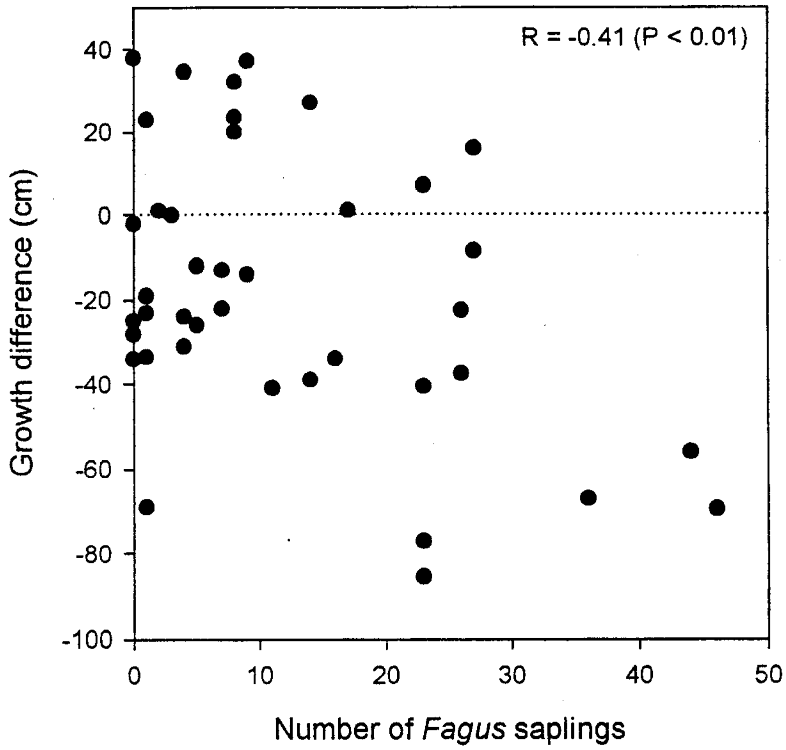

Figure 3. The difference between *Acer* juvenile total growth (1999-2001) and *Fagus* juvenile total growth (1999-2001) is plotted against the number of *Fagus* saplings in a plot. Significant negative correlation was found between the two variables (Pearson correlation $R = -0.41$, $P<0.01$).

figure shows that, as the number of *Fagus* saplings in a plot increases, the difference between *Acer* and *Fagus* juveniles become negative. This indicates that *Acer* juveniles are not growing as well as their *Fagus* counterparts in plots where *Fagus* saplings occur more frequently.

Discussion

Height growth comparison of *Fagus*-sprouts and *Acer* in gap light conditions

Under gap light conditions following an ice storm, sprout-origin *Fagus* seedlings/juveniles grew faster upward than *Acer*. This result is inconsistent with the previous notion that *Acer* grows faster than *Fagus* under increased light, which is based on comparisons between seed-origin *Fagus* and *Acer* (Poulson and Platt 1989, 1996). Our finding implies that one of the important premises of the disturbance hypothesis (i.e. *Acer* grows faster than *Fagus* under higher light levels; Poulson and Platt 1996) is not met when sprouts are considered.

Previous studies have paid relatively little attention to *Fagus* sprouts in understanding the *Fagus-Acer* forest dynamics (Abrell and Jackson 1977, Kupfer and Runkle 1996, Foré et al. 1997, Forrester and Runkle 2000); this, in large part, is because *Fagus* sprouts do not occur frequently at locations where previous studies were conducted. Low occurrence of *Fagus* sprouts may be due to one or a combination of the following factors: 1) geographical location, 2) soil depth and structure, and 3) genetic differences. Firstly, Held (1983) suggested that the relative importance of sprouts in *Fagus* regeneration increases with increasing latitude. Compared to the sites of other studies (e.g. southern Michigan, Indiana, Ohio), our site is located near the northern limit of the distribution range of *Fagus*; this may have contributed to the frequent occurrence of *Fagus* sprouts at our study site. In another old-growth *Fagus-Acer* forest in southwestern Quebec (approximately 100 km southwest of our study site), *Fagus* sprouts were found to dominate (Brisson et al 1994, Beaudet et al. 1999). Secondly, exposed root systems at the soil surface have been suggested to favor sprout production (Jones and Raynal 1988). Our study site has very rocky, shallow soil, where the root systems of *Fagus* trees are often found exposed at the soil surface. The differences in soil depth and structure may have contributed to the differences in *Fagus* sprout occurrence among the study sites. Thirdly, there may be genetic differences in the populations of *Fagus* among study sites. Previous study on geographical variations of *Fagus* has suggested that the root sprout populations have genetic similarity (Kitamura and Kawano 2001). This implies that whether or not sprout-origin individuals are found intensively at a particular location may depend on which genotype is dominant at that site (i.e. if genotypes that actively forms root sprouts are dominant, sprout-origin *Fagus* will be found abundantly). Note that the genetic differences may also explain the latitudinal differences in the intensity of root sprout formation (see Kitamura and Kawano 2001). Thus due to any one or any combinations of factors listed above, *Fagus* sprouts could have occurred less frequently at locations where previous studies were conducted, which may explain why earlier studies did not consider *Fagus* sprouts to be and important component of *Fagus-Acer* forest dynamics.

Another reason why *Fagus* sprouts received less attention in previous studies is because there have been questions of whether or not *Fagus* sprouts can attain canopy tree size (Woods 1979, 1984). For example, Tubbs and Houston (1990) report a study where all root sprouts died after 60-year-old *Fagus* trees were cut, indicating that sprouts do not become independent from the parent tree and that they will die soon after their parent dies;

this will prevent the sprouts from reaching the canopy. In contrast, some studies have clearly shown that sprouts play an important role in *Fagus* regeneration (Ward 1961, Jones and Raynal 1986, Brisson et al. 1994, Beaudet et al. 1999). Brisson et al. (1994) found sprout-origin *Fagus* saplings that were still attached to the old decaying stump of the parent tree at their study site. Genetic studies have also shown that some groups of overstory trees are in fact clones (Tubbs and Houston 1990, Kitamura et al. 2000).

Thus there is no doubt that, in some geographical locations, sprouts play an important role in *Fagus-Acer* forest dynamics. Our study is not the first to point out the potential importance of *Fagus* sprouts in *Fagus-Acer* forests (e.g. De Steven et al. 1991, Brisson et al. 1994, Beaudet et al. 1999). However, the importance of *Fagus* sprouts have been mainly argued in terms of their ability of grow faster upward and survive for longer periods of time under closed canopy conditions. Our finding that sprout-origin *Fagus* can also grow faster than *Acer* in elevated light conditions further emphasizes the potential importance of *Fagus* sprouts, as they are likely to out-compete *Acer*, whatever the light (disturbance) regime may be.

The effect of number of *Fagus* saplings on *Acer* growth

Acer growth in gap light conditions was lower at locations where *Fagus* saplings occurred abundantly; these were also locations where greater basal area of *Fagus* trees in the canopy is found (Arii and Lechowicz 2002b). At our study site, plots that have greater number of *Fagus* saplings typically have lower soil Ca availability (Arii and Lechowicz 2002b). Previous studies have shown that chemical properties of *Fagus* leaf litter can play a significant role in modifying the Ca availability. Leaf litter of *Fagus* has low pH (Finzi et al., 1998), and as the pH is lowered, availability of Ca also decreases (Ellis & Mellor, 1995). Additionally, *Fagus* leaf-litter is low in Ca concentration (Gosz et al., 1973; Côté & Fyles, 1994a,b) and also has high levels of lignin and polyphenols (Melillo et al., 1982; Aber et al., 1990), which can complex with Ca and increase leaching of these elements (Davies, 1971).

The deficiency in soil Ca availability has been shown to lower growth rates and cause mortality in *Acer*. Kobe et al. (2002) found that *Acer* seedlings show significant increases in growth (radial) after Ca fertilization (Kobe et al. 2002). Kobe et al. (1995) and Kobe (1996) also found that mortality of *Acer* saplings was higher in non-calcareous soils than calcareous soils under deep shade. *Acer* has been found to achieve maximum dominance at sites with high Ca content in northern Michigan (Woods 2001), and deficiencies in Ca have also been linked to the wide spread dieback of *Acer* in southern Quebec and adjacent regions (Liu et al., 1997). Thus, given the negative effects of Ca deficiency on *Acer*, it is not surprising to find lower *Acer* growth at sites with low Ca availability (i.e. sites where *Fagus* saplings occurred frequently) at our study site.

An argument could be made that the variability in *Acer* growth is a consequence of variation in gap light conditions following the ice storm (gap fraction values in the summer immediately after the ice storm ranged between 15-45%: Arii and Lechowicz 2002c). Low *Acer* growth may have been found at plots with less canopy gaps. However, the ice storm of 1998 significantly created more gaps at sites where *Fagus* saplings occurred in greater number (Arii and Lechowicz 2002c), and as a consequence, these sites had more gap fraction in the summer immediately following the storm. Thus, if *Acer* seedling/juvenile growth was strongly influenced by the variability in light after the storm, *Acer* growth should have correlated positively, not negatively, with the number of *Fagus* saplings.

Conclusion

In summary, we introduce a successional pathway, which can lead to *Fagus* dominance in *Fagus-Acer* forests. Based on our results and on a previous study (Beaudet et al. 1999), *Fagus* sprouts have an advantage in height growth over *Acer*, regardless of the light regime. This increases the chances of *Fagus* growing into a larger tree and reaching the canopy, which in turn can lead to more production of root sprouts. The increase in stem density of *Fagus* modifies the soil Ca availability, which amplifies the disadvantage of height growth for *Acer*. This cycle can result in a gradual exclusion of *Acer* from the stand, leading to *Fagus* dominance. As *Fagus* dominance is gradually being established, low light availability during the inter-gap phase period may also limit successful establishment of *Acer*. *Fagus* canopy has been shown to cast deeper shade than that of *Acer* (Beaudet et al. 2002), and at such low light conditions growth and survivorship of *Acer* are lowered, particularly in combination with low Ca availability (Kobe et al 1995, Kobe 1996). Thus once *Fagus* dominance is reached, the chances of *Acer* to re-establish at these locations will be quite low, unless a major disturbance (e.g. windthrow, fire, beech-bark disease) removes a significant portion of *Fagus* from the site. This successional pathway may explain why *Fagus* saplings dominated the understory at some locations at our study site, despite the frequent occurrence of ice storm disturbance.

Obviously, to initialize this pathway, seed-origin *Fagus* must establish and grow successfully to attain a size that can produce root sprouts. Seed-origin *Fagus* do not have the height growth advantage over *Acer* in gap light conditions (Poulson and Platt 1989, 1996, Canham 1988), so if the disturbance is frequent, they may not be able to reach a size that can produce sprouts. Additionally, if the conditions are unfavorable for seed-origin *Fagus* to establish (e.g. low soil moisture), *Acer* is likely to dominate a stand. Nonetheless, even under unfavorable conditions, seed-origin *Fagus* may successfully establish and grow by chance (e.g. long periods without major canopy disturbances, *Fagus* seed lands in a depression where soil moisture is higher). At our study site, we have observed small patches of *Fagus* in an area completely surrounded by stands dominated by *Acer*. These sites may represent locations where some seed-origin *Fagus* established by chance and have gradually expanded by roots sprouts (most of the *Fagus* saplings at these sites appear to be of sprout origin – personal observation).

The pathway that we invoke is limited to locations, or perhaps populations, where root sprouts are a primary mode of *Fagus* regeneration. Even if the *Fagus* sprouts are the primary mode of regeneration, the expansion by root sprouts is likely to be very gradual, as sprout extension is estimated to be approximately 7.5-10 m from the parent tree (Jones and Raynal 1986, Kitamura et al. 2000). Nonetheless, at locations where they occur, *Fagus* sprouts are likely to play an important role in the long-term dynamics of *Fagus-Acer* forests.

Acknowledgements

This work was supported by grants from Natural Sciences and Engineering Research Council of Canada (NSERC) and Fonds pour la Formation de Chercheurs et l'Aide à la Recherche (FCAR). We thank McGill University for access to the Gault Nature Reserve. We thank Jeff Yuen, Julie Girouard, and Mariam Futer for their assistance.

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Summary and original contributions

Acer saccharum and *Fagus grandifolia* are the two most dominant canopy tree species found at Mont St. Hilaire (MSH). The two species dominate sites with gentler slope and lower amounts of direct solar radiation received during the growing season, both of which are good indicators of higher soil moisture (Original contribution – Chapter 1). Soil moisture also had the strongest influence on the overall pattern of the saplings found at MSH; *Acer* and *Fagus* saplings dominated the understory at the mesic sites (Original contribution – Chapter 3). Because these mesic sites are also where *Fagus* and *Acer* dominate the canopy, *Fagus-Acer* stands that are currently found at MSH are likely to persist in the long-term.

At these sites dominated by *Acer* and *Fagus*, early successional, shade-intolerant species such as *Populus grandidentata* and *Betula papyrifera*, were rarely found in either the overstory or the understory. Given the frequent occurrence of major ice storms that cause considerable canopy damage (Hooper et al. 2001, Proulx and Green 2001), and given the known strategy of early successional species to establish and grow rapidly under gap light conditions, I had expected early successional species to be more abundant at MSH, even at sites where *Fagus* and *Acer* dominate. My study that examined the changes in light regimes following the 1998 ice storm showed that gaps created by this storm returned to pre-storm levels within three years after the storm (Original contribution – Chapter 2). This recovery time is much shorter than what is expected for a treefall gap (approximately 10 years). This suggests gaps created by ice storms may not provide sufficient duration of gap light conditions to allow successful establishment and growth of the earlier successional species. Conversely, these ice storm-origin gaps may promote the shade-tolerant species to reach canopy height more quickly, as the shade-tolerant species are known to require multiple gap encounters before they can reach the canopy (Canham 1990). Thus, in contrary to the general notion that disturbances play an important role in maintaining species diversity by allowing less shade-tolerant species to establish, disturbances in the form of ice storms may in fact accelerate succession, and also help maintain the dominance of later successional, shade-tolerant species (i.e. *Fagus* and *Acer*) (Original contribution – Chapter 2).

Based on the disturbance hypothesis by Poulson and Platt (1996), frequent occurrence of disturbances is predicted to be more advantageous for *Acer* recruitment than for *Fagus* in *Fagus-Acer* forests. At MSH, relatively large-scale ice storms have occurred approximately every 20 years for the past 60 years (major storms have been recorded in 1942, 1961, 1983, 1998). Because a single ice storm can damage canopy trees over an extensive area, *Acer* was expected to be the dominant species in the understory at most localities at MSH. An examination of sapling distribution pattern at MSH showed that *Acer* saplings occur in high density at most locations, as expected under the disturbance hypothesis (Chapter 3). However, there were some localities at the study site where *Acer* saplings were almost absent; *Fagus* saplings dominated the understory at these sites (Chapter 3). The presence of sites dominated by *Fagus* saplings is inconsistent with the predictions made by the disturbance hypothesis given the disturbance frequency at MSH. But these sites dominated by *Fagus* saplings were particularly peculiar because they were mostly found under *Fagus* dominated canopy (Original contribution – Chapter 3). Because *Fagus* trees are more prone to ice storm damage than *Acer* (Melancon and Lechowicz 1987, Hooper 2001), sites where *Fagus* trees dominate the overstory have greater increase in gap opening after the storm (Original contribution – Chapter 2). Thus the light conditions at sites where *Fagus* trees dominate the canopy should be particularly suitable for the recruitment of *Acer*, yet it is in this situation that *Acer* saplings are absent.

The absence of *Acer* at sites where *Fagus* dominate both the understory and the overstory, can be explained by a combination of two factors. Firstly, *Acer* growth is retarded at sites where *Fagus* saplings occur in greater abundance (Original contribution - Chapter 4). Sites where *Fagus* saplings occur in higher density typically have lower soil Ca-availability (Original contribution - Chapter 3), and Ca deficiency has been shown to cause negative effects on *Acer* (e.g. Kobe et al. 2002). This may explain why *Acer* growth is retarded at sites dominated by *Fagus* saplings, which contradicts the prediction made by the reciprocal replacement hypothesis (Fox 1981, Woods 1979, 1984, Runkle 1981). Secondly, *Fagus* seedlings and saplings of sprout-origin, which occur extensively at our study site, grow upward faster than *Acer*, even in gap light conditions (Original contribution - Chapter 4). Therefore, at sites where *Fagus* trees dominate, the combined effects of low Ca-availability and *Fagus* sprouts can give *Fagus* saplings substantial growth advantage over *Acer* even in gap light conditions. This may explain why *Acer* was absent at sites dominated by *Fagus*, which led to sapling distribution pattern that is inconsistent predictions made by the disturbance hypothesis.

Summarizing the findings in this thesis, I propose a conceptual model that can lead to *Fagus* dominance despite the frequent occurrence of disturbances (Fig. 1 – Original contribution). The model can be viewed as an extension of the model proposed by Poulson and Platt (1996) for locations with frequent disturbances. Consider an early successional forest where *Fagus* and *Acer* begin to establish (seed-origin). Because the light level is still relatively high beneath the early successional canopy, *A. saccharum* seedlings/saplings generally will perform better than *Fagus* seedlings/saplings, except at sites where the conditions are less favorable for *Acer* (e.g. acidic soil with low Ca-availability). At sites where *Acer* saplings perform well, they will eventually dominate the overstory, and frequent canopy

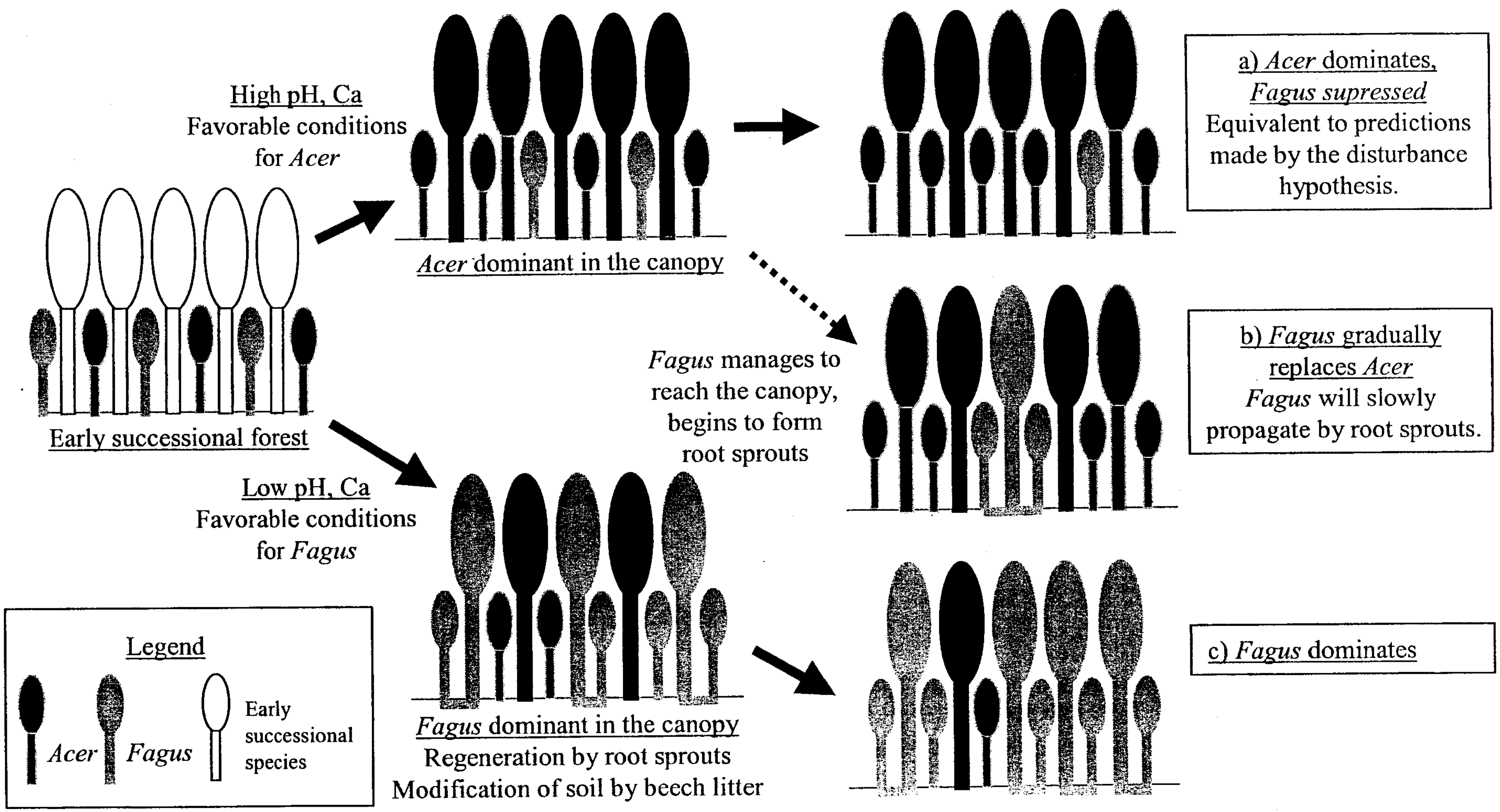

Figure 1. Illustration of a conceptual model that can lead to *Fagus* dominance even under high disturbance frequency, which under the disturbance hypothesis should favor only *Acer*. See text for details.

disturbance will ensure continuous regeneration of *Acer* at these locations (Fig.1 path (a)). This is what is predicted by the disturbance hypothesis for areas where disturbances occur frequently. At locations where *Acer* seedling/sapling growth is retarded, *Fagus* saplings will capture a greater proportion of the overstory (Fig. 1 path (c)). These overstory *Fagus* trees will produce sprouts and modify the soil conditions to be less suitable for *Acer* recruitment (*Fagus* leaf litter lowers the pH and Ca availability – see Discussion in Chapter 3). These combined effects of sprouts and edaphic factors will prevent *Acer* from successfully establishing and growing at these sites, which will lead to dominance by *Fagus* saplings/trees, and ensure "self-replacement" of *F. grandifolia* at these sites (Fig. 1 path (c)). Even at locations where conditions are favorable for *Acer*, if several *Fagus* seedling/saplings of seed-origin are able to establish by chance and grow into overstory trees, they may be able to slowly replace the neighboring *Acer* stands over time by the same mechanism as path (c) (Fig. 1 path (b)). At our study site, we have observed numbers of small patches consisting of almost pure *Fagus grandifolia* stands, both in the overstory and understory, in an area where *Acer* dominates. These patches may suggest the rare occasions where *F. grandifolia* of seed-origin were able to establish by chance and grow into overstory trees at sites more favorable for *Acer*.

Overall, this thesis demonstrates that edaphic factors and regeneration of *Fagus* by root sprouts can play a significant role in the dynamics of *Fagus-Acer* forests. Although the effects of edaphic factors and *Fagus* spouts are not considered in the disturbance hypothesis, several previous studies have suggested that these two factors can potentially affect the community pattern and dynamics in *Fagus-Acer* forests (Forcier 1975, Melancon and Lechowicz 1987, Brisson et al. 1994, Beaudet et al. 1999). This study contributes significantly to the growing evidence that edaphic factors and *Fagus* sprouts play a role in the dynamics of *Fagus-Acer* forests.

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Appendix 1

Note on the use of hemispherical photographs to measure canopy gap fraction

In Chapter 2, hemispherical photography was used to measure canopy gap fraction values. As the name implies, hemispherical photography requires an extreme wide-angle (180-degree fisheye) lens to capture a complete picture of the canopy from the forest floor (the lens used in this study was a Samigon 180-degree Fisheye Lens, owned by Prof. Lechowicz). Once the pictures are taken with the fisheye lens, these images are digitized with a scanner and analyzed with an image analysis software. Although there are only few steps involved from taking the hemispherical photographs to analyzing them, there are many details that one must be cautious of during this process.

Here, rather than outlining the step-by-step procedure of taking and analyzing hemispherical photographs, I cite a reference that describes the essentials of the procedures in detail. Note that the references cited in this appendix are all included in Prof. Lechowicz's reprint collection. In my opinion, the most useful and comprehensive reference related to hemispherical photography is a report by Frazer et al. (1997). Detailed procedures and some caution on how to take and analyze the hemispherical photographs are given in pages 15-21 of this report. This report also has a very good reference list.

There are several image analysis computer programs that can be used to analyze the digitized photographs (e.g. GLI/C, HEMIPHOT – see Frazer et al. 1997). The programs that I used for my study are,

1) GLA ver. 2.0 (http://www.rem.sfu.ca/forestry/gla/)
2) LIA32 (http://hp.vector.co.jp/authors/VA008416/index.html)

Both of these programs are freewares that can be downloaded from the internet (web addresses are given above – Last checked July 21, 2002). Both programs are Windows-based. GLA is very good program that comes with an excellent manual, which is also available from the internet (Frazer et al. 1999). Not only does the software calculate gap fraction values, it also computes effective leaf area index (L_e), sunfleck-frequency distribution and daily duration, and the amount of above- and below-canopy (transmitted) direct, diffuse, and total solar radiation incident on a horizontal or arbitrarily inclined receiving surface. An overview of this program is given in Frazer et al. (2000). LIA32 is a program developed by a Japanese researcher; unfortunately the program is available only in Japanese. The program computes less output than GLA, but it is a much smaller program, with simple steps to analyze the image. Perhaps there will be an English version in the future.

The hemispherical photographs I used for my study were taken with a 35 mm film. With this method, the film must be developed, printed, and scanned (i.e. digitized), before it can be used in the image analysis software. These steps can potentially become sources of error if not done carefully. Additionally, if the images are taken on film, one cannot determine if the photographs taken are acceptable images until they are printed. The use of digital cameras can solve both of these problems. Images taken with a digital camera can be directly exported to an image analysis software, and the LCD screen on the camera can be used to check the quality of the image instantly.

Currently, several companies offer hemispherical photography systems for use in studies investigating canopy structure and light regimes in forested ecosystems. These systems come with all the essential instruments/programs necessary to take and analyze hemispherical photographs (i.e. digital camera, fisheye-lens, and image analysis software). The system requires fewer steps than the conventional methods because of the use of digital cameras. Additionally, because these instruments/programs come in a bundle, there are fewer details that one needs to be concerned about when taking and analyzing hemispherical photographs. If these systems are available, I would strongly recommend using these systems over the conventional method I used for my study. Below I give two examples of such all-in-one systems.

Hemiview 2.1
The image analysis software was developed by Helios Environmental Modeling Institute, LLC. (Kansas City, U.S.A.; http://www.hemisoft.com), and the software, which is bundled with a digital camera and fisheye lens, are available commercially at AT Delta-T Devices, Ltd. (Cambridge, U.K.; http://www.delta-t.co.uk).

SCANOPY
An all-in-one system available from Regents Instruments Inc. (Quebec, Canada; http://www.regent.qc.ca)

The images taken with these systems can also be analyzed with software other than the ones provided with the system. Note that such all-in-one system is currently available at GREFi (Groupe de recherche en écologie forestière interuniversitaire).

Recommended References (All available in Prof. Lechowicz's reprint collection).

Gives overall information on the use of hemispherical photography. Also has a good reference list for publications prior to 1997.

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Appendix 2

Location of plots used at MSH

		Plot number	Plot number	GPS measurements		
Plot number	Setup year	1997	Permanent tags	Latitude	Longitude	Elevation
P001	1997	T2-P01	KA020	45.54017000	-73.14775000	265.87200000
P002	1997	T2-P02	KA021	45.53991000	-73.14767000	251.49900000
P003	1997	T2-P03	KA022	45.53956000	-73.14755000	230.61400000
P004	1997	T2-P04	KA023	45.53931000	-73.14741000	214.90500000
P005	1997	T2-P05	KA024	45.53902000	-73.14726000	200.62300000
P006	1997	T2-P06	KA025	45.53881000	-73.14699000	186.79100000
P007	1997	T2-P07				
P008	1997	T2-P08	KA026	45.54166000	-73.13944000	144.90000000
P009	1997	T2-P09	KA027	45.54196000	-73.13970000	166.97000000
P010	1997	T2-P10	KA028	45.54212000	-73.13981000	172.00700000
P011	1997	T2-P11	KA029	45.54247000	-73.13988000	186.35100000
P012	1997	T2-P12	KA030	45.54274000	-73.14003000	199.98100000
P013	1997	T2-P13	KA031	45.54310000	-73.14022000	219.50200000
P014	1997	T2-P14	KA032	45.54351000	-73.14033000	241.42300000
P015	1997	T2-P15	KA033	45.54373000	-73.14048000	250.46300000
P016	1997	T2-P16	KA034	45.54392000	-73.14049000	259.28700000
P017	1997	T2-P17	KA035	45.54424000	-73.14069000	275.44200000
P018	1997	T2-P18	KA036	45.54460000	-73.14095000	287.66700000
P027	1997	T2-P27	KA037	45.55640000	-73.16391000	403.60600000
P028	1997	T2-P28	KA038	45.55588000	-73.16372000	394.53500000
P029	1997	T2-P29	KA039	45.55544000	-73.16351000	377.42100000
P030	1997	T2-P30	KA040	45.55484000	-73.16314000	359.58100000
P031	1997	T2-P31	KA041	45.55467000	-73.16306000	349.42000000
P032	1997	T2-P32	KA042	45.55429000	-73.16295000	337.97800000
P033	1997	T2-P33	KA043	45.55387000	-73.16288000	320.81900000
P034	1997	T2-P34	KA044	45.55370000	-73.16286000	318.86600000
P035	1997	T2-P35	KA045	45.55346000	-73.16294000	301.59400000
P036	1997	T2-P36	KA046	45.55314000	-73.16283000	288.78100000
P037	1997	T2-P37	KA047	45.55278000	-73.16276000	268.21000000
P038	1997	T2-P38	KA048	45.55240000	-73.16267000	261.22200000
P039	1997	T2-P39	KA049	45.55186000	-73.16251000	234.11700000
P040	1997	T2-P40	KA050	45.55135000	-73.16237000	218.03100000
P041	1997	T2-P41	KA051	45.55096000	-73.16224000	211.89600000
P042	1997	T2-P42	KA052	45.55051000	-73.16212000	210.58300000
P043	1997	T2-P43	KA053	45.55023000	-73.16198000	206.61100000
P044	1997	T2-P44	KA054	45.54964000	-73.16178000	205.00100000
P045	1997	T2-P45	KA055	45.54931000	-73.16147000	209.08300000
P046	1997	T2-P46	KA056	45.54896000	-73.16130000	200.62200000
P047	1997	T2-P47	KA057	45.54882000	-73.16108000	202.38800000
P048	1997	T2-P48	KA058	45.54810000	-73.16081000	203.48100000
P049	1997	T2-P49	KA059	45.54786000	-73.16076000	198.83100000
P050	1997	T2-P50	KA060	45.54730000	-73.16047000	206.64700000
P051	1997	T2-P51	KA061	45.54699000	-73.16036000	204.25400000
P052	1997	T2-P52	KA062	45.54656000	-73.15996000	208.84000000
P053	1997	T2-P53	KA063	45.54596000	-73.15957000	204.02900000
P054	1997	T2-P54	KA064	45.54574000	-73.15940000	202.43400000
P055	1997	T2-P55	KA065	45.55662000	-73.16977000	296.44800000
P056	1997	T2-P56	KA066	45.55690000	-73.16915000	311.24400000
P057	1997	T2-P57	KA067	45.55688000	-73.16874000	325.65500000
P058	1997	T2-P58	KA068	45.55702000	-73.16845000	335.14000000
P059	1997	T2-P59	KA069	45.55710000	-73.16771000	353.04700000

		Plot number	Plot number	GPS measurements		
Plot number	Setup year	1997	Permanent tags	Latitude	Longitude	Elevation
P060	1997	T2-P60	KA070	45.55722000	-73.16731000	357.57800000
P061	1997	T2-P61	KA071	45.55739000	-73.16671000	364.61200000
P062	1997	T2-P62	KA072	45.55736000	-73.16637000	370.97500000
P063	1997	T2-P63	KA073	45.55762000	-73.16516000	381.96500000
P064	1997	T2-P64	KA074	45.55777000	-73.16471000	382.56600000
P065	1997	T2-P65	KA075	45.55780000	-73.16448000	382.83500000
P066	1997	T2-P66	KA076	45.55796000	-73.16396000	375.55700000
P067	1997	T2-P67	KA077	45.55791000	-73.16365000	373.70400000
P068	1997	T2-P68	KA078	45.55806000	-73.16279000	359.86700000
P069	1997	T2-P69	KA079	45.55820000	-73.16230000	339.47800000
P070	1997	T2-P70	KA080	45.55844000	-73.16126000	298.84800000
P071	1997	T2-P71	KA081	45.55853000	-73.16086000	288.65000000
P072	1997	T2-P72	KA082	45.55870000	-73.16030000	271.99300000
P073	1997	T2-P73	KA083	45.55887000	-73.15939000	267.93000000
P074	1997	T2-P74	KA084	45.55901000	-73.15858000	259.25800000
P075	1997	T2-P75	KA085	45.55489000	-73.13767000	243.10400000
P076	1997	T2-P76	KA086	45.55480000	-73.13817000	262.11500000
P077	1997	T2-P77				
P078	1997	T2-P78	KA087	45.55455000	-73.13884000	275.65800000
P079	1997	T2-P79	KA088	45.55445000	-73.13948000	292.37800000
P080	1997	T2-P80	KA089	45.55436000	-73.14002000	290.33400000
P081	1997	T2-P81	KA090	45.55422000	-73.14041000	300.75500000
P082	1997	T2-P82	KA091	45.55410000	-73.14101000	309.84400000
P083	1997	T2-P83	KA092	45.55396000	-73.14164000	317.02000000
P084	1997	T2-P84	KA093	45.55371000	-73.14293000	339.63700000
P085	1997	T2-P85	KA094	45.55357000	-73.14340000	345.20300000
P086	1997	T2-P86				
P087	1997	T2-P87	KA095	45.55317000	-73.14533000	330.60800000
P088	1997	T2-P88	KA096	45.55288000	-73.14617000	316.24500000
P089	1997	T2-P89	KA097	45.55258000	-73.14717000	306.61800000
P090	1997	T2-P90	KA098	45.55238000	-73.14768000	303.61600000
P091	1997	T2-P91	KA099	45.55217000	-73.14827000	288.84100000
P092	1997	T2-P92	KA100	45.55200000	-73.14916000	282.40600000
P093	1997	T2-P93	KA101	45.55178000	-73.14986000	261.84200000
P094	1997	T2-P94	KA102	45.55161000	-73.15062000	261.11800000
P095	1997	T2-P95	KA103	45.55140000	-73.15128000	231.68700000
P096	1997	T2-P96	KA104	45.55124000	-73.15172000	204.68800000
P097	1997	T2-P97				
P098	1997	T2-P98				
P099	1997	T2-P99	KA105	45.54554000	-73.16028000	207.61000000
P100	1997	T2-P100	KA106	45.54586000	-73.15970000	203.16700000
P101	1997	T2-P101	KA107	45.54591000	-73.15954000	206.70600000
P102	1997	T2-P102	KA108	45.54610000	-73.15855000	195.47900000
P103	1997	T2-P103	KA109	45.54622000	-73.15829000	194.91000000
P104	1997	T2-P104	KA110	45.54655000	-73.15741000	199.71800000
P105	1997	T2-P105	KA111	45.54658000	-73.15720000	205.51100000
P106	1997	T2-P106	KA112	45.54689000	-73.15646000	202.43700000
P107	1997	T2-P107	KA113	45.54712000	-73.15570000	193.86600000
P108	1997	T2-P108	KA114	45.54733600	-73.15519100	194.32645000
P109	1997	T2-P109	KA115	45.54739000	-73.15459000	192.94000000

		Plot number	Plot number	GPS measurements		
Plot number	Setup year	1997	Permanent tags	Latitude	Longitude	Elevation
P110	1997	T2-P110	KA116	45.54721600	-73.15418100	188.25452000
P111	1997	T2-P111	KA117	45.54718000	-73.15368000	177.52104500
P112	1997	T2-P112	KA118	45.54717500	-73.15347100	176.36958600
P201	1997	T1-P1	KA001	45.53679000	-73.15092000	147.09000000
P202	1997	T1-P2	KA002	45.53728000	-73.15124000	160.43600000
P203	1997	T1-P3	KA003	45.53797000	-73.15160000	177.77100000
P204	1997	T1-P4	KA004	45.53846000	-73.15182000	191.07800000
P205	1997	T1-P5	KA005	45.53885000	-73.15185000	209.22100000
P206	1997	T1-P6	KA006	45.53897000	-73.15199000	209.12900000
P207	1997	T1-P7	KA007	45.53947000	-73.15230000	214.51500000
P208	1997	T1-P8	KA008	45.53979000	-73.15250000	213.31500000
P209	1997	T1-P9	KA009	45.53994000	-73.15260000	211.57900000
P210	1997	T1-P10	KA010	45.54049000	-73.15269000	212.34100000
P211	1997	T1-P11	KA011	45.54064000	-73.15284000	206.25200000
P212	1997	T1-P12				
P213	1997	T1-P13	KA012	45.54288000	-73.14890000	183.57800000
P214	1997	T1-P14	KA013	45.54265000	-73.14869000	202.21800000
P215	1997	T1-P15	KA014	45.54228000	-73.14854000	229.44600000
P216	1997	T1-P16	KA015	45.54188000	-73.14851000	247.48400000
P217	1997	T1-P17	KA016	45.54149000	-73.14842000	265.37600000
P218	1997	T1-P18	KA017	45.54104000	-73.14811000	271.97500000
P219	1997	T1-P19	KA018	45.54073000	-73.14814000	279.59500000
P220	1997	T1-P20	KA019	45.54051000	-73.14805000	277.40000000
P301	2000	n/a	KA119	45.53991700	-73.16020100	193.45004600
P302	2000	n/a	KA120	45.54010200	-73.15935700	193.33671900
P303	2000	n/a	KA121	45.54018200	-73.15887400	188.78148200
P304	2000	n/a	KA122	45.54059800	-73.16325200	241.53647200
P305	2000	n/a	KA123	45.54117700	-73.16442900	269.57504600
P306	2000	n/a	KA124	45.54251100	-73.16748800	293.81117600
P307	2000	n/a	KA125	45.54308500	-73.16974700	280.65004300
P308	2000	n/a	KA126	45.54390500	-73.17088000	267.02409700
P309	2000	n/a	KA127	45.54695200	-73.17095300	297.93339800
P310	2000	n/a	KA128	45.54281600	-73.16248200	216.98147900
P311	2000	n/a	KA129	45.54191000	-73.16172700	223.44375900
P320	2000	n/a	KA137	45.54396000	-73.14479400	186.40813000
P321	2000	n/a	KA138	45.54397200	-73.14450000	176.39740300
P322	2000	n/a	KA139	45.54379000	-73.14259600	189.69717700
P323	2000	n/a	KA140	45.55210300	-73.13065700	139.19162300
P324	2000	n/a	KA141	45.55155700	-73.13073900	134.60259600
P325	2000	n/a	KA142	45.55136300	-73.13049800	143.01982400
P326	2000	n/a	KA143	45.55519400	-73.13655500	245.69208100
P327	2000	n/a	KA144	45.55566500	-73.13650400	264.72980000
P328	2000	n/a	KA145	45.55602800	-73.13695400	256.08859600

		Plot number	Plot number	GPS measurements		
Plot number	Setup year	1997	Permanent tags	Latitude	Longitude	Elevation
P329	2000	n/a	KA146	45.55638600	-73.13700600	267.72223200
P330	2000	n/a	KA147	45.55601700	-73.13776600	248.91714800
P331	2000	n/a	KA148	45.55680500	-73.13893100	262.34477500
P332	2000	n/a	KA149	45.55850100	-73.14161700	242.73338600
P333	2000	n/a	KA150	45.56063000	-73.15134900	211.85116900
P334	2000	n/a	KA151	45.56030200	-73.15043800	214.49843100
P335	2000	n/a	KA152	45.55359800	-73.15288200	191.30314900
P336	2000	n/a	KA153	45.54926400	-73.15208400	184.88400600
P340	2000	n/a	KA132	45.55332300	-73.16920500	285.95566100
P341	2000	n/a	KA136	45.56107900	-73.17461700	355.45006100
P342	2000	n/a	KA135	45.56042900	-73.17497500	348.96419100
P343	2000	n/a	KA134	45.56027600	-73.17242300	329.22114900
P344	2000	n/a	KA133	45.55915600	-73.17199200	317.72652000
P345	2000	n/a	KA130	45.54495100	-73.16178900	189.12634600
P346	2000	n/a	KA131	45.54434400	-73.16141700	186.24888900

Appendix 3

Archival data files

This appendix provides a list of computer data files that were used in the thesis. Electronic copies of the files are kept by Professor Martin J. Lechowicz and myself (contact addresses are given at the end of this appendix). The directory names are given in bold, files names are underlined.

Overall Directory structure

I) Subdirectory: **DATA**

This directory has files containing **raw data**.

II) Subdirectory: **CHAPTER 1**

Files related to Chapter 1 – text and files used for analyses.

III) Subdirectory: **CHAPTER 2**

Files related to Chapter 2 – text and files used for analyses.

IV) Subdirectory: **CHAPTER 3**

Files related to Chapter 3 – text and files used for analyses.

V) Subdirectory: **CHAPTER 4**

Files related to Chapter 4 – text and files used for analyses.

VI) Subdirectory: **OTHERS**

Contains files that were used to compile the thesis (Introduction, conclusions, appendix, etc.)

Directory details

I) Subdirectory **DATA**

1) <u>Environmental_data.xls</u>

   Contains environmental data (e.g. physical factors, nutrient data, moisture data, etc.)

2) <u>Species_abb.xls</u>

   Contains an acronym conversion table for trees found in the permanent plots. This file is needed for <u>Tree_data.xls (the next file)</u>.

3) <u>Tree_data.xls</u>

   Contains tree data that were surveyed in 1997 and 2000. All trees and saplings DBH>=1 cm.

4) Subdirectory **Hemispherical photo related**

   a) Gaps_data.xls

      Gap fraction data from 1997 to 2000. Changes in gap fraction values are also given. Refer to the GLA manual (Frazer et al. 1999) for column headings.

   b) solar radiation.zip

      Estimated values of solar radiation – calculated with GLA ver. 2. This file contains four separate files, each file representing different years between 1997-2000. Refer to GLA manual (Frazer et al. 1999) for column headings.

   c) Subdirectory **F-e photos**

      Contains Fish eye photographs used in the thesis (1997-2000). These are black-white images (images saved after the threshold procedure). Also contains configuration files for each year (caution! the parameters differ for each year and each plot).

   d) Subdirectory **GLA parameters**

      1. 1997.zip

         This contains configuration files to be used separately for each plot (for 1997 images). The next three files are for different years, from 1998-2000. The input parameters are estimated from the last four files in this directory.

      2. 1998.zip
      3. 1999.zip
      4. 2000.zip
      5. ENV_used for GLA configuration of each plot.xls

         Contains the physical environmental variables used in the GLA configuration file.

      6. Radiation_data.xls

         This file contains radiation data that are modified from the original file (**/Radiation Data Purchased from Env. Canada/**Radition.zip). For legends and units, check the files in Radiation.zip.

      7. daily_average_97-00_calculation of parameters.xls

         This file contains radiation data used to calculate the parameters for the GLA configuration file. Refer to GLA manual (Frazer et al. 1999) for abbreviations.

      8. GLA parameter summary.xls

         This file contains summary tables of parameters to be used in GLA. Refer to GLA manual (Frazer et al. 1999) for abbreviations.

5) Subdirectory **Radiation data purchased from Env. Canada**

   a) Radiation.zip

      Contains data files purchased from Environment Canada (1963-2001). Another copy of the data file is kept in Professor Lechowicz's lab archival data. The contents should be self-explanatory.

II) Subdirectory **Chapter 1**

1) subdirectory **analysis**

a) data files.zip - This file contains data files used in chapter 1 (CANOCO format). This file contains the following ".dte" files.

1. Cca_ax1.dte – Axis 1 scores are added to ENV.dte. Use to do the Monte Carlo permutation test.
2. CCA_FW_AX1.dte – Axis 1 scores added to ENV.dte. Use to the Monte Carlo permutation test. This is the file used for CCA with forward selection.
3. ENV.dte – environmental (explanatory) data.
4. For_tree.dte – Tree data for forest cover analysis. Data obtained from the Mont St. Hilaire Nature Centre, modified by B. Hamel..
5. Tree_IV.dte – Tree data (importance value).

b) solution files.dte – This file contains output files (solution file) of CANOCO. Contains ".sol" files

1. CCA.sol – solution file for the original CCA.
2. CCA_ax1.sol – solution file for the original CCA – effect of first axis removed by using Cca_ax1.dte.
3. CCA_FW.sol – solution file for CCA with forward selection.
4. CCA_FW_ax1.sol – solution file for CCA with forward selection - effect of first axis removed by using CCA_FW_AX1.dte.

2) Subdirectory **text**

03-chapter1.doc – Chapter 1 text, figures, and tables

III) Subdirectory **Chapter 2**

1) Subdirectory **analysis**

a) data files1.zip - This file contains data files used in this chapter. (".xls" files - Microsoft Excel).

1. 1_matrix.xls – Sapling data.
2. 2_matrix.xls – Environmental data (Overstory tree + abiotic)
3. 2_matrix_axis.xls - Environmental data (Overstory tree + abiotic) and first axis scores from all analysis (use in Mote Carlo Permutation test)

b) data files2.zip - This file contains data files used in this chapter (CANOCO format). This file contains the following ".dte" files.

1. 1_matrix.dte – Sapling data.
2. 2_matrix.dte – Environmental data (Overstory tree + abiotic)
3. 2_matrix_axis.dte - Environmental data (Overstory tree + abiotic) and first axis scores from all analysis (use in Mote Carlo Permutation test)
4. 2_matrix_standardized.xls – same as "2_matrix.dte" but all the environmental variables are standardized.

c) solution files.zip – This file contains output files (solution file) of CANOCO. Contains ".sol" files.

1. AM-mix.sol; AM-mix-axis.sol – CCA with abiotic variables as explanatory variables. "-axis.col" also has the first axis scores (use for Monte Carlo permutation test on the second axis).
2. AM-pure.sol; AM-pure-axis.sol – CCA with abiotic variables as explanatory variables, and overstory tree variables as covariables. "-axis.col" also has the first axis scores (use for Monte Carlo permutation test on the second axis).
3. TM-mix.sol; TM-mix-axis.sol – CCA with overstory tree variables as explanatory variables. "-axis.col" also has the first axis scores (use for Monte Carlo permutation test on the second axis).
4. TM-pure.sol; TM-pure-axis.sol – CCA with overstory tree variable as explanatory variables, and abiotic variables as covariables. "-axis.col" also has the first axis scores (use for Monte Carlo permutation test on the second axis).
5. Total.sol; Total-axis.sol – CCA with both abiotic and overstory tree variables as explanatory variables. "-axis.col" also has the first axis scores (use for Monte Carlo permutation test on the second axis).
6. Total_2matrix_standardized.sol - CCA with both abiotic and overstory tree variables as explanatory variables. All the explanatory variables are transformed (mean of 0, S.D. of 1).
7. Total_wo-asp - CCA with both abiotic and overstory tree variables as explanatory variables. *Acer spicatum* removed from analysis.

8. Total_wo-oak - CCA with both abiotic and overstory tree variables as explanatory variables. *Quercus rubra* removed from analysis.

2) Subdirectory **text**

04-Chpater2.doc - Chapter 1 text, figures, and tables.

IV) Subdirectory **Chapter 3**

1) Subdirectory **analysis**

a) Changes_gaps_for manuscript data.xls – gap fraction data from 1997 to 2000.

b) Changes_gaps_for manuscript data_correlation.xls – gap fraction data – comparison of partial gap fraction (angle 0-60 degrees) and total gap fraction (angle 0-90 degrees).

c) BA_U10.dte – basal area of saplings below 10 cm in DBH. For use with CANOCO.

d) BA_U10.sol – CANOCO output file. (PCA on BA_U10.dte)

e) BA_U10_axis.xls – site scores of four PCA axes. Also included in Changes_gaps_for manuscript data.xls.

3) Subdirectory **text**

a) 05-Chapter3.doc – Chapter 3 text.

b) 06-Chapter3_figures.doc – Chapter3 figures.

c) 07-Chapter3_tables.doc – Chapter 3 tables.

V) Subdirectory **Chapter 4**

1) Subdirectory **analysis**

a) sd-regression.zip – Contains files related to seedling analysis. Following files are included.

1. Environmental data_correlation with sapling number.xls – Used to correlate number of *Fagus* saplings with various environmental variables.

2. sd-for Fig1.xls; sd-Data modified to make Figure data sheet.xls – Used to make Fig. 1 in Chapter 4. Also used to do the regression analysis.

3. sd-ACER00-RESID.xls – Original data and residuals from regression (*Acer* seedling growth 2000).

4. sd-ACER99-RESID.xls - Original data and residuals from

regression (*Acer* seedling growth 1999).

5. sd-FAGUS00-RESID.xls - Original data and residuals from regression (*Fagus* seedling growth 2000).
6. sd-FAGUS99-RESID.xls - Original data and residuals from regression (*Fagus* seedling growth 1999).
7. sd-regression results.txt – Regression results.

b) sapling.xls – juvenile growth data (paired)

c) sapling-paired-sample-t.xls – juvenile growth data used for paired sample t-test.

d) Environmental data_correlation with sapling number – file used to correlate number of *Fagus* saplings in a plot with various environmental variables.

2) Subdirectory **text**

08-Chapter4.doc – Chapter 4 text, figures and tables.

VI) Subdirectory **Others**

1) 01-Preface.doc – abstract, table of contents, preface and acknowledgements
2) 02-General Introduction.doc – general introduction
3) 09-Summary.doc – summary and original contributions
4) 10-appendix.doc – appendix

(Contact addresses)

Professor Martin J. Lechowicz
Department of Biology, McGill University
1205 Ave. Docteur Penfield, Montreal, QC, H3A 1B1, Canada
martin.lechowicz@mcgill.ca

Ken Arii (permanent address)
1-30-6, Seijo, Setagaya-ku, Tokyo, 157-0066, Japan
ken.arii@mail.mcgill.ca

The forest cover map of Mont St. Hilaire can be obtained at Mont St. Hilaire Nature Centre.